Multivariate Real Analysis

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Contents

In here, we extend the results of univariate real analysis to multivariate and/or vector-valued functions. In practice, multivariate calculus is used, and there are many new results that arise in the multivariate case. The case for computing limits and continuity on multivariate functions is very straightforward, since these are topological properties. Generally, to prove that a limit is something, we just use ϵ-δ, and to actually *compute* what the limit might be, we can just compute $f(a)$ assuming that it is continuous, or take some sort of path p to take the univariate limit. We can also show that the multivariate limit doesn't exist by taking two sequences or two path functions where the limits do not equal each other.

However, the definition of the derivative and the integral will need to be generalized. Note that both the domain and codomain of a function $f : \mathbb{R}^n \to \mathbb{R}^m$ are vector spaces. However, Euclidean spaces have a lot of structure on them, and it is nice to identify the essential properties we need from these sets. It is essential that we work in vector spaces because when we define a derivative, we usually see some form that looks like

$$\frac{f(x+h) - f(x)}{h} \tag{1}$$

Note the operations used here. First, we want a notion of addition in the domain $(x + h)$ and the codomain $f(x + h) - f(x)$, along with some scalar multiplication when we multiply by $1/h$. A vector space precisely supports these operations and therefore is a natural choice. It is immediate that to define convergence, we definitely need a topology. We will see later that we want to define multivariate derivatives by adding a norm to this term, requiring the use of a normed vector space. Completion is clearly essential as we have seen in single-variable analysis.

Definition 0.1 (Banach Space)

A **Banach space** is a normed complete vector space.

Note that by extending the dimension, we have essentially lost the ordering $\leq$ on these spaces, along with the field properties. Therefore, we will need to adapt our definitions accordingly.

1 Differentiation

1.1 Partial, Directional, and Path Derivatives

A simple way to extract derivatives is to fix all of the input components except for one, and treat f as a single-variable function. This results in—for now—a completely separate notion of a derivative.

Definition 1.1 (Partial Derivative)

Given function $f : E \subset \mathbb{R}^n \to \mathbb{R}^m$, the ith **partial derivative** of f at $a \in E$ is defined

$$\frac{\partial f}{\partial x_i}\bigg|_a := \lim_{h \to 0} \frac{f(a + he_i)}{h} \tag{2}$$

where e_i is the ith basis vector of $\mathbb{R}^n$.

Since we are fixing every other parameter except for the ith one, we can find the partial derivative by differentiating the function w.r.t. x_i and fixing all other variables.

Example 1.1 (Partial Derivatives of Saddle Points)

To find all partial derivatives of the function

$$f(x, y) = x^2 - y^2 \tag{3}$$

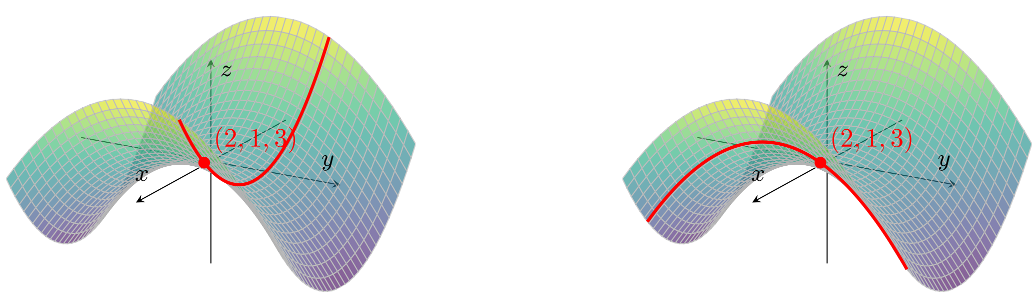
We compute

$$\frac{\partial f}{\partial x} = 2x \tag{4}$$

$$\frac{\partial f}{\partial y} = -2y. \tag{5}$$

At the point $(2, 1)$, these become

$$\frac{\partial f}{\partial x}(2, 1) = 4 \qquad \text{and} \qquad \frac{\partial f}{\partial y}(2, 1) = -2. \tag{6}$$



(a) Slicing the saddle surface along the plane $y = 1$, the resulting curve has slope 4 at $(2, 1, 3)$.

(b) Slicing along the plane $x = 2$ gives a curve with slope -2 at the same point.

Figure 1: Two cross-sections of the saddle surface $z = x^2 - y^2$ through the point $(2, 1, 3)$. Each slice isolates one partial derivative by holding the other variable fixed.

Example 1.2 (Partial Derivatives Using Differentiation Rules)

To find all partial derivatives of the function

$$f(x, y, z) = x \ln(y^2 + z^2), \tag{7}$$

We compute

$$\frac{\partial f}{\partial x} = \ln(y^2 + z^2) \tag{8}$$

$$\frac{\partial f}{\partial y} = x \cdot \frac{1}{y^2 + z^2} \cdot (2y) = \frac{2xy}{y^2 + z^2} \tag{9}$$

$$\frac{\partial f}{\partial z} = x \cdot \frac{1}{y^2 + z^2} \cdot (2z) = \frac{2xz}{y^2 + z^2} \tag{10}$$

Example 1.3 (Partial Derivatives Using Limit Definition)

Let

$$f(x, y) = \begin{cases} \dfrac{xy}{x^2 + y^2} & (x, y) \neq (0, 0), \\ 0 & (x, y) = (0, 0). \end{cases} \tag{11}$$

We compute the partial derivatives $f_x(0, 0)$ and $f_y(0, 0)$ using the limit definition.
First, for the partial derivative with respect to x, we have

$$\begin{aligned} f_x(0, 0) &= \lim_{h \to 0} \frac{f(h, 0) - f(0, 0)}{h} && (12) \\ &= \lim_{h \to 0} \frac{\frac{h(0)}{h^2 + 0^2} - 0}{h} && (13) \\ &= \lim_{h \to 0} \frac{0}{h} && (14) \\ &= 0. && (15) \end{aligned}$$

Similarly, for the partial derivative with respect to y, we have

$$\begin{aligned} f_y(0, 0) &= \lim_{h \to 0} \frac{f(0, h) - f(0, 0)}{h} && (16) \\ &= \lim_{h \to 0} \frac{\frac{0(h)}{0^2 + h^2} - 0}{h} && (17) \\ &= \lim_{h \to 0} \frac{0}{h} && (18) \\ &= 0. && (19) \end{aligned}$$

Therefore,

$$f_x(0, 0) = 0 \qquad \text{and} \qquad f_y(0, 0) = 0. \tag{20}$$

Note that even though the limit of $f(x, y)$ as $(x, y) \to (0, 0)$ does not exist, the partial derivatives at $(0, 0)$ still exist.

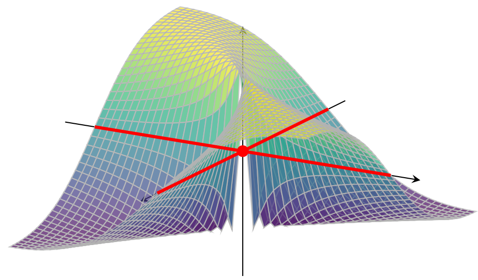


Figure 2: Both coordinate cross-sections of $f(x,y) = \frac{xy}{x^2+y^2}$ through the origin lie on the line $z = 0$, which shows why $f_x(0,0) = 0$ and $f_y(0,0) = 0$ even though the full two-variable limit does not exist.

Theorem 1.1 (Bounded Partials Implies Continuity)

If f is a real-valued function defined in an open set $E \subset \mathbb{R}^n$ and the partial derivatives $\partial_1 f, \ldots, \partial_n f$ are bounded in E, then f is continuous in E.

Proof. Since the partials exist, this means that the cross-sections are differentiable, so we take a stepwise approach. Let the partials be bounded by M. Fix $\epsilon > 0$ and $x \in E$. Then, for $y \in E$, let $\phi_k = (x_1, \ldots, x_k, y_{k+1}, \ldots, y_n)^T$. So,

$$\|f(x) - f(y)\| = \bigg\| \sum_{k=1}^n f(\phi_k(x)) - f(\phi_{k-1}(x)) \bigg\| \tag{21}$$

$$\leq \sum_{k=1}^n \|f(\phi_k(x)) - f(\phi_{k-1}(x))\| \tag{22}$$

$$\leq \sum_{k=1}^n M|y_k - x_k| \tag{23}$$

So setting $\delta = \frac{\epsilon}{nM}$ gives the desired result.

The partial derivative looks at the function as it is approaching a along an axis, but we can consider any direction in the domain.

Definition 1.2 (Directional Derivative)

Given function $f : E \subset \mathbb{R}^n \to \mathbb{R}^m$, the **directional derivative** of f at $a \in E$ in direction $u \in \mathbb{R}^n$ is defined

$$D_v f_a \coloneqq \lim_{h \to 0} \frac{f(a + hu)}{h} \tag{24}$$

It should be obvious that if all directional derivatives exist, then the partials exist (since we can just set the directional vectors to be the unit vectors).

When computing directional derivatives, it is convenient to normalize the directional vector v to be unit length so that it coincides with the partial derivatives. We don't technically need to set $||v|| = 1$, but if we have two vectors v and a scaled cv, then the directional derivatives will also be scaled ($\nabla_{cv} f(a) = c\nabla_v f(a)$), so we will only work with unit directional vectors. Some say that this restriction is undesirable, since it loses the linearity of the function $v \mapsto \partial_v f(a)$.

Example 1.4

Let

$$f(x,y) = x^2y + 4y^2. \tag{25}$$

We compute the directional derivative of f at the point $(2,1)$ in the direction of the vector

$$v = (1,1).$$

First, we normalize v. Since

$$\|v\| = \sqrt{1^2+1^2} = \sqrt{2}, \tag{26}$$

the unit direction vector is

$$u = \frac{v}{\|v\|} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right). \tag{27}$$

By the limit definition of the directional derivative,

$$D_u f(2,1) = \lim_{h\to 0} \frac{f\big((2,1)+hu\big) - f(2,1)}{h} \tag{28}$$

$$= \lim_{h\to 0} \frac{f\left(2+\frac{h}{\sqrt{2}}, 1+\frac{h}{\sqrt{2}}\right) - f(2,1)}{h}. \tag{29}$$

Now,

$$f(2,1) = 2^2(1) + 4(1)^2 = 8. \tag{30}$$

Also,

$$f\left(2+\frac{h}{\sqrt{2}}, 1+\frac{h}{\sqrt{2}}\right) = \left(2+\frac{h}{\sqrt{2}}\right)^2\left(1+\frac{h}{\sqrt{2}}\right) + 4\left(1+\frac{h}{\sqrt{2}}\right)^2 \tag{31}$$

$$= 8 + \frac{16h}{\sqrt{2}} + \frac{9h^2}{2} + \frac{h^3}{2\sqrt{2}}. \tag{32}$$

Therefore,

$$D_u f(2,1) = \lim_{h\to 0} \frac{8 + \frac{16h}{\sqrt{2}} + \frac{9h^2}{2} + \frac{h^3}{2\sqrt{2}} - 8}{h} \tag{33}$$

$$= \lim_{h\to 0}\left(\frac{16}{\sqrt{2}} + \frac{9h}{2} + \frac{h^2}{2\sqrt{2}}\right) \tag{34}$$

$$= \frac{16}{\sqrt{2}} \tag{35}$$

$$= 8\sqrt{2}. \tag{36}$$

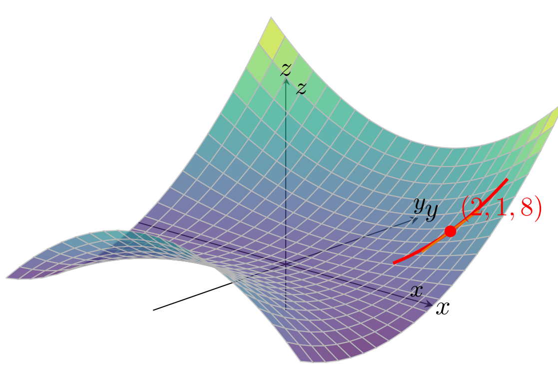


Figure 3: The surface $z = x^2y + 4y^2$ together with the red curve obtained by moving from $(2,1)$ in the direction $u = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$. The orange line is the tangent line, whose slope is the directional derivative $D_u f(2,1) = 8\sqrt{2}$.

Example 1.5

Let

$$f(x, y, z, w) = x^2 y + z \cos(w). \tag{37}$$

We compute the directional derivative of f at the point $(1, 2, 0, 0)$ in the direction of the vector

$$v = (1, 1, 1, 1).$$

First, we normalize v. Since

$$\|v\| = \sqrt{1^2 + 1^2 + 1^2 + 1^2} = 2, \tag{38}$$

the unit direction vector is

$$u = \frac{v}{\|v\|} = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right). \tag{39}$$

By the limit definition of the directional derivative,

$$\begin{aligned} D_u f(1, 2, 0, 0) &= \lim_{h \to 0} \frac{f\big((1, 2, 0, 0) + hu\big) - f(1, 2, 0, 0)}{h} && (40) \\ &= \lim_{h \to 0} \frac{f\left(1 + \frac{h}{2}, 2 + \frac{h}{2}, \frac{h}{2}, \frac{h}{2}\right) - f(1, 2, 0, 0)}{h}. && (41) \end{aligned}$$

Now,

$$f(1, 2, 0, 0) = 1^2(2) + 0 \cos(0) = 2. \tag{42}$$

Also,

$$\begin{aligned} f\left(1 + \frac{h}{2}, 2 + \frac{h}{2}, \frac{h}{2}, \frac{h}{2}\right) &= \left(1 + \frac{h}{2}\right)^2 \left(2 + \frac{h}{2}\right) + \frac{h}{2} \cos\left(\frac{h}{2}\right) && (43) \\ &= \left(1 + h + \frac{h^2}{4}\right)\left(2 + \frac{h}{2}\right) + \frac{h}{2} \cos\left(\frac{h}{2}\right) && (44) \\ &= 2 + \frac{5h}{2} + h^2 + \frac{h^3}{8} + \frac{h}{2} \cos\left(\frac{h}{2}\right). && (45) \end{aligned}$$

Therefore,

$$\begin{aligned} D_u f(1, 2, 0, 0) &= \lim_{h \to 0} \frac{2 + \frac{5h}{2} + h^2 + \frac{h^3}{8} + \frac{h}{2} \cos\left(\frac{h}{2}\right) - 2}{h} && (46) \\ &= \lim_{h \to 0} \left(\frac{5}{2} + h + \frac{h^2}{8} + \frac{1}{2} \cos\left(\frac{h}{2}\right)\right) && (47) \\ &= \frac{5}{2} + \frac{1}{2} && (48) \\ &= 3. && (49) \end{aligned}$$

But be careful! The existence of partials does not imply the existence of all directional derivatives!

Example 1.6 (Existence of Partials $\not\Longrightarrow$ Existence of Directional Derivatives)

Consider the function

$$f(x_1, x_2) = \begin{cases} \frac{x_1 x_2}{x_1^2 + x_2^2} & \text{if } (x_1, x_2) \neq (0, 0) \\ 0 & \text{if } (x_1, x_2) = (0, 0) \end{cases}$$

The partial derivatives exist everywhere. Away from the origin we can simply compute

$$\frac{\partial f}{\partial x_1} = \frac{x_2(x_1^2 + x_2^2) - x_1 x_2 \cdot 2x_1}{(x_1^2 + x_2^2)^2} = \frac{-x_1^2 x_2 + x_2^3}{(x_1^2 + x_2^2)^2}$$
$$\frac{\partial f}{\partial x_2} = \frac{x_1(x_1^2 + x_2^2) - x_1 x_2 \cdot 2x_2}{(x_1^2 + x_2^2)^2} = \frac{x_1^3 - x_1 x_2^2}{(x_1^2 + x_2^2)^2}$$

As for the partials at the origin, we must compute using the limit rule.

$$\partial_{x_1} f(0) = \lim_{h \to 0} \frac{f(0 + he_1) - f(0)}{h} = \lim_{h \to 0} \frac{f(h, 0)}{h} = 0$$
$$\partial_{x_2} f(0) = \lim_{h \to 0} \frac{f(0 + he_2) - f(0)}{h} = \lim_{h \to 0} \frac{f(0, h)}{h} = 0$$

However, the directional derivative taken in direction $v = (1, 1)$ gives

$$\begin{aligned} \nabla_{(1,1)} f(0,0) &= \lim_{h \to 0} \frac{f(0 + h(1,1)) - f(0)}{h} \\ &= \lim_{h \to 0} \frac{f(h,h)}{h} \\ &= \lim_{h \to 0} \frac{1}{2h} \end{aligned}$$

which does not have a limit as $h \to 0$. To visualize this, let's look at the values of f along various lines in $\mathbb{R}^2$.

1. $f = 0$ at the line $x_1 = 0$ and $x_2 = 0$, which is why the partials are 0.
2. $f = \frac{1}{2}$ at the line where $x_1 = x_2$, except for the point $(0, 0)$, where $f = 0$, which is why the limit in the direction doesn't exist.

Definition 1.3 (Path Derivative)

Let $f : E \subset \mathbb{R}^n \to \mathbb{R}^m$, $a \in E$, and $p : \mathbb{R} \to \mathbb{R}^n$ be a path function where $p(t_0) = a$. Then, the **path derivative** of f at a with respect to path p is defined

$$D_p f(a) \coloneqq \lim_{t \to t_0} \frac{f(p(t)) - f(p(t_0))}{h} \tag{50}$$

Example 1.7

Let

$$f(x, y) = x^2 + y, \tag{51}$$

and let

$$p(t) = (1 + t, 1 + t^2). \tag{52}$$

Since

$$p(0) = (1, 1), \tag{53}$$

we compute the path derivative of f at $a = (1, 1)$ with respect to the path p.
By the definition of the path derivative,

$$D_p f(1, 1) = \lim_{t \to 0} \frac{f(p(t)) - f(p(0))}{t - 0} \tag{54}$$
$$= \lim_{t \to 0} \frac{f(1 + t, 1 + t^2) - f(1, 1)}{t}. \tag{55}$$

Now,

$$f(1,1) = 1^2 + 1 = 2, \tag{56}$$

and

$$\begin{aligned} f(1+t, 1+t^2) &= (1+t)^2 + (1+t^2) && (57)\\ &= 1 + 2t + t^2 + 1 + t^2 && (58)\\ &= 2 + 2t + 2t^2. && (59) \end{aligned}$$

Therefore,

$$\begin{aligned} D_p f(1,1) &= \lim_{t\to 0} \frac{2 + 2t + 2t^2 - 2}{t} && (60)\\ &= \lim_{t\to 0} \frac{2t + 2t^2}{t} && (61)\\ &= \lim_{t\to 0} (2 + 2t) && (62)\\ &= 2. && (63) \end{aligned}$$

Example 1.8

Let

$$f(x,y) = (xy, x^2 - y), \tag{64}$$

and let

$$p(t) = (\cos t, \sin t). \tag{65}$$

Since

$$p(0) = (1,0), \tag{66}$$

we compute the path derivative of f at $a = (1,0)$ with respect to the path p.
By the definition of the path derivative,

$$\begin{aligned} D_p f(1,0) &= \lim_{t\to 0} \frac{f(p(t)) - f(p(0))}{t - 0} && (67)\\ &= \lim_{t\to 0} \frac{f(\cos t, \sin t) - f(1,0)}{t}. && (68) \end{aligned}$$

Now,

$$f(1,0) = (0,1), \tag{69}$$

and

$$f(\cos t, \sin t) = \big(\cos t \sin t, \cos^2 t - \sin t\big). \tag{70}$$

Therefore,

$$\begin{aligned} D_p f(1,0) &= \lim_{t\to 0} \frac{\big(\cos t \sin t, \cos^2 t - \sin t\big) - (0,1)}{t} && (71)\\ &= \lim_{t\to 0} \left(\frac{\cos t \sin t}{t}, \frac{\cos^2 t - \sin t - 1}{t} \right). && (72) \end{aligned}$$

We compute each component separately. For the first component,

$$\begin{aligned} \lim_{t\to 0} \frac{\cos t \sin t}{t} &= \lim_{t\to 0} \cos t \cdot \frac{\sin t}{t} && (73)\\ &= 1. && (74) \end{aligned}$$

For the second component,

$$\lim_{t\to 0}\frac{\cos^2 t-\sin t-1}{t}=\lim_{t\to 0}\left(\frac{\cos^2 t-1}{t}-\frac{\sin t}{t}\right) \tag{75}$$

$$=\lim_{t\to 0}\left(\frac{-\sin^2 t}{t}-\frac{\sin t}{t}\right) \tag{76}$$

$$=0-1 \tag{77}$$

$$=-1. \tag{78}$$

Hence,

$$D_p f(1,0)=(1,-1). \tag{79}$$

1.2 Total Derivatives

Definition 1.4 (Total/Frechet Derivative)

Let $f : E \subset \mathbb{R}^n \to \mathbb{R}^m$ and $a \in E$. f is **differentiable** at a if it satisfies either of the equivalent conditions.

1. *Linear Approximation.* There exists a linear map $Df_a : \mathbb{R}^n \longrightarrow \mathbb{R}^m$[a] such that

$$\lim_{h\to 0}\frac{\|f(a+h)-f(a)-Df_a h\|}{\|h\|}=0 \tag{80}$$

2. *Fundamental Increment Lemma.* There exists a linear map $Df_a : \mathbb{R}^n \to \mathbb{R}^m$ and an infinitesimal[b] function $\Phi : \mathbb{R}^n \to \mathbb{R}^m$ defined for sufficiently small nonzero $h \in \mathbb{R}^n$ such that

$$f(a+h)=f(a)+Df_a h+\Phi(h)\|h\|, \qquad \lim_{h\to 0}\Phi(h)=0 \tag{81}$$

If such a map Df_a exists, then it is unique, and we call this the **total derivative**—or **Frechet derivative**—of f at a.

[a]Note that when $m=1$, then Df_a is a linear functional, i.e. a dual vector.
[b]As in $\Phi(h) \to 0$ as $h \to 0$.

Proof. We'll first prove that the two are equivalent.

1. $(\to)$. Given that linear approximation holds, we can construct the map

$$\Phi(h) := \frac{f(a+h)-f(a)-Df_a h}{\|h\|} \tag{82}$$

Therefore, by our assumption and absolute homogeneity of the norm, we have

$$0=\lim_{h\to 0}\frac{\|f(a+h)-f(a)-Df_a h\|}{\|h\|} \tag{83}$$

$$=\lim_{h\to 0}\left\|\frac{f(a+h)-f(a)-Df_a h}{\|h\|}\right\| \tag{84}$$

$$=\lim_{h\to 0}\|\Phi(h)\| \tag{85}$$

Since the norm $v \mapsto \|v\|$ is continuous, this implies that

$$\lim_{h\to 0}\Phi(h)=0 \tag{86}$$

2. ($\leftarrow$). Given that the fundamental increment lemma holds, we can see that for nonzero h,

$$\Phi(h) = \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \tag{87}$$

Therefore, by taking the limit of both sides, we must have

$$0 = \lim_{h \to 0} \Phi(h) = \lim_{h \to 0} \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \tag{88}$$

Therefore, by continuity of the norm and absolute homogeneity,

$$\begin{align}
0 = \|0\| &= \left\| \lim_{h \to 0} \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \right\| \tag{89} \\
&= \lim_{h \to 0} \left\| \frac{f(a+h) - f(a) - Df_a h}{\|h\|} \right\| \tag{90} \\
&= \lim_{h \to 0} \frac{\|f(a+h) - f(a) - Df_a h\|}{\|h\|} \tag{91}
\end{align}$$

and we are done.

Finally, to prove uniqueness of the derivative, it seems more convenient to use the fundamental increment lemma. Say that there are two derivatives, A_1 and A_2, each with their own infinitesimal functions Φ_1, Φ_2 defined in the intersection of the open neigborhoods of a, which is also an open neighborhood. Then,

$$\begin{align}
f(a+h) &= f(a) + A_1 h + \Phi_1(h)\|h\| \tag{92} \\
f(a+h) &= f(a) + A_2 h + \Phi_2(h)\|h\| \tag{93}
\end{align}$$

So, by subtracting them and rearranging, we have

$$0 = (A_1 - A_2)h + (\Phi_1 - \Phi_2)(h) \tag{94}$$

So for $h \neq 0$, we have

$$(A_2 - A_1)\frac{h}{\|h\|} = (\Phi_1 - \Phi_2)(h) \tag{95}$$

The left hand side is a linear map of h and is in fact a constant c since $\frac{h}{\|h\|}$ is a unit vector. So by taking the limits of both sides, we get

$$c = \lim_{h \to 0}(A_2 - A_1)\frac{h}{\|h\|} = \lim_{h \to 0}(\Phi_1 - \Phi_2)(h) = 0 \tag{96}$$

Therefore, $c = 0 \implies A_2 - A_1 = 0$. So $A_1 = A_2$.

The reason we want E to be open is that we require that $x + h$ to also be in D for sufficiently small h, and we can guarantee this since an ϵ-ball around x is guaranteed to be in E. As in univariate analysis, we first establish the relation to continuity.

Theorem 1.2 (Differentiability Implies Continuity)

If f is differentiable at x, then it is continuous at x.

Proof. By assumption, we have

$$\lim_{h \to 0} \frac{\|f(x+h) - f(x) - Df_x h\|}{\|h\|} = 0 \implies \lim_{h \to 0} f(x+h) - f(x) - Df_x h = 0 \tag{97}$$

since the norm is a continuous mapping. But $\lim_{h \to 0} Df_x h = 0$ since Df_x is linear, and so

$$\lim_{h \to 0} f(x+h) - f(x) = 0 \tag{98}$$

Therefore, we have 4 separate notions of derivatives. A natural question to ask is whether the existence of one derivative implies the existence of another.

Theorem 1.3 (Differentiability $\implies$ Existence of All Directionals)

If $f : E \subset \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at $a \in E$, then all of its directional derivatives exist. Furthermore, given direction vector $u \in \mathbb{R}^n$,

$$D_u f_a = Df_a(u) \tag{99}$$

That is, we can compute the directional derivative by plugging in the direction vector into the linear map Df_a.

Proof. Since f is differentiable at a, the limit

$$\lim_{h \to 0} \frac{\|f(a+h) - f(a) - Df_a(h)\|}{\|h\|} = 0 \tag{100}$$

for all paths through 0. Therefore, let us choose the path

$$\lim_{h \to 0} \frac{\|f(a+hu) - f(a) - Df_a(hu)\|}{\|hu\|} = \lim_{h \to 0} \left\| \frac{f(a+hu) - f(a) - Df_a(hu)}{h} \right\| = 0 \in \mathbb{R} \tag{101}$$

Since the norm is a continuous mapping, this means that

$$0 = \lim_{h \to 0} \frac{f(a+hu) - f(a) - Df_a(hu)}{h} \tag{102}$$

$$= \lim_{h \to 0} \frac{f(a+hu) - f(a) - hDf_a(u)}{h} \tag{103}$$

$$= \lim_{h \to 0} \frac{f(a+hu) - f(a)}{h} - Df_a(u) \tag{104}$$

Therefore,

$$Df_a(u) = \lim_{h \to 0} \frac{f(a+hu) - f(a)}{h} = D_u f_a \tag{105}$$

Therefore, we know that differentiability (i.e. existence of the total derivative) is the strongest. If this is the case, then the partial and directional derivatives exist, and we also know how the derivative acts on each basis vector. Therefore, we can immediately know that

Lemma 1.4 (Action of Derivative on Basis Vectors)

Let $(e_i)_i$ and (e'_j) be the standard basis vectors of $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. If $f : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at a, then the partial derivatives $\frac{\partial f_j}{\partial x_i}$ exist, and

$$Df_a(e_i) = \sum_{j=1}^m \bigg(\frac{\partial f_j}{\partial x_i}(a) \bigg) e_j' \tag{106}$$

Proof. The partial derivatives existing is an immediate consequence of the previous theorem. As for the identity, we can see that

$$\begin{aligned} Df_a(e_i) = D_{e_i} f_a = D_{e_i} \bigg(\sum_{j=1}^m f_j e_j' \bigg)_a & \qquad (107) \\ = \sum_{j=1}^m D_{e_i} (f_j e_j')_a & \qquad (108) \\ = \sum_{j=1}^m D_{e_i} (f_j)_a e_j' & \qquad (109) \\ = \sum_{j=1}^m \frac{\partial f_j}{\partial x_i}(a)\, e_j' & \qquad (110) \end{aligned}$$

Since our vector spaces come with a basis, with this theorem, we can realize Df_x in a matrix form.

Definition 1.5 (Jacobian)

Given $f : E \subset \mathbb{R}^n \to \mathbb{R}^m$ differentiable at x, the **Jacobian** of f at x is the matrix realization of the total derivative, with entries being precisely the partial derivatives.

$$Jf_a = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \ldots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \ldots & \frac{\partial f_m}{\partial x_n} \end{pmatrix} \tag{111}$$

Therefore, we can see that the derivative maps the ith basis vector $e_i \in \mathbb{R}^n$ to the ith column of our Jacobian matrix.

Analogous to the univariate case, a nice way to visualize the derivative is by looking at the tangent *plane*. In general, we have an *affine hyperplane*. Define this here. Unfortunately, the only types of functions that we can meaningful visualize are those mapping $f : \mathbb{R}^2 \to \mathbb{R}$.

Definition 1.6 (Tangent Hyperplane)

If $f : E \subset \mathbb{R}^n \to \mathbb{R}$ is differentiable at $a \in \mathbb{R}^n$, then the affine hyperplane tangent to the graph of f at $(a, f(a))$ is the set

$$P = \{(x, f(x) \colon z = f(a) + Df_a(x - a)\}. \tag{112}$$

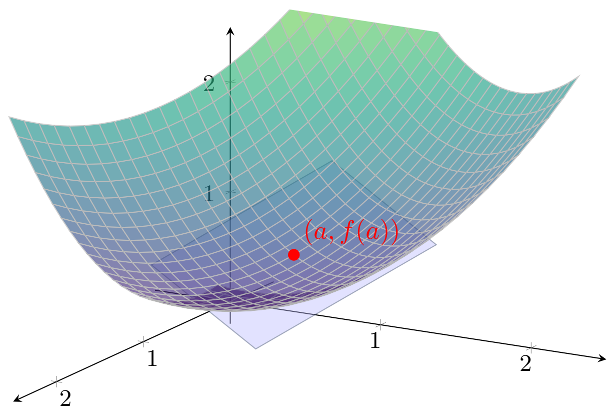


Figure 4: The tangent plane to the surface $z = f(x, y)$ at the point $(a, f(a))$. Here $f(x, y) = \frac{x^2+y^2}{2}$ and $a = (1, 1)$, so the tangent plane is $z = x + y - 1$.

So, to prove that a function $f : \mathbb{R}^n \longrightarrow \mathbb{R}$ is differentiable at a, and if so, what its total derivative is, there are essentially two steps.

1. We find a candidate for M by evaluating the partials.
$$M = \big(\partial_{x_1} f(a) \quad \partial_{x_2} f(a) \quad \ldots \quad \partial_{x_n} f(a)\big) \tag{113}$$

2. We check to see if the limit is true.
$$\lim_{h \to 0} \frac{f(a + h) - f(a) - Mh}{||h||} = 0 \tag{114}$$
If it is, then $Df_a = M$, and the "tangent plane" of f at a is defined by the equation $y = f(a) + Mh$.

Example 1.9 (Computing Total Derivative)

The function $f(x_1, x_2) = x_1^2 + x_2^2$ is differentiable at $(1, 1)$. We let $M = \big(\partial_{x_1} f(1, 1), \partial_{x_2} f(1, 1)\big) = \big(2, 2\big)$ and see that
$$\lim_{h\to 0} \frac{f(a + h) - f(a) - Mh}{||h||} = \lim_{h\to 0} \frac{f(1 + h_1, 1 + h_2) - f(1, 1) - 2h_1 - 2h_2}{\sqrt{h_1^2 + h_2^2}} \tag{115}$$
$$= \lim_{h\to 0} \frac{(1 + h_1)^2 + (1 + h_2)^2 - 2 - 2h_1 - 2h_2}{\sqrt{h_1^2 + h_2^2}} \tag{116}$$
$$= \lim_{h\to 0} \sqrt{h_1^2 + h_2^2} = 0 \tag{117}$$
So, $Df(1, 1) = (2, 2)$.

Example 1.10 (Computing Tangent Plane)

Let us find the equation of the tangent plane to $f(x, y) = \ln(2x + y)$ at $(-1, 3)$. Our total derivative, if it exists, is the covector of partials
$$Df = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{pmatrix} = \big(\tfrac{2}{2x+y} \quad \tfrac{1}{2x+y} \big) \tag{118}$$
which is indeed continuous at a neighborhood of $(-1, 3)$ (in fact, in every neighborhood not containing 0). By continuity of partials, f is differentiable at $(-1, 3)$, with $Df_{(-1,3)} = (2, 1)$. The equation of the

plane is then

$$z = f(-1,3) + Df_{(-1,3)} \begin{pmatrix} x+1 \\ y-3 \end{pmatrix} = 2(x+1) + 1(y-3) \implies z = 2x + y - 1 \tag{119}$$

However, the converse is not true for either statements. You cannot just evaluate all the partial derivatives and assume that the total derivative exists!

Example 1.11 (Partial Derivatives Exist, but Not Differentiable)

Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by:

$$f(x,y) = \begin{cases} \frac{xy}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases} \tag{120}$$

At the origin $(0,0)$, the partial derivatives are:

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h \to 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h \to 0} \frac{0-0}{h} = 0. \tag{121}$$

Similarly, $\frac{\partial f}{\partial y}(0,0) = 0$. Both partial derivatives exist. However, the total derivative **cannot** exist because the function is not continuous at $(0,0)$. As we saw in the limits section, approaching along $y = x$ gives a limit of $1/2$, while the function value is 0. A fundamental property of differentiability is that it implies continuity; therefore, f is not differentiable at the origin.

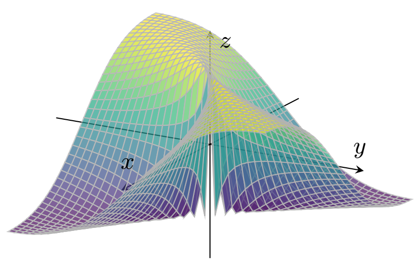


Figure 5: A function where partial derivatives exist at the origin, but the surface is "torn" there.

Example 1.12 (All Directional Derivatives Exist, but Not Differentiable)

Consider $f(x,y) = \begin{cases} \frac{x^2 y}{x^4+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$. For any direction $u = (u_1, u_2)$, the directional derivative at the origin exists and is given by $D_u f(0,0) = \frac{u_1^2}{u_2}$ (if $u_2 \neq 0$) or 0 (if $u_2 = 0$).
Despite having directional derivatives in every single direction, the total derivative does not exist. First, f is not continuous at the origin (approaching along $y = x^2$ gives $1/2 \neq 0$). Second, even if it were continuous, the total derivative requires that the linear map $Df(a)$ approximates the function well in *every* manner of approach, not just along straight lines. Here, the "peak" of the function follows the parabola $y = x^2$, which escapes the linear approximation provided by the directional derivatives.

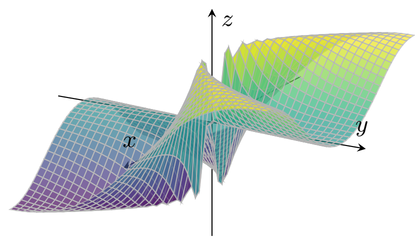


Figure 6: A function where all directional derivatives exist at the origin, yet the surface is non-differentiable.

1.3 Rules of Differentiation

Just like single variable calculus, the total derivative behaves in predictable ways: it is linear, product/quotient rules, and the chain rule.

Lemma 1.5 (Linearity of Total Derivatives)

Let $f, g : E \subset \mathbb{R}^n \longrightarrow \mathbb{R}^m$ be differentiable at $a \in E$. Then, the total derivative at a is linear w.r.t. the input functions.

1. *Addition.* $D(f+g)_a = Df_a + Dg_a$
2. *Scalar Multiplication.* $D(cf)_a = cDf_a$

Proof. Let f, g be differentiable at a and $c \in \mathbb{R}$.

1. *Addition.* There exists linear functions Df_a, Dg_a and infinitesimal functions Φ_f, Φ_g such that

$$f(a+h) = f(a) + Df_a(h) + \Phi_f(h)\|h\| \tag{122}$$
$$g(a+h) = g(a) + Dg_a(h) + \Phi_g(h)\|h\| \tag{123}$$

By adding the two, we get

$$(f+g)(a+h) = (f+g)(a) + (Df_a + Dg_a)(h) + (\Phi_f + \Phi_g)(h)\|h\| \tag{124}$$

where $Df_a + Dg_a$ is also linear and $\Phi_f + \Phi_g$ is also infinitesimal. So $f+g$ is differentiable with derivative $D(f+g)_a = Df_a + Dg_a$.

2. *Scalar Multiplication.* There exists a linear function Df_a and infintesimal Φ such that

$$f(a+h) = f(a) + Df_a(h) + \Phi(h)\|h\| \tag{125}$$

By multiplying both sides by c, we get

$$(cf)(a+h) = (cf)(a) + (cDf_a)(h) + (c\Phi)(h)\|h\| \tag{126}$$

where cDf_a is also linear and Φ also infinitesimal. So, cf is differentiable with derivative $D(cf)_a = cDf_a$.

Note that for the product and quotient rules, our scope is only for scalar valued functions since products and quotients aren't defined over codomains that are vector spaces.

Lemma 1.6 (Product Rule)

Let $f, g : E \subset \mathbb{R}^n \longrightarrow \mathbb{R}$ be differentiable at a. Then,

$$D(fg)_a = Df_a g(a) + f(a) Dg_a \tag{127}$$

Proof. By differentiability, we have

$$f(a+h) = f(a) + Df_a(h) + \Phi_f(h)\|h\| \tag{128}$$
$$g(a+h) = g(a) + Dg_a(h) + \Phi_g(h)\|h\| \tag{129}$$

By multiplying and expanding,[a] we get

$$\begin{aligned}(fg)(a+h) &= f(a)g(a) + f(a)Dg_a(h) + g(a)Dg_a(h) \quad (130)\\ &+ Df_a(h)Dg_a(h) + f(a)\Phi_g(h)\|h\| + g(a)\Phi_g(h)\|h\| + Df_a(h)\Phi_g(h)\|h\| \quad (131)\\ &+ Dg_a(h)\Phi_f(h)\|h\| + \Phi_f(h)\|h\|\Phi_g(h)\|h\| \quad (132)\end{aligned}$$

The map $fDg_a + Df_a f$ is linear, so it suffices to prove that the rest of the terms can be turned into a form of $\Phi(h)\|h\|$ where Φ is infinitesimal. We can see that by linearity,

$$Df_a(h)Dg_a(h) = \frac{Df_a(h)Dg_a(h)\|h\|}{\|h\|} = Df_a(h)Dg_a(\frac{h}{\|h\|})\|h\| \tag{133}$$

where $\lim_{h\to 0} Df_a(h)Dg_a(\frac{h}{\|h\|}) = \big(\lim_{h\to 0} Df_a(h)\big)\big(\lim_{h\to 0} Dg_a(\frac{h}{\|h\|})\big) = 0 \cdot \big(\lim_{h\to 0} Dg_a(\frac{h}{\|h\|})\big) = 0$. By the limit rules, the rest of the terms are obvious.

[a] I wish there was a cleaner solution such as the proof for the univariate case, but I wasn't able to derive one.

Lemma 1.7 (Quotient Rule)

Let $f, g : E \subset \mathbb{R}^n \longrightarrow \mathbb{R}$ be differentiable at a with $g(a) \neq 0$. Then,

$$D\Big(\frac{f}{g}\Big)_a = \frac{Df_a g(a) - f(a)Dg_a}{g(a)^2} \tag{134}$$

Proof. Just like the univariate case, we may make an educated guess that $D(1/g)_a = -\frac{Dg_a}{g^2}$. Therefore,

$$\begin{aligned}\frac{(1/g)(a+h) - (1/g)(a) - \big(-\frac{Dg_a}{g^2}\big)}{\|h\|} &= \frac{\frac{g(a)^2 - g(a)g(a+h) + Dg_a(h)g(a+h)}{g(a+h)g(a)^2}}{\|h\|} \quad (135)\\ &= \frac{1}{g(a)^2}\frac{g(a)^2 - g(a)g(a+h) + Dg_a(h)g(a+h)}{\|h\|g(a+h)} \quad (136)\end{aligned}$$

We would like to turn this into a form that represents the limit definition of the derivative of g.

$$\begin{aligned}&= \frac{1}{g(a)^2}\frac{g(a)^2 - g(a+h)^2 + g(a+h)^2 - g(a)g(a+h) + Dg_a(h)g(a+h)}{\|h\|g(a+h)} \quad (137)\\ &= \frac{1}{g(a)^2}\Bigg(\underbrace{\frac{g(a)^2 - g(a+h)^2}{\|h\|g(a+h)}}_{\to 0} + \frac{g(a+h)\big[g(a+h) - g(a) - Dg_a(h)\big]}{\|h\|g(a+h)}\Bigg) \quad (138)\end{aligned}$$

The left term goes to 0 since g is differentiable and hence continuous. The right term can be simplified by canceling out the $g(a+h)$, which can be done since g is not 0 in a neighborhood of a. So, adding the limit back in, the whole term simplifies to

$$\begin{aligned}\lim_{h\to 0}\frac{(1/g)(a+h) - (1/g)(a) - \big(-\frac{Dg_a}{g^2}\big)}{\|h\|} &= \lim_{h\to 0}\frac{1}{g(a)^2}\frac{g(a+h) - g(a) - Dg_a(h)}{\|h\|} \quad (139)\\ &= \frac{1}{g(a)^2}\lim_{h\to 0}\frac{\|g(a+h) - g(a) - Dg_a(h)\|}{\|h\|} = 0 \quad (140)\end{aligned}$$

by the definition of g being differentiable at a.

Theorem 1.8 (Chain Rule)

Let $f : E \subset \mathbb{R}^n \longrightarrow \mathbb{R}^m$ be differentiable at $a \in E$ and $g : E' \subset \mathbb{R}^m \longrightarrow \mathbb{R}^p$ be differentiable at $g(a) \in E'$. Then, $g \circ f$ is differentiable at a, and

$$D(g \circ f)_a = Dg_{f(a)} \circ Df_a \tag{141}$$

Proof. Since f is differentiable at a and g at $f(a)$, we can write

$$f(a+h) = f(a) + Df_a(h) + \Phi_f(h)\|h\| \tag{142}$$

$$g(f(a)+t) = g(f(a)) + Dg_{f(a)}(t) + \Phi_g(t)\|t\| \tag{143}$$

Therefore, we wish to show that

$$\lim_{h \to 0} \frac{g(f(a+h)) - g(f(a)) - (Dg_{f(a)} \circ Df_a)(h)}{\|h\|} = 0 \tag{144}$$

Let's remove the limit and focus on the term. First substitute $f(a+h)$ and expand the term for $g(f(a)+t)$.

$$= \frac{g(f(a + \overbrace{Df_a(h) + \Phi_f(h)\|h\|}^{t})) - g(f(a)) - (Dg_{f(a)} \circ Df_a)(h)}{\|h\|} \tag{145}$$

$$= \frac{g(f(a)) + Dg_{f(a)}\big(Df_a(h) + \Phi_f(h)\|h\|\big) + \Phi\big(Df_a(h) + \Phi_f(h)\|h\|\big)\|h\| - g(f(a)) - Dg_{f(a)}(Df_a(h))}{\|h\|} \tag{146}$$

$$= \frac{(Dg_{f(a)} \circ Df_a)(h) + Dg_{f(a)}\big(\Phi_f(h)\|h\|\big) + \Phi_g\big(Df_a(h) + \Phi_f(h)\|h\|\big)\|h\| - (Dg_{f(a)} \circ Df_a)(h)}{\|h\|} \tag{147}$$

$$= Dg_{f(a)}\big(\Phi_f(h)\big) + \Phi_g\big(Df_a(h) + \Phi_f(h)\|h\|\big) \to 0 \tag{148}$$

where the final step follows from linearity of $Dg_{f(a)}$. As $h \to 0$, the limit of the left term is 0 since linear maps are continuous, and the right term goes to 0 since Φ_g is infinitesimal. We are done.

Therefore, given the composition of function $f \circ g$, we have two methods of finding the derivative matrix of $f \circ g$ at point x_0. First is to explicitly compute $f \circ g$ and find its $m \times p$ derivative matrix $D(f \circ g)$, and plug in a to get $D(f \circ g)_a$. The second way is to use the chain rule to find the individual total derivatives $Df_{g(a)}$ and Dg_a, and multiply them together.

1.4 Continuously Differentiable Functions

An even stronger condition beyond differentiability is continuous partials, and we often prove continuity of partials to prove differentiability.

Definition 1.7 (Continuously Differentiable)

A function $f : E \subset \mathbb{R}^n \to \mathbb{R}^m$ is **continuously differentiable** at $a \in E$ if it satisfies either of the equivalent definitions:

1. f is differentiable at a and the map $a \in \mathbb{R}^n \to Df_a \in \mathbb{R}^{m \times n}$ is continuous.

2. f is differentiable at a and all the partial derivatives $a \in \mathbb{R}^n \to \frac{\partial f_j}{\partial x_i}(a) \in \mathbb{R}$ is continuous.

Proof. We prove bidirectionally.

1. $(1 \to 2)$. Let $\phi_{ij} : \mathbb{R}^{m \times n} \to \mathbb{R}$ be the projection of the matrix onto the ijth element. This is continuous and we can see that
$$\frac{\partial f_j}{\partial x_i}(a) = \phi(Df_a) \tag{149}$$
so all partial derivatives are continous.
2. $(2 \to 1)$. Let $\epsilon > 0$. If all partial derivatives are continuous, then there exists a $\delta > 0$ such that
$$\|x - a\| < \delta \implies \bigg\| \frac{\partial f_j}{\partial x_i}(x) - \frac{\partial f_j}{\partial x_i}(a) \bigg\| < \frac{\epsilon}{mn} \implies \|Df_a\|_1 = \sum_{i,j} (Df_a)_{ij} < \epsilon \tag{150}$$
So $a \to Df_a$ is continuous with respect to the L_1 norm on $\mathbb{R}^{m \times n}$. But since all L_p norms are equivalent in finite-dimensional Euclidean space, it is continuous with respect to all norms.

Theorem 1.9 (Continuous Partials $\implies$ Differentiability)

Given a function $f : E \subset \mathbb{R}^n \longrightarrow \mathbb{R}$ and a point $a \in E$, if all the partials $\partial_{x_i} f$ exist and are continuous at a, then f is differentiable at a.

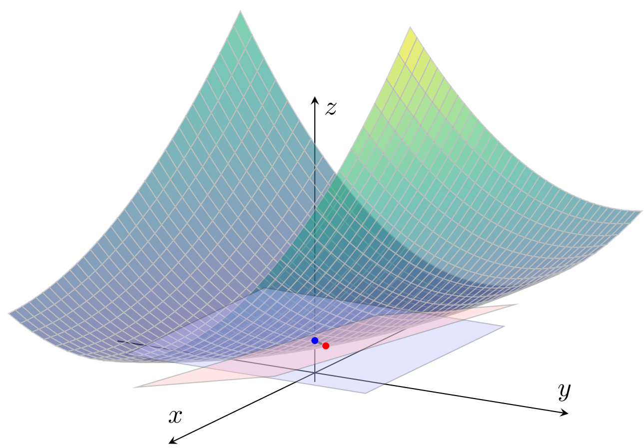


Figure 7: We can also visualize this theorem. Since the partials are continuous, then the tangent subspace, which is determined by the span of the tangent vectors determined by the partials, also changes continuously, and therefore the total derivative within a neighborhood of a exists.

Proof. The idea is to bound the perturbations in each component using the mean value theorem. Let us consider real valued $f : \mathbb{R}^n \to \mathbb{R}$ for now. Given $a, a + h \in \mathbb{R}^n$, let us construct a stepwise path from a to $a + h$. Let us denote $\phi_k(h) = (h_1, h_2, \ldots, h_k, 0, \ldots, 0)$. Then, we can construct the sequence
$$a = a + \phi_0(h), a + \phi_1(h), a + \phi_2(h), \ldots, a + \phi_k(h) = a + h \tag{151}$$
We take the matrix of partials
$$A = \begin{pmatrix} \frac{\partial f}{\partial x_1}(a) & \ldots & \frac{\partial f}{\partial x_n}(a) \end{pmatrix} \tag{152}$$

and wish to show that this is indeed the total derivative. We see that

$$f(a+h) - f(a) - Ah = f(a+h) - f(a) - \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a) h_i \tag{153}$$

$$= \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a) h_i + \big(f(a+\phi_i(h)) - f(a+\phi_{i-1}(h))\big) \tag{154}$$

Since $h_i \mapsto f(a+\phi_i(h)) - f(a+\phi_{i-1}(h))$ is a univariate differentiable function, by MVT, there exists a $c_i \in (a_i, a_i + h_i)$ s.t. $\frac{\partial f}{\partial x_i}(c_i) = \frac{f(a+\phi_i(h)) - f(a+\phi_{i-1}(h))}{h_i}$. Therefore, the above equals

$$= \sum_{i=1}^n \bigg(\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i)\bigg) h_i \tag{155}$$

So,

$$\|f(a+h) - f(a) - Ah\| = \bigg\|\sum_{i=1}^n \bigg(\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i)\bigg) h_i\bigg\| \tag{156}$$

Since $|h_i| \leq \|h\|$,

$$\frac{\|f(a+h) - f(a) - Ah\|}{\|h\|} = \bigg\|\sum_{i=1}^n \bigg(\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i)\bigg) \frac{h_i}{\|h\|}\bigg\| \tag{157}$$

$$= \bigg\|\sum_{i=1}^n \bigg(\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i)\bigg)\bigg\| \tag{158}$$

$$\leq \sum_{i=1}^n \bigg\|\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i)\bigg\| \tag{159}$$

But from continuity of partials, we know that for $\epsilon > 0$, there exists a $\delta > 0$ s.t.

$$\|h\| < \delta \implies \bigg|\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(a - \phi_k(h))\bigg| < \frac{\epsilon}{n} \tag{160}$$

$$\implies \bigg|\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i)\bigg| < \frac{\epsilon}{n} \tag{161}$$

since $c_i \in (a, a + \phi_k(h))$. Therefore,

$$\frac{\|f(a+h) - f(a) - Ah\|}{\|h\|} = \sum_{i=1}^n \bigg|\frac{\partial f}{\partial x_i}(a) - \frac{\partial f}{\partial x_i}(c_i)\bigg| < \epsilon \tag{162}$$

While continuous partial derivatives guarantee differentiability, the converse is not true. A function can be differentiable even if its partial derivatives are not continuous. However, if they are not continuous, we cannot use the previous theorem to determine differentiability and must revert to the definition.

Example 1.13 (Non-Continuous Partials and Not Differentiable)

Recall $f(x,y) = \begin{cases} \frac{xy}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$. The partial derivative $\frac{\partial f}{\partial x}$ for $(x,y) \neq (0,0)$ is:

$$\frac{\partial f}{\partial x} = \frac{y(x^2+y^2) - xy(2x)}{(x^2+y^2)^2} = \frac{y^3 - x^2 y}{(x^2+y^2)^2}.$$

At the origin, we found $\frac{\partial f}{\partial x}(0,0) = 0$. However, $\lim_{(x,y)\to(0,0)} \frac{y^3 - x^2 y}{(x^2+y^2)^2}$ does not exist (it depends on the path). Thus, the partial derivatives are not continuous at the origin. As shown before, the function is not differentiable because it is not continuous.

Example 1.14 (Non-Continuous Partials and Differentiable)

Consider $f(x,y) = \begin{cases} (x^2+y^2)\sin\left(\frac{1}{\sqrt{x^2+y^2}}\right) & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$. At the origin, the partial derivatives exist and are 0:

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h\to 0} \frac{h^2 \sin(1/|h|) - 0}{h} = \lim_{h \to 0} h \sin(1/|h|) = 0.$$

Away from the origin, $\frac{\partial f}{\partial x}$ involves a $\cos(1/\sqrt{x^2+y^2}) \cdot \frac{1}{x^2+y^2}$ term which oscillates wildly as $(x,y) \to (0,0)$. Thus, the partials are not continuous.
However, f **is** differentiable at the origin. Using the definition with $Df(0,0) = 0$:

$$\lim_{h \to 0} \frac{|(h_1^2 + h_2^2)\sin(1/\|h\|) - 0 - 0|}{\|h\|} = \lim_{h\to 0} \|h\| \sin(1/\|h\|) = 0.$$

This satisfies the definition of the total derivative.

Theorem 1.10 (C^1 Space)

The set of all continuously differentiable functions $f : E \subset \mathbb{R}^n \to \mathbb{R}$, denoted $C^1(E)$, is a vector space.

Proof. The proof is trivial since Df is continuous. Therefore, sums and scalar multiples of are continuous as well.

Note that from now, whenever we talk about differentiating a function f, we will assume that it is C^1. This is overkill, since the set of all k-times differentiable functions is a subset of C^k, but it is conventional to work with C^k functions.

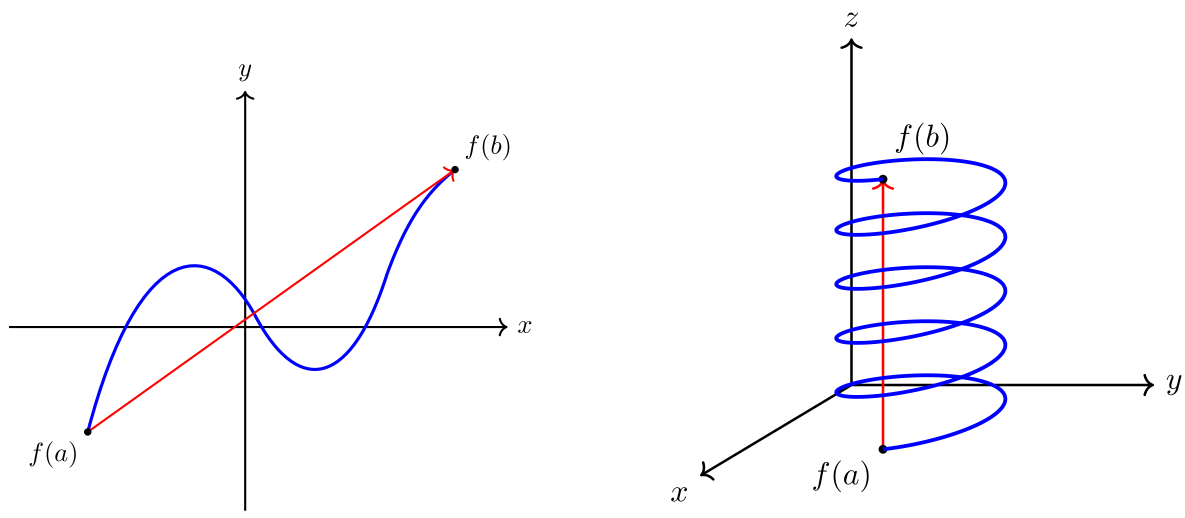
1.5 Mean Value Theorem

Lemma 1.11 (Mean Value Theorem for Paths)

If $f : E \subset \mathbb{R} \to \mathbb{R}^m$ is continuous on $[a,b]$ and differentiable on (a,b), then there exists $x \in (a,b)$ s.t.

$$\|f(b) - f(a)\| \leq (b-a)\|f'(x)\| \tag{163}$$

Proof. The general idea is that we want to convert this into a scalar-valued function in a clever way.



(a) (b)

Figure 8: If we sketch out a path in $\mathbb{R}^2$, we might speculate that at some point, the tangent line of the curve must be parallel to the secant vector $f(b) - f(a)$. Perhaps defining $\phi(x) = \langle f(b) - f(a), \frac{Df_x}{\|Df_x\|}\rangle$ and trying to maximize it using EVT or something might help. However, there is no guarantee that Df_x is continuous, and unfortunately, this assumption turns out not to be true as seen in the solenoid.

After some thought (see figure caption), we might think of looking at the inner product of the actual vectors on the curve with the secant vector, defining

$$\phi(x) = \langle f(b) - f(a), f(x) \langle \tag{164}$$

ϕ is differentiable on (a, b) and continuous on $[a, b]$, so by MVT, there exists some $x \in (a, b)$ such that

$$\phi(b) - \phi(a) = D\phi_x(b - a) = \langle f(b) - f(a), Df_x \rangle (b - a) \tag{165}$$

But $\phi(b) - \phi(a) = \langle f(b) - f(a), f(b) - f(a)\rangle = \|f(b) - f(a)\|^2$, so by Cauchy Schwarz, we get

$$\|f(b) - f(a)\|^2 = \langle f(b) - f(a), Df_x \rangle (b - a) \leq \|f(b) - f(a)\| \|Df_x\| (b - a) \tag{166}$$

Therefore, $\|f(b) - f(a)\| \leq \|Df_x\|(b - a)$.

There is a nice reason why equality doesn't hold in higher dimensions. For my first attempt at this proof, I tried to replicate my work for proving that C^1 implies differentiable by dividing f into component-wise paths. I defined

$$\phi_k(f) = \big(f_1(a), \ldots, f_k(a), f_{k+1}(b), \ldots, f_m(b)\big)^T \tag{167}$$

Therefore, I was hoping to bound $\|f(b) - f(a)\| \leq \sum_{k=1}^m \|\phi_k(f) - \phi_{k-1}(f)\| = \sum_{k=1}^m |f_k(b) - f_k(a)|$. But invoking the univariate MVT for the real-valued function f_k leads to finding multiple $x_k \in (a, b)$. This is a problem since we need to find one x.

Example 1.15 (MVT Inequality Does Not Hold on Circle)

Let

$$f : [0, 2\pi] \to \mathbb{R}^2, \quad f(x) = \big(\cos(x), \sin(x)\big) \tag{168}$$

Then, $f(2\pi) - f(0) = 0$, but

$$Df_x = \begin{pmatrix} -\sin(x) \\ \cos(x) \end{pmatrix} \tag{169}$$

is never 0 over $x \in [0, 2\pi]$.

Theorem 1.12 (Mean Value Theorem)

Let $f : E \subset \mathbb{R}^n \to \mathbb{R}^m$ be differentiable on convex open set E. If there exists a $M \in \mathbb{R}$ s.t. $\|Df_x\| \leq M$ for all $x \in E$, then

$$\|f(b) - f(a)\| \leq M\|b - a\| \tag{170}$$

for all $a, b \in E$.

Proof. Since E is convex, for any $a, b \in E$, we can consider

$$\gamma : [0,1] \to E, \quad \gamma(t) \coloneqq ta + (1-t)b \tag{171}$$

Then, $f \circ \gamma$ is a path function that is continuous on $[0,1]$ and differentiably on $(0,1)$. Therefore, there exists a $t_0 \in (0,1)$ (and hence, corresponding $x_0 = \gamma(t_0) \in E$) such that

$$\begin{aligned} \|(f\circ\gamma)(1) - (f\circ\gamma)(0)\| &\leq \|D(f\circ\gamma)_{t_0}\|\|1-0\| && (172)\\ &= \|Df_{x_0} D\gamma_{t_0}\| && (173)\\ &\leq \|Df_{x_0}(a-b)\| && (174)\\ &\leq \|Df_{x_0}\|\|a-b\| && (175)\\ &\leq M\|b-a\| && (176) \end{aligned}$$

where in the second step, we used the chain rule, in the third step, we computed the derivative $D\gamma = a - b$, and in the fourth step, we used the operator norm bound. The left hand side is really $\|f(b) - f(a)\|$, and we have our desired result.

The following corollary is pretty obvious.

Corollary 1.13 (Vanishing Derivative Implies Constant)

Let $f : E \subset \mathbb{R}^n \to \mathbb{R}^m$ be differentiable on convex open set E. If $\|Df_x\| = 0$ for all $x \in E$, then f is constant on E.

Proof. Set $M = 0$ in the proof above.

1.6 Inverse Function Theorem

Definition 1.8 (Contraction)

Given a metric space (X, d) and some $c < 1$, a **contraction** is a mapping $\phi : X \to X$ such that

$$d(\phi(x), \phi(y)) \leq c d(x, y) \tag{177}$$

Theorem 1.14 (Banach Fixed Point Theorem)

Given a complete metric space (X, d) and a contraction ϕ on X, there exists a unique point $x \in X$ such that $\phi(x) = x$.

Proof. We give a constructive proof on how to find such a point. Let us pick any $x_0 \in X$ and set $x_k = \phi^k(x_0)$ to be the output of the k-fold composition of ϕ. Then,

$$d(x_k, x_{k+1}) \leq c^k d(x_0, x_1) \tag{178}$$

So given any $m, n \in \mathbb{N}$, WLOG let $m < n$. Then,

$$\begin{aligned}
d(x_m, x_n) &\leq \sum_{k=m}^{n-1} d(x_k, x_{k+1}) && (179)\\
&\leq \sum_{k=m}^{n-1} c^k d(x_0, x_1) && (180)\\
&= d(x_0, x_1) \sum_{k=m}^{n-1} c^k && (181)\\
&= d(x_0, x_1) \sum_{k=m}^{\infty} c^k && (182)
\end{aligned}$$

Since the series converges from the geometric test, the tail sum $\sum_{k=m}^{\infty} c^k$ converges to 0 as $m \rightarrow +\infty$. So, $(x_n)_n$ is a Cauchy sequence and therefore converges to some $x \in X$.
Since ϕ is a contraction, ϕ is continuous. In fact it is uniformly continuous, since given $\epsilon > 0$, we can set $\delta = \frac{\epsilon}{c}$ and see that for all $x, y \in X$,

$$d(x, y) < \delta = \frac{\epsilon}{c} \implies d(\phi(x), \phi(y)) < c\frac{\epsilon}{c} = \epsilon \tag{183}$$

Therefore,

$$\phi(x) = \lim_{n \rightarrow +\infty} \phi(x_n) = \lim_{n \rightarrow +\infty} x_{n+1} = x \tag{184}$$

To prove uniqueness, assume that $x, y \in X$ are both fixed points. Then,

$$d(x, y) = d(\phi(x), \phi(y)) \leq c d(x, y) \implies d(x, y) = 0 \implies x = y \tag{185}$$

It is a bit strange that we just introduce this fixed point theorem all of a sudden when talking about invertibility, but this becomes useful in the proof for the inverse function theorem. When we have a point y and want to prove an inverse x exists such that $y = f(x)$, we can use the fixed point theorem to guarantee existence of such a point x.

Theorem 1.15 (Inverse Function Theorem)

Let $E \subset \mathbb{R}^n$ be an open neighborhood of a, and let $f : E \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ be C^1 over E. If Df_a is invertible, then

1. there exists open neighborhoods U of a and V of $b = f(a)$ such that

$$f|_U : U \longrightarrow V \tag{186}$$

is a bijection.

2. Moreover, the inverse map $(f|_U)^{-1} : V \to U$ is C^1 over V, and for every $y \in W$,
$$D(f^{-1})_y = (Df_x)^{-1}. \tag{187}$$
where $f(x) = y$.

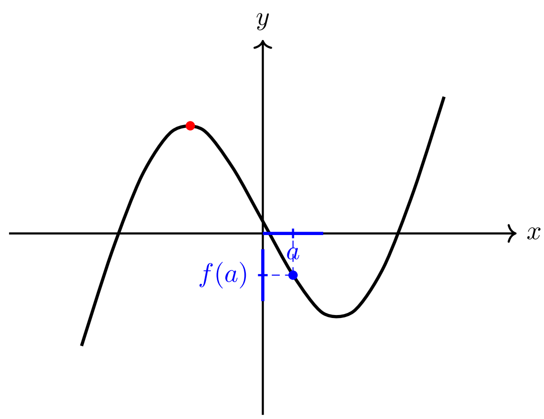


Figure 9: A cubic can fail to be locally invertible at a turning point, but behave regularly at nearby non-turning points.

Proof. We basically want to invert f, i.e. for each $y_0 \in \mathbb{R}^n$, we want to find a x_0 s.t. $f(x_0) = y_0$. How do we do this constructively? Since this may be difficult for an arbitrary function f, the next best thing to do is linearize $f(x)$ and find the inverse of the best linear approximation. By definition of differentiability at $a \in E$, there exists a linear map $Df_a : \mathbb{R}^n \to \mathbb{R}^n$ and infinitesimal function $\Phi : \mathbb{R}^n \to \mathbb{R}^n$ such that $f(x) = f(a) + Df_a(x-a) + \Phi(x-a)\|x-a\|$ within some neighborhood U_1 of a. Therefore, we can approximate $f(x)$ with a linear approximation $f_a(x)$, where the subscript is used to denote that it is centered at a.
$$y = f_a(x) \approx f(a) + Df_a(x-a) \tag{188}$$
After inverting this, we get the equation
$$x = g_a(y) = a + Df_a^{-1}(y - f(a)) \tag{189}$$
This function g basically takes the difference $y - f(x)$ in the codomain, inverts it to $Df_a^{-1}(y - f(x))$ as the difference in the domain, and finally adds it to x. This equation is a very natural approximation of the inverse, but the problem is that is that it is not an exact inverse. One quick fix is to set the center $a = x$ so that $f(x) = y$. Then, we have[a b]
$$f(x) = y \iff g_x(y) = x + Df_a^{-1}(y - f(x)) = x \tag{190}$$
Therefore, we have an equivalent notion of y being the image of x. Now rather than thinking about g_x as a function of y, we can think of it as a function of x. Define
$$\phi_y(x) \coloneqq g_x(y) \tag{191}$$
to be the "linear inverse of y centered at x." Then, note that
$$f(x_0) = y_0 \iff \phi_{y_0}(x_0) = g_{x_0}(y_0) = x_0 \iff x_0 \text{ is a fixed point of } \phi_{y_0} \tag{192}$$
The idea is to prove that ϕ_{y_0} is a contraction of some subset that is a complete metric space, and so (by the Banach fixed point theorem) the unique fixed point—call it $x_0 \in \mathbb{R}^n$—is the inverse. Let's do this step by step.

1. ϕ_0 *is a contraction.* $\phi_{y_0}(x) = x + Df_a^{-1}(y_0 - f(x))$, which implies that
$$D(\phi_{y_0})_x = I - Df_a^{-1} Df_x = Df_a^{-1}(Df_a - Df_x) \tag{193}$$
But since $x \to Df_x$ is continuous at a, there exists an open neighborhood U_2 containing a s.t.
$$x \in U_2 \implies \|Df_x - Df_a\| < \frac{1}{2\|Df_a^{-1}\|} \tag{194}$$
Therefore, let $U = U_1 \cap U_2$, so when $x \in U$,
$$\|D(\phi_{y_0})_x\| = \|Df_a^{-1}\| \|Df_a - Df_x\| \leq \|Df_a^{-1}\| \frac{1}{2\|Df_a^{-1}\|} = \frac{1}{2} \tag{195}$$
Therefore, by the MVT, for all $x_1, x_2 \in U$,
$$\|\phi_{y_0}(x_1) - \phi_{y_0}(x_2)\| \leq \frac{1}{2}\|x_1 - x_2\| \tag{196}$$
and so, ϕ_{y_0} is a contraction on U.
2. f *is injective over* U. Say that $x_1, x_2 \in U$ such that $f(x_1) = f(x_2) = y_0$. Then, from here, x_1 and x_2 must both be fixed points of U. Then,
$$\|x_1 - x_2\| = \|\phi_{y_0}(x_1) - \phi_{y_0}(x_2)\| \leq \frac{1}{2}\|x_1 - x_2\| \implies x_1 = x_2 \tag{197}$$
Therefore, there can at most be one such x_0 that maps to y_0. Hence f is injective and therefore $f : U \to f(U)$ is bijective, hence is invertible. Note that we did *not* use Banach fixed point theorem yet.
3. $V = f(U)$ *is open.* This is much trickier than it looks. First, pick $y_1 \in V$ and so $y_1 = f(x_1)$ for some $x_1 \in U$. We wish to show that there exists a δ-neighborhood around y_1, which establishes that V is open. The choose δ, let us choose $\epsilon > 0$ so small that $\overline{B_\epsilon(x_1)} \subset U$. Then, if $x \in B_\epsilon(x_1)$, then *for all* $y_0 \in V$, we have
$$\begin{aligned}\|\phi_{y_0}(x) - x_1\| &\leq \|\phi_{y_0}(x) - \phi_{y_0}(x_1)\| + \|\phi_{y_0}(x_1) - x_1\| && (198)\\ &\leq \frac{1}{2}\|x - x_1\| + \|Df_a^{-1}(y_0 - f(x_1))\| && (199)\\ &\leq \frac{\epsilon}{2} + \|Df_a^{-1}\|\|y_0 - f(x_1)\| && (200)\\ &= \frac{\epsilon}{2} + \|Df_a^{-1}\|\|y_0 - y_1\| && (201)\end{aligned}$$
[c] Therefore, this hints that we need to bound $\|y_0 - y_1\|$, so we can choose y_0 (and hence ϕ_{y_0}) s.t.
$$\|y_0 - y_1\| < \frac{\epsilon}{2\|Df_a^{-1}\|} = \delta \tag{202}$$
By plutting this into the inequality above, we can see that
$$x \in B(x_1, \epsilon), \|y_0 - y_1\| < \delta \implies \phi_{y_0}(x) \in B(x_1, \epsilon) \tag{203}$$
[d] Since ϕ_{y_0} is a contraction over U, it is also a contraction over $\overline{B_\epsilon(x_1)}$. Furthemore, since $\overline{B_\epsilon(x_1)}$ is closed in $\mathbb{R}^n$, it is complete, so by Banach fixed point theorem, ϕ_{y_0} has a unique fixed point, call it $x_0 \in \overline{B_\epsilon(x_1)}$. Since x_0 being a fixed point of $\phi_{y_0}(x)$ is equivalent to $y_0 = f(x_0)$, we can therefore say that the inverse $x_0 = f^{-1}(y_0) \in \overline{B(x_1, \epsilon)}$. Finally, we have
$$x_0 \in \overline{B_\epsilon(x_1)} \implies y_0 = f(x_0) \in f(B_\epsilon(x_1)) \subset f(U) = V \tag{204}$$

Great, so we have established that $f : U \to V$ is bijective and thus invertible. Now we wish to show that for $f : U \to V$, $D(f^{-1})_b = (Df_a)^{-1}$. Let $f(a+h) = f(a) + k$. Then , we can see that

$$\begin{aligned} f^{-1}(b+k) - f^{-1}(b) - Df_a^{-1}k &= (a+h) - a - Df_a^{-1}k && (205) \\ &= h - Df_a^{-1}k && (206) \\ &= -Df_a^{-1}(k - Df_a h) && (207) \\ &= -Df_a^{-1}\big(f(a+h) - f(a) - Df_a h\big) && (208) \end{aligned}$$

Therefore,

$$\|f^{-1}(b+k) - f^{-1}(b) - Df_a^{-1}k\| \leq \|Df_a^{-1}\| \|f(a+h) - f(a) - Df_a h\| \tag{209}$$

We hope to divide the left hand side by $\|k\|$, the right hand side by $\|h\|$, and take their limits to 0. So, we would like to find a bound between k and h. Recall that since ϕ_{y_0} is a contraction for any y_0,

$$\begin{aligned} \frac{1}{2}\|h\| &\geq \|\phi_{y_0}(x+h) - \phi_{y_0}(x)\| && (210) \\ &= \|h + Df_a^{-1}\big(f(x) - f(x+h)\big) && (211) \\ &= \|h - Df_a^{-1}k\| && (212) \\ &\geq \|h\| - \|Df_a^{-1}k\| && (213) \end{aligned}$$

Therefore, $\|h\| \leq 2\|Df_a^{-1}k\| \leq 2\|Df_a^{-1}\|\|k\|$, which implies that as $k \to 0$, then $h \to 0$. Therefore, by combining the two inequalities, we have

$$\frac{\|f^{-1}(b+k) - f^{-1}(b) - Df_a^{-1}k\|}{\|k\|} \leq 2\|Df_a^{-1}\|^2 \frac{\|f(a+h) - f(a) - Df_a h\|}{\|h\|} \tag{214}$$

Taking both sides as $k \to 0$ makes the right hand side vanish, since this implies that we are letting $h \to 0$. Therefore, the left hand limit vanishes, and by definition this means that $D(f^{-1})_b = Df_a^{-1}$. Finally, since inversion is a continuous operator, the chain of mappings $a \mapsto Df_a \mapsto Df_a^{-1}$ is continuous, and therefore the composition

$$a \mapsto D(f^{-1})_a = Df_a^{-1} \tag{215}$$

is also continuous, making f^{-1} continuously differentiable over V.

[a] where the reverse implication follows from Df_a being nonsingular.

[b] Note that the a in Df_a is *fixed* and does not change with respect to x.

[c] When I tried reproving it, I tried to bound $\|\varphi_{y_0}(x) - x_1\| = \|x - x_1 + Df_a^{-1}(y_0 - f(x))\| \leq \|x - x_1\| + \|Df_a^{-1}\|\|y_0 - f(x)\| < \epsilon + \|Df_a^{-1}\|\|y_0 - f(x)\|$. However, there still has a dependence on x, and we can't reduce it to just one choice of y_0.

[d] That is, as long as the center x is close enough to x_1 *and* y_0 is close enough to y_1, the approximate inverse of y_0 is close to x_1.

Example 1.16 (A Univariate Example)

Consider the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ defined by

$$f(x) = x^2. \tag{216}$$

Its derivative is

$$Df_x = f'(x) = 2x. \tag{217}$$

Therefore, at any point $x_0 \neq 0$, we have

$$Df_{x_0} = 2x_0 \neq 0. \tag{218}$$

Hence Df_{x_0} is invertible as a linear map from $\mathbb{R}$ to $\mathbb{R}$.

By the inverse function theorem, for every $x_0 \neq 0$, there exists an open neighborhood V of x_0 such that

$$f|_V : V \longrightarrow f(V) \tag{219}$$

is invertible.
For example, at $x_0 = 1$, we can take a small interval around 1, such as

$$V = \left(\frac{1}{2}, \frac{3}{2}\right). \tag{220}$$

On this interval, $f(x) = x^2$ is injective, so it has a well-defined inverse

$$f^{-1}(y) = \sqrt{y}. \tag{221}$$

However, f is not invertible on all of $\mathbb{R}$, since

$$f(1) = 1 = f(-1). \tag{222}$$

Thus the inverse function theorem guarantees local invertibility near points where $f'(x_0) \neq 0$, but it does not guarantee global invertibility.

Example 1.17

Consider the vector-valued function $f : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ defined by

$$f(x_1, x_2) = \begin{pmatrix} e^{x_1} \cos(x_2) \\ e^{x_1} \sin(x_2) \end{pmatrix}$$

The total derivative/Jacobian matrix is

$$Jf(x_1, x_2) = \begin{pmatrix} e^{x_1} \cos(x_2) & -e^{x_1} \sin(x_2) \\ e^{x_1} \sin(x_2) & e^{x_1} \cos(x_2) \end{pmatrix} \implies \det Jf(x_1, x_2) = e^{2x_1} \cos^2(x_2) + e^{2x_1} \sin^2(x_2) = e^{2x_1}$$

Since the determinant e^{2x_1} is nonzero everywhere, Df_x is nonsingular. Thus, the theorem guarantees that for every point $x_0 \in \mathbb{R}^2$, there exists a neighborhood about x_0 over which f is invertible. However, this does not mean f is invertible over its entire domain: in this case f isn't even injective since it is periodic: e.g. the preimage of $(e, 0)$ contains $(1, 0)$ and $(1, 2\pi)$.

Example 1.18 (A Function from $\mathbb{R}^3$ to $\mathbb{R}^3$)

Consider the function $f : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ defined by

$$f(x_1, x_2, x_3) = \begin{pmatrix} e^{x_1} \cos(x_2) \\ e^{x_1} \sin(x_2) \\ x_3 \end{pmatrix}. \tag{223}$$

The total derivative/Jacobian matrix is

$$Jf(x_1, x_2, x_3) = \begin{pmatrix} e^{x_1} \cos(x_2) & -e^{x_1} \sin(x_2) & 0 \\ e^{x_1} \sin(x_2) & e^{x_1} \cos(x_2) & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{224}$$

Therefore,

$$\det Jf(x_1, x_2, x_3) = e^{2x_1} \cos^2(x_2) + e^{2x_1} \sin^2(x_2). \tag{225}$$

Hence
$$\det Jf(x_1, x_2, x_3) = e^{2x_1}. \tag{226}$$
Since $e^{2x_1} \neq 0$ for all $x_1 \in \mathbb{R}$, the derivative Df_x is invertible at every point $x \in \mathbb{R}^3$.
Thus, by the inverse function theorem, for every point $x_0 \in \mathbb{R}^3$, there exists an open neighborhood V of x_0 such that
$$f|_V : V \longrightarrow f(V) \tag{227}$$
is invertible.
However, f is not invertible on all of $\mathbb{R}^3$, since it is periodic in the x_2-variable. For example,
$$f(1, 0, 5) = \begin{pmatrix} e \\ 0 \\ 5 \end{pmatrix} = f(1, 2\pi, 5). \tag{228}$$
Therefore, f is locally invertible everywhere, but not globally invertible.

1.7 Implicit Function Theorem

Remember that given an explicit representation of a set $y = f(x)$, we can easily find the implicit form as $F(x, y) = y - f(x) = 0$. What about the other way around? That is, given an implicit representation of some surface, what conditions must be met so that it can be represented as the graph of a function? The implicit function theorem is a tool that allows relations between points in $\mathbb{R}^n$ to be converted to functions of several real variables. That is, it states that for sufficiently "nice" points on a n-dimensional surface defined as $F(x, y) = 0$ (where $F : \mathbb{R}^{n+m} \longrightarrow \mathbb{R}^m$), we can locally pretend that this surface is a graph of a function $g : \mathbb{R}^n \longrightarrow \mathbb{R}^m$ whose graph $\big(x, g(x)\big)$ is precisely the set of all (x, y) s.t. $f(x, y) = 0$. When $m = 1$, it basically states that if an implicit surface suffices the vertical line test in a neighborhood, then it can be written as a function.

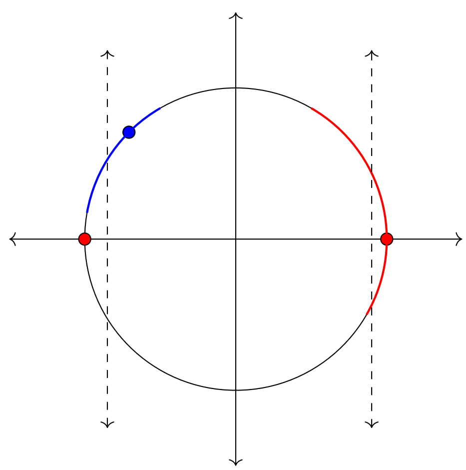


Figure 10: Let $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$ be defined by $f(x, y) = x^2 + y^2 - 1$. The level set at $z = 0$ would be the set of points satisfying $x^2 + y^2 - 1 = 0$, i.e. the unit circle. The derivative of f with respect to y is 0 at the points $(-1, 0)$ and $(1, 0)$, meaning that in any neighborhood of these points, we cannot define a function of y with respect to x. This is true, indeed, since any such function would fail the vertical line test, which can be seen in the red neighborhood around $(1, 0)$. However, the blue neighborhood of the point $(-\sqrt{2}/2, \sqrt{2}/2)$ does indeed define a function of y with respect to x satisfying the vertical line test.

Theorem 1.16 (Implicit Function Theorem)

Let $F : E \subset \mathbb{R}^{n+m} \to \mathbb{R}^m$ be C^1 over an open set E. Write points in $\mathbb{R}^{n+m}$ as (x, y), where $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$. Given $(a, b) \in E$, if

1. $F(a, b) = 0$,
2. $D_y F_{(a,b)} \in \mathbb{R}^{m\times m}$ is invertible,

then there exist open neighborhoods $U \ni a$ and $V \ni b$ where $U \times V \subset E$, and a unique C^1 function $f : U \longrightarrow V$ such that $f(a) = b$ and, for every $(x, y) \in U \times V$,

$$F(x, y) = 0 \iff y = f(x). \tag{229}$$

Moreover, for every $x \in U$,

$$Df_x = -\left(D_y F_{(x,f(x))}\right)^{-1} D_x F_{(x,f(x))}. \tag{230}$$

Proof. The idea is quite simple. We augment F to a locally invertible function G, then take the inverse of it, and finally project it back to the y coordinates. Let us define

$$G : E \subset \mathbb{R}^{n+m} \to \mathbb{R}^{n+m}, \quad G(x, y) = (x, F(x, y)) \tag{231}$$

Note that $G(a, b) = (a, F(a, b)) = (a, 0)$, and to evaluate the derivative of G, we can find its Jacobian in block form:

$$DG_{(a,b)} = \begin{pmatrix} I_n & 0 \\ D_x F_{(a,b)} & D_y F_{(a,b)} \end{pmatrix}. \tag{232}$$

This is a block lower triangular matrix, with both block diagonal elements invertible, hence $\det DG_{(a,b)} = \det(I_n)\det(D_y F_{(a,b)}) = \det(D_y F_{(a,b)}) \neq 0$, and so $DG_{(a,b)}$ is invertible. By the inverse function theorem, there exist open neighborhoods W of (a, b) and O of $(a, 0)$ such that $G|_W : W \to O$ is a bijection with a C^1 inverse $(G|_W)^{-1} : O \to W$.

1. Since W is an open neighborhood of (a, b), we may choose open neighborhoods U_0 of a and V of b such that
$$U_0 \times V \subset W. \tag{233}$$
2. Since $(G|_W)^{-1}(a, 0) = (a, b) \in U_0 \times V$ and $U_0 \times V$ is open, we can take the preimage of it with respect to the continuous inverse $(G|_W)^{-1}$ to see that
$$\left((G|_W)^{-1}\right)^{-1}(U_0 \times V) = G|_W(U_0 \times V) \tag{234}$$
is an open neighborhood of $G|_W(a, b) = (a, 0)$. Therefore, you can choose an open neighborhood $U \ni a$ s.t.
$$U \times \{0\} \subset G|_W(U_0 \times V) \tag{235}$$
Then for every $x \in U$, we see that $(G|_W)^{-1}(x, 0) \in U_0 \times V$. This will be useful for setting up our proper codomain for our function f later.

Now that we have our desired open set U, we define[a]

$$f : U \to V, \qquad f(x) \coloneqq \pi_y\big((G|_W)^{-1}(x, 0)\big) \tag{236}$$

Note that since $x \in U$, $G|_W^{-1}(x, 0) \in U_0 \times V$ and so its y-projection $f(x) \in V$. It is also clear that f is C^1 since $(G|_W)^{-1}$ is C^1. Note that $f(a) = b$ because $G(a, b) = (a, 0)$, and $f(x)$ is the unique vector such that $(G|_W)^{-1}(x, 0) = (x, f(x))$. Now we must show that $y = f(x) \iff F(x, y) = 0$.

1. $(\to)$. Let $y = f(x)$, then $y = \pi_y\big((G|_W)^{-1}(x, 0)\big)$. By definition of projection, we must have $(G|_W)^{-1}(x, 0) = (*, y)$, but $* = x$ since neither G nor the inverse affects the x element. So,
$$(G|_W)^{-1}(x, 0) = (x, y) \implies (x, 0) = G|_W(x, y) = (x, F(x, y)) \implies 0 = F(x, y) \tag{237}$$

2. ($\leftarrow$). Now let $F(x, y) = 0$. Then, for $(x, y) \in U \times V$, we have

$$G|_W(x, y) = \big(x, F(x, y)\big) = (x, 0) \tag{238}$$

Since $G|_W$ is invertible, we can apply $(G|_W)^{-1}$ to both sides and so $(x, y) = (G|_W)^{-1}(x, 0)$. By applying the projection operator to both sides, we have

$$y = \pi_y(x, y) = \pi_y\big((G|_W)^{-1}(x, 0)\big) = f(x) \tag{239}$$

This also proves uniqueness. Indeed, if another function $\widetilde{f} : U \to V$ satisfies $F(x, \widetilde{f}(x)) = 0$ for all $x \in U$, then by the equivalence above, $\widetilde{f}(x) = f(x)$ for all $x \in U$.
It remains to compute the derivative of f. Since $F(x, f(x)) = 0$ for every $x \in U$, we differentiate both sides with respect to x. By the chain rule,

$$D_x F_{(x,f(x))} + D_y F_{(x,f(x))} Df_x = 0 \implies D_y F_{(x,f(x))} Df_x = -D_x F_{(x,f(x))}. \tag{240}$$

Since $D_y F_{(a,b)}$ is invertible, the set of invertible maps in $\mathcal{L}(\mathbb{R}^m, \mathbb{R}^m)$ is an open set, and the map $(x, y) \mapsto D_y F_{(x,y)}$ is continuous, there exists a neighborhood around (a, b) such that if $(x, f(x))$ is close enough to (a, b), then $D_y F_{(x,f(x))}$ is invertible. Therefore, we can shrink U accordingly to be in this neighborhood. Thus, in this new U, V, we have for all $x \in U$

$$Df_x = -\big(D_y F_{(x,f(x))}\big)^{-1} D_x F_{(x,f(x))}. \tag{241}$$

[a] π_y denotes projection onto the $\mathbb{R}^m$ component.

Example 1.19 (Circle Example)

Let $n = c = 1$, and consider the function $F : \mathbb{R} \times \mathbb{R} \longrightarrow \mathbb{R}$ defined by

$$F(x_1, x_2) = x_1^2 + x_2^2 - 1. \tag{242}$$

The level set

$$F(x_1, x_2) = 0 \tag{243}$$

is the unit circle

$$x_1^2 + x_2^2 = 1. \tag{244}$$

We would like to determine at which points of the circle we can locally write x_2 as a function of x_1. The total derivative of F is

$$DF_{(x_1,x_2)} = \begin{pmatrix} \partial_{x_1} F & \partial_{x_2} F \end{pmatrix} = \begin{pmatrix} 2x_1 & 2x_2 \end{pmatrix}. \tag{245}$$

In this case, if we want to solve for x_2 as a function of x_1, then we look at the partial derivative with respect to x_2:

$$D_{x_2} F_{(x_1,x_2)} = 2x_2. \tag{246}$$

This is invertible as a 1×1 matrix precisely when

$$x_2 \neq 0. \tag{247}$$

Therefore, by the implicit function theorem, at every point of the circle with $x_2 \neq 0$, we can locally write the circle in the form

$$x_2 = g(x_1). \tag{248}$$

For example, consider the point

$$\left(\frac{4}{5}, \frac{3}{5}\right) \tag{249}$$

on the unit circle. Since $\frac{3}{5} \neq 0$, we can locally solve for x_2 as a function of x_1. Near this point, the appropriate branch is

$$x_2 = g(x_1) = \sqrt{1 - x_1^2}. \tag{250}$$

On the other hand, at the points $(1, 0)$ and $(-1, 0)$, we have

$$D_{x_2} F_{(x_1, x_2)} = 2x_2 = 0. \tag{251}$$

So the implicit function theorem does not allow us to solve for x_2 as a function of x_1 near these points. Indeed, near $x_1 = 1$ or $x_1 = -1$, the circle fails to be the graph of a function of x_1, since nearby x_1-values correspond to two possible x_2-values.

The derivative of the implicit function g is given by the theorem:

$$Dg = -\left(D_{x_2} F\right)^{-1} D_{x_1} F. \tag{252}$$

Therefore,

$$Dg = -(2x_2)^{-1}(2x_1) = -\frac{x_1}{x_2}. \tag{253}$$

Since $x_2 = g(x_1)$, this becomes

$$g'(x_1) = -\frac{x_1}{g(x_1)}. \tag{254}$$

Similarly, if we want to solve for x_1 as a function of x_2, then we look at the partial derivative with respect to x_1:

$$D_{x_1} F_{(x_1, x_2)} = 2x_1. \tag{255}$$

This is invertible precisely when

$$x_1 \neq 0. \tag{256}$$

Hence, at every point of the circle with $x_1 \neq 0$, we can locally write the circle in the form

$$x_1 = h(x_2). \tag{257}$$

The derivative of h is

$$Dh = -\left(D_{x_1} F\right)^{-1} D_{x_2} F. \tag{258}$$

Therefore,

$$Dh = -(2x_1)^{-1}(2x_2) = -\frac{x_2}{x_1}. \tag{259}$$

Since $x_1 = h(x_2)$, this becomes

$$h'(x_2) = -\frac{x_2}{h(x_2)}. \tag{260}$$

Example 1.20 (Sphere Example)

Let $n = 2$ and $c = 1$, and consider the function $F : \mathbb{R}^2 \times \mathbb{R} \longrightarrow \mathbb{R}$ defined by

$$F(x_1, x_2, y) = x_1^2 + x_2^2 + y^2 - 1. \tag{261}$$

The level set

$$F(x_1, x_2, y) = 0 \tag{262}$$

is the unit sphere

$$x_1^2 + x_2^2 + y^2 = 1. \tag{263}$$

We would like to determine at which points of the sphere we can locally write y as a function of (x_1, x_2).

The total derivative of F is

$$DF_{(x_1,x_2,y)} = \begin{pmatrix} \partial_{x_1} F & \partial_{x_2} F & \partial_y F \end{pmatrix} = \begin{pmatrix} 2x_1 & 2x_2 & 2y \end{pmatrix}. \tag{264}$$

If we want to solve for y as a function of (x_1, x_2), then we look at the partial derivative with respect to y:

$$D_y F_{(x_1,x_2,y)} = 2y. \tag{265}$$

This is invertible as a 1×1 matrix precisely when

$$y \neq 0. \tag{266}$$

Therefore, by the implicit function theorem, at every point of the sphere with $y \neq 0$, we can locally write the sphere in the form

$$y = g(x_1, x_2). \tag{267}$$

For example, consider the point

$$\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}} \right) \tag{268}$$

on the unit sphere. Since $\frac{1}{\sqrt{2}} \neq 0$, we can locally solve for y as a function of (x_1, x_2). Near this point, the appropriate branch is

$$y = g(x_1, x_2) = \sqrt{1 - x_1^2 - x_2^2}. \tag{269}$$

On the other hand, at points on the equator where $y = 0$, the implicit function theorem does not allow us to solve for y as a function of (x_1, x_2). Geometrically, near such points, the sphere has both an upper and a lower branch over the same nearby (x_1, x_2)-values.
The derivative of the implicit function g is given by

$$Dg = -\left(D_y F\right)^{-1} D_x F. \tag{270}$$

In this case,

$$D_x F = \begin{pmatrix} 2x_1 & 2x_2 \end{pmatrix} \tag{271}$$

and

$$D_y F = 2y. \tag{272}$$

Therefore,

$$Dg = -(2y)^{-1} \begin{pmatrix} 2x_1 & 2x_2 \end{pmatrix} = \begin{pmatrix} -\frac{x_1}{y} & -\frac{x_2}{y} \end{pmatrix}. \tag{273}$$

Since $y = g(x_1, x_2)$, this becomes

$$Dg_{(x_1,x_2)} = \begin{pmatrix} -\frac{x_1}{g(x_1,x_2)} & -\frac{x_2}{g(x_1,x_2)} \end{pmatrix}. \tag{274}$$

Example 1.21 (A System of Two Equations)

Let $n = 2$ and $c = 2$, and consider the function $F : \mathbb{R}^2 \times \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ defined by

$$F(x_1, x_2, y_1, y_2) = \begin{pmatrix} y_1^2 + y_2 - x_1 \\ y_1 + e^{y_2} - x_2 \end{pmatrix}. \tag{275}$$

We would like to locally solve the system

$$F(x_1, x_2, y_1, y_2) = 0 \tag{276}$$

for (y_1, y_2) as a function of (x_1, x_2).

First observe that
$$F(1,2,1,0) = \begin{pmatrix} 1^2 + 0 - 1 \\ 1 + e^0 - 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \tag{277}$$
Thus the point
$$(x_1, x_2, y_1, y_2) = (1, 2, 1, 0) \tag{278}$$
lies on the zero set of F.
The derivative of F with respect to the y-variables is
$$D_y F_{(x_1,x_2,y_1,y_2)} = \begin{pmatrix} \partial_{y_1} F_1 & \partial_{y_2} F_1 \\ \partial_{y_1} F_2 & \partial_{y_2} F_2 \end{pmatrix} = \begin{pmatrix} 2y_1 & 1 \\ 1 & e^{y_2} \end{pmatrix}. \tag{279}$$
At the point $(1, 2, 1, 0)$, this becomes
$$D_y F_{(1,2,1,0)} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}. \tag{280}$$
Its determinant is
$$\det D_y F_{(1,2,1,0)} = 2 \cdot 1 - 1 \cdot 1 = 1. \tag{281}$$
Therefore $D_y F_{(1,2,1,0)}$ is invertible.
By the implicit function theorem, there exist open neighborhoods U of $(1, 2)$ and V of $(1, 0)$, and a unique C^1 function
$$h : U \longrightarrow V \tag{282}$$
such that
$$h(x_1, x_2) = \begin{pmatrix} h_1(x_1, x_2) \\ h_2(x_1, x_2) \end{pmatrix} \tag{283}$$
and
$$F(x_1, x_2, h_1(x_1, x_2), h_2(x_1, x_2)) = 0 \tag{284}$$
for all $(x_1, x_2) \in U$.
In other words, near $(x_1, x_2) = (1, 2)$, the system
$$\begin{cases} y_1^2 + y_2 = x_1, \\ y_1 + e^{y_2} = x_2 \end{cases} \tag{285}$$
implicitly defines (y_1, y_2) as a function of (x_1, x_2).
To compute the derivative of h, we use
$$Dh_x = -\left(D_y F_{(x,h(x))}\right)^{-1} D_x F_{(x,h(x))}. \tag{286}$$
In this example,
$$D_x F_{(x_1,x_2,y_1,y_2)} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{287}$$
Therefore,
$$Dh_x = \left(D_y F_{(x,h(x))}\right)^{-1}. \tag{288}$$
At the point $(x_1, x_2) = (1, 2)$, we have $h(1, 2) = (1, 0)$, so
$$Dh_{(1,2)} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}^{-1}. \tag{289}$$
Hence
$$Dh_{(1,2)} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}. \tag{290}$$

1.8 Extrema

Recall what a local extrema is from topology. We restate it here.

Definition 1.9 (Local Extrema)

Given a function $f : E \subset \mathbb{R}^n \longrightarrow \mathbb{R}$,

1. $a \in D$ is a **local minimum** if there exists a neighborhood U of a such that
$$f(x) \geq f(a) \text{ for every } x \in U \tag{291}$$
2. $a \in D$ is a **local maximum** if there exists a neighborhood U of x such that
$$f(x) \leq f(a) \text{ for every } x \in U \tag{292}$$

Theorem 1.17 (First Derivative Test)

Let $E \subset \mathbb{R}^n$ be open, with $f : E \to \mathbb{R}$ differentiable at $a \in E$. If a is a local extremum of f, then $Df_a = 0$.

Proof. Unlike the sequential proof we did for the univariate case, we prove the contrapositive. WLOG, let a be a maximum and $Df_a \neq 0$. Then, we basically have a nontrivial dual vector $Df_a \in (\mathbb{R}^n)^*$, and so there exists a $v \in \mathbb{R}^n$ s.t. $Df_a v > 0$. Now by definition of differentiability, we have $f(a + v) = f(a) + Df_a v + \Phi(\|v\|)\|v\|$, so let us make the norm of v so small such that
$$\frac{Df_a v}{\|v\|} > |\Phi(\|v\|)| \tag{293}$$
which is possible since $\frac{Df_a v}{\|v\|}$ is a constant while $\Phi(\|v\|) \to 0$ as $\|v\| \to 0$. Let us also shrink v so that $a + v \in E$ as well. Therefore, we can find a $v \in \mathbb{R}^n$ so that $Df_a v > |\Phi(\|v\|)|\|v\|$, and so
$$f(a + v) = f(a) + \underbrace{Df_a v + \Phi(\|v\|)\|v\|}_{>0} > f(a) \tag{294}$$
This can be done for an arbitrary neighborhood of a, contradicting the fact that a is a local maximum.

Note that even though the converse of this theorem is not true, we can use the contrapositive to determine that every point that has a nonzero derivative cannot be a extremum.

Example 1.22 (Function with an Infinite Number of Critical Points)

A function may also have an infinite amount of critical points (e.g. if they lie in a circle).

Unfortunately, determining whether a point a is a local extremum requires us to define an open neighborhood around a. This means that we can only determine local extrema within open sets in $\mathbb{R}^n$. Therefore, we must modify our procedure when looking for extrema on functions defined over closed bounded sets. We now describe a method of computing the global extrema. Let $f : E \subset \mathbb{R}^n \longrightarrow \mathbb{R}$ be a multivariable function defined on a closed and bounded set $E = E^\circ \cup \partial E$, where the E° is the interior and ∂U is the boundary of E. To find the global extrema on E, we find all local extrema of

1. f defined over the interior of E°, which is open, by locating all points where $Df = 0$, and
2. f defined over ∂E.

In the second point, we could reparamaterize ∂E cleverly by another function $\phi : \mathbb{R}^k \to \mathbb{R}^n$ (e.g. a path function for 1-dimensional manifolds ∂E) where the image coincides with ∂E, but this can be tedious. A more flexible method is to introduce a system of equality constraints of the form

$$g : \mathbb{R}^n \to \mathbb{R}^c, \quad g(x) = \begin{pmatrix} g_1(x) \\ \vdots \\ g_c(x) \end{pmatrix} = 0 \tag{295}$$

which really just represents a level set of g at 0, i.e. a set described by an implicit representation.[1]

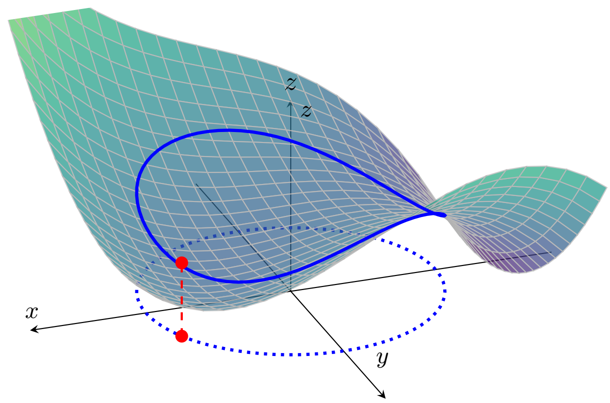


Figure 11: An example of a function $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$ constrained to the unit circle, where $g(x, y) = x^2 + y^2 - 1 = 0$. The blue curve in the xy-plane is the constraint set, and the upper blue curve is its image on the graph of f.

To solve this constraint problem, we use the method of Lagrange multipliers. The basic idea is to convert a constrained problem into a form such that the derivative test of an unconstrained problem can still be applied.

Theorem 1.18 (Lagrange Multipliers Theorem)

Let $f : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}^c$ be C^1 functions, and define $M = \{x \in \mathbb{R}^n : g(x) = 0\}$. Suppose that

1. $a \in M$ is a local maximum of f restricted to M,
2. $\mathrm{rank}(Dg_a) = c$.

Then there exists a unique row vector $\lambda \in \mathbb{R}^{1\times c}$, called the **Lagrange multiplier vector**, such that

$$Df_a = \lambda Dg_a. \tag{296}$$

Proof. Since $\mathrm{rank}(Dg_a) = c$, the $c \times n$ matrix Dg_a contains an invertible $c \times c$ submatrix. After reordering the coordinates of $\mathbb{R}^n$ if necessary, write

$$x = (u, v) \in \mathbb{R}^{n-c} \oplus \mathbb{R}^c, \tag{297}$$

Then, we can write $a = (\alpha, \beta)$, so that $D_v g_{(\alpha,\beta)}$ is invertible. Since $a \in M$, we have $g(\alpha, \beta) = g(a) = 0$. Therefore, by the implicit function theorem, there exist open neighborhoods $U \ni \alpha$ and $V \ni \beta$, together with a unique C^1 function

$$h : U \to V, \tag{298}$$

such that $h(\alpha) = \beta$ and, for every $(u, v) \in U \times V$, $g(u, v) = 0 \iff v = h(u)$. Moreover, for all $u \in U$,

$$Dh_u = -\big(D_v g_{(u,h(u))}\big)^{-1} D_u g_{(u,h(u))}. \tag{299}$$

[1]In physics, these types of "well-behaved" constraints are known as *holonomic constraints*.

and in particular, $Dh_\alpha = -\big(D_v g_a\big)^{-1} D_u g_a$. Define the restriction of f to this local parameterization of M by

$$\varphi : U \to \mathbb{R}, \qquad \varphi(u) = f(u, h(u)). \tag{300}$$

Since $g(u, h(u)) = 0$, every point $(u, h(u))$ lies in M. Furthermore,

$$(\alpha, h(\alpha)) = (\alpha, \beta) = a. \tag{301}$$

Because a is a local maximum of f restricted to M, after shrinking U if necessary, α is a local maximum of φ. Therefore,

$$D\varphi_\alpha = 0. \tag{302}$$

By the chain rule, we have $D\varphi_\alpha = D_u f_a + D_v f_a D h_\alpha$, and substituting the formula for Dh_α gives

$$0 = D_u f_a - D_v f_a \big(D_v g_a\big)^{-1} D_u g_a \implies D_u f_a = D_v f_a \big(D_v g_a\big)^{-1} D_u g_a. \tag{303}$$

If we define $\lambda \coloneqq D_v f_a \big(D_v g_a\big)^{-1} \in \mathbb{R}^{1\times c}$, then we can see that

1. From above, we have $D_u f_a = \lambda D_u g_a$.
2. By the definition of λ, $\lambda D_v g_a = D_v f_a \big(D_v g_a\big)^{-1} D_v g_a = D_v f_a$.

Combining these two into matrix form, we have

$$D f_a = \begin{pmatrix} D_u f_a & D_v f_a \end{pmatrix} = \begin{pmatrix} \lambda D_u g_a & \lambda D_v g_a \end{pmatrix} = \lambda \begin{pmatrix} D_u g_a & D_v g_a \end{pmatrix} = \lambda D g_a. \tag{304}$$

It remains to prove uniqueness. Suppose that $\lambda, \widetilde{\lambda} \in \mathbb{R}^{1\times c}$ both satisfy

$$D f_a = \lambda D g_a = \widetilde{\lambda} D g_a \implies (\lambda - \widetilde{\lambda}) D g_a = 0. \tag{305}$$

Restricting this equality to the v-columns gives $(\lambda - \widetilde{\lambda}) D_v g_a = 0$. Since $D_v g_a$ is invertible, it follows that $\lambda - \widetilde{\lambda} = 0$, and we are done.

By transposing Equation 296, we can get a matrix realization of

$$\begin{pmatrix} \frac{\partial f}{\partial x_1}(a) \\ \vdots \\ \frac{\partial f}{\partial x_n}(a) \end{pmatrix} = \begin{pmatrix} \frac{\partial g_1}{\partial x_1}(a) & \dots & \frac{\partial g_c}{\partial x_1}(a) \\ \vdots & \ddots & \vdots \\ \frac{\partial g_1}{\partial x_n}(a) & \dots & \frac{\partial g_c}{\partial x_n}(a) \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_c \end{pmatrix} \tag{306}$$

$$= \lambda_1 \begin{pmatrix} \frac{\partial g_1}{\partial x_1}(a) \\ \vdots \\ \frac{\partial g_1}{\partial x_n}(a) \end{pmatrix} + \lambda_2 \begin{pmatrix} \frac{\partial g_2}{\partial x_1}(a) \\ \vdots \\ \frac{\partial g_2}{\partial x_n}(a) \end{pmatrix} + \ldots + \lambda_c \begin{pmatrix} \frac{\partial g_c}{\partial x_1}(a) \\ \vdots \\ \frac{\partial g_c}{\partial x_n}(a) \end{pmatrix} \tag{307}$$

This equation tells us that at any critical points a of f evaluated under the equality constraints, the total derivative of f at a can be expressed as a unique linear combination of the total derivatives of the constraints $D(g_i)_a$, with the Lagrange multipliers acting as coefficients. Therefore, finding the critical points a of f constrained with g is equivalent to solving the system of $c + n$ equations for the n unknowns in x and c unknowns in λ.

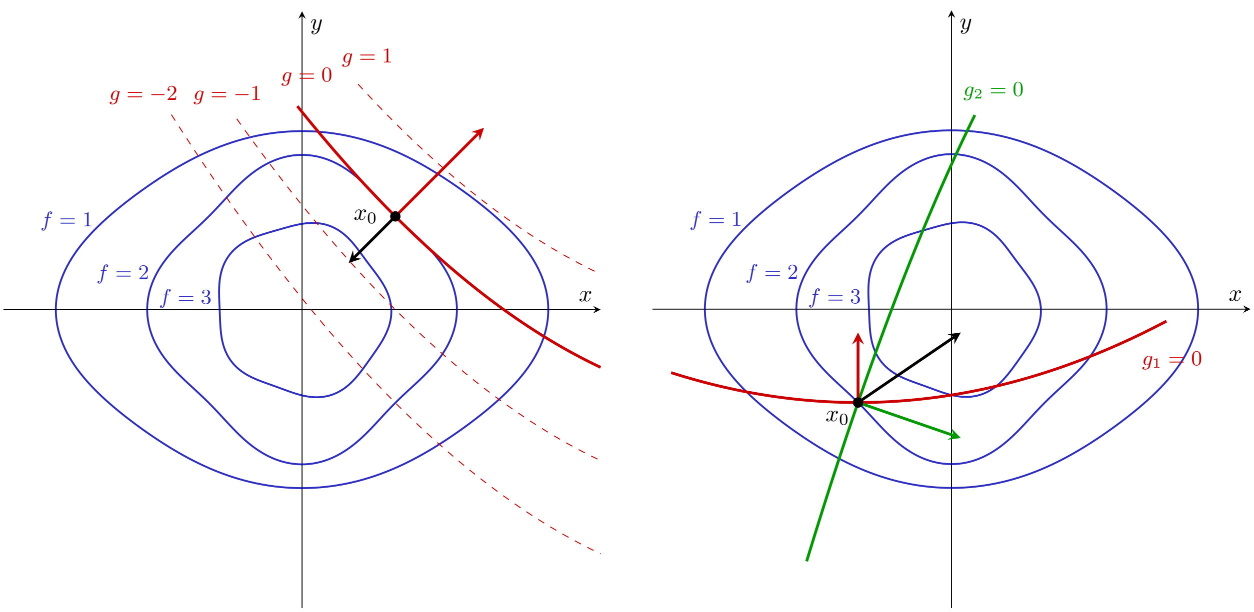
$$g(x) = 0 \iff \begin{cases} g_1(x) & = 0 \\ \dots & = 0 \\ g_c(x) & = 0 \end{cases} \tag{308}$$

$$D f_a = \lambda D g_a \iff \begin{cases} \frac{\partial f}{\partial x_1}(a) & = \lambda_1^* \frac{\partial g_1}{\partial x_1}(a) + \lambda_2^* \frac{\partial g_2}{\partial x_1}(a) + \ldots + \lambda_c^* \frac{\partial g_c}{\partial x_1}(a) \\ \dots & = \dots \\ \frac{\partial f}{\partial x_n}(a) & = \lambda_1^* \frac{\partial g_1}{\partial x_n}(a) + \lambda_2^* \frac{\partial g_2}{\partial x_n}(a) + \ldots + \lambda_c^* \frac{\partial g_c}{\partial x_n}(a) \end{cases} \tag{309}$$

This relationship between the gradient of the function and gradients of the constraints rather naturally leads to a reformulation of the original problem, known as the *Lagrangian function*. That is, in order to find the maximum/minimum of f subjected to the equality constraint $g(x) = 0$, we form the Lagrangian function

$$\mathcal{L}(x, \lambda) = f(x) - \lambda^T g(x) \tag{310}$$

and find the stationary points of $\mathcal{L}$ considered as a function of $x \in \mathbb{R}^n$ and the Lagrange multiplier $\lambda \in \mathbb{R}^c$. The main advantage to this method is that it allows the optimization to be solved without explicit parameterization in terms of the constraints.



(a) When there is a single constraint $g : \mathbb{R}^n \longrightarrow \mathbb{R}$, $\nabla f(x_0)$ can be expressed as a linear combination of $\nabla g(x_0)$.

(b) When there are multiple constraints g_1, g_2, $\nabla f(x_0)$ can be expressed as a linear combination of $\nabla g_1(x_0)$ and $\nabla g_2(x_0)$.

Figure 12: From the properties of the gradient, $\nabla f(x_0)$ is orthogonal to the level set of points satisfying $f(x) = f(x_0)$ at point x_0. We can see that the point where the contour line of $g(x) = 0$ tangentially touches the contours of f is the maximum. Since it intersects it tangentially, the gradient vector at that point $\nabla g(x_0)$ is parallel to $\nabla f(x_0)$.

From the properties of the gradient introduced before, $\nabla f(x_0)$ is orthogonal to the level set of points satisfying $f(x) = c$ at the point x_0. But this level set $f(x) = c$ actually intersects the level set determined by $g(x) = c$ at the point x_0 and is indistinguishable from each other at x_0. This means that $\nabla g(x_0)$ is normal the level set of $g(x) = c$ at $x_0 \iff$ it is normal to the level set of $f(x) = c$ at x_0. But $\nabla f(x_0)$ is also normal at that point, so $\nabla f(x_0)$ must be parallel to $\nabla g(x_0)$.

1.9 Exercises

Exercise 1.1 (Rudin 9.6)

If $f(0,0) = 0$ and

$$f(x, y) = \frac{xy}{x^2 + y^2} \qquad \text{if } (x, y) \neq (0, 0), \tag{311}$$

prove that $(D_1 f)(x, y)$ and $(D_2 f)(x, y)$ exist at every point of $\mathbb{R}^2$, although f is not continuous at $(0, 0)$.

Solution. Easy.

Exercise 1.2 (Rudin 9.7)

Suppose that f is a real-valued function defined in an open set $E \subset \mathbb{R}^n$, and that the partial derivatives $D_1 f, \ldots, D_n f$ are bounded in E. Prove that f is continuous in E.

Solution. Proved in this theorem. If we had stronger assumptions like the function was differentiable over convex set, we can say that since the partials are bounded, the total derivative Df is bounded—by, say M—over E. If E is convex, then by MVT, we have $f(y) - f(x) \leq M\|x - y\|$, and so f is uniformly continuous.

Exercise 1.3 (Rudin 9.8)

Suppose that f is a differentiable real function in an open set $E \subset \mathbb{R}^n$, and that f has a local maximum at a point $x \in E$. Prove that

$$f'(x) = 0. \tag{312}$$

Solution. Proved in first derivative test.

Exercise 1.4 (Rudin 9.9)

If f is a differentiable mapping of a connected open set $E \subset \mathbb{R}^n$ into $\mathbb{R}^m$, and if $f'(x) = 0$ for every $x \in E$, prove that f is constant in E.

Solution. It would be nice if we can upgrade connectedness to path connectedness since then we can just work with path functions. Luckily, open sets in $\mathbb{R}^n$ are locally path connected, so they are also path connected. Therefore, for any $x, y \in E$, find a path $\gamma : [0,1] \to E$ s.t. $\gamma(0) = x, \gamma(1) = y$. Then, $(f \circ \gamma)'(t) = Df_{\gamma(t)}\gamma'(t) = 0\gamma'(t) = 0$.[a] So, by the univariate MVT applied to each component, $(f \circ \gamma)(t) = c$ for some constant c. γ was chosen with arbitrary $x, y \in E$, so this implies that $f(x) = c$.

[a] Technically, γ must be differentiable, but we can just say that γ is piecewise linear, aka polygonally connected.

Exercise 1.5 (Rudin 9.10)

If f is a real function defined in a convex open set $E \subset \mathbb{R}^n$, such that $(D_1 f)(x) = 0$ for every $x \in E$, prove that $f(x)$ depends only on $x_2, \ldots, x_n$.

Show that the convexity of E can be replaced by a weaker condition, but that some condition is required. For example, if $n = 2$ and E is shaped like a horseshoe, the statement may be false.

Exercise 1.6 (Rudin 9.11)

If f and g are differentiable real functions in $\mathbb{R}^n$, prove that

$$\nabla(fg) = f\nabla g + g\nabla f, \tag{313}$$

and that

$$\nabla\left(\frac{1}{f}\right) = -f^{-2}\nabla f \tag{314}$$

wherever $f \neq 0$.

Exercise 1.7 (Rudin 9.12)

Fix two real numbers a and b, $0 < a < b$. Define a mapping $f = (f_1, f_2, f_3)$ of $\mathbb{R}^2$ into $\mathbb{R}^3$ by

$$f_1(s,t) = (b + a\cos s)\cos t, \tag{315}$$
$$f_2(s,t) = (b + a\cos s)\sin t, \tag{316}$$
$$f_3(s,t) = a\sin s. \tag{317}$$

Describe the range K of f. (It is a certain compact subset of $\mathbb{R}^3$.)

(a) Show that there are exactly 4 points $p \in K$ such that

$$(\nabla f_1)(f^{-1}(p)) = 0. \tag{318}$$

Find these points.

(b) Determine the set of all $q \in K$ such that

$$(\nabla f_3)(f^{-1}(q)) = 0. \tag{319}$$

(c) Show that one of the points p found in part (a) corresponds to a local maximum of f_1, one corresponds to a local minimum, and that the other two are neither (they are so-called "saddle points").
Which of the points q found in part (b) correspond to maxima or minima?

(d) Let λ be an irrational real number, and define $g(t) = f(t, \lambda t)$. Prove that g is a one-to-one mapping of $\mathbb{R}^1$ onto a dense subset of K. Prove that

$$|g'(t)|^2 = a^2 + \lambda^2(b + a\cos t)^2. \tag{320}$$

Exercise 1.8 (Rudin 9.13)

Suppose that f is a differentiable mapping of $\mathbb{R}^1$ into $\mathbb{R}^3$ such that $|f(t)| = 1$ for every t. Prove that

$$f'(t) \cdot f(t) = 0. \tag{321}$$

Interpret this result geometrically.

Solution. We must have $\|f(t)\|^2 = 1$. By differentiating both sides, we get $2f(t) \cdot f(t) = 0$, which is the desired result. This means that if f lives in the unit sphere, then the direction of its change must always be tangent to the sphere.

Exercise 1.9 (Rudin 9.14)

Define $f(0,0) = 0$ and

$$f(x,y) = \frac{x^3}{x^2 + y^2} \qquad \text{if } (x,y) \neq (0,0). \tag{322}$$

(a) Prove that $D_1 f$ and $D_2 f$ are bounded functions in $\mathbb{R}^2$. (Hence f is continuous.)

(b) Let u be any unit vector in $\mathbb{R}^2$. Show that the directional derivative $(D_u f)(0,0)$ exists, and that its absolute value is at most 1.

(c) Let γ be a differentiable mapping of $\mathbb{R}^1$ into $\mathbb{R}^2$ (in other words, γ is a differentiable curve in $\mathbb{R}^2$), with $\gamma(0) = (0,0)$ and $|\gamma'(0)| > 0$. Put $g(t) = f(\gamma(t))$ and prove that g is differentiable for every $t \in \mathbb{R}^1$.
If $\gamma \in \mathscr{C}'$, prove that $g \in \mathscr{C}'$.

(d) In spite of this, prove that f is not differentiable at $(0,0)$.

Hint: Formula (40) fails.

Exercise 1.10 (Rudin 9.15)

Define $f(0,0) = 0$, and put

$$f(x,y) = x^2 + y^2 - 2x^2 y - \frac{4x^6 y^2}{(x^4 + y^2)^2} \tag{323}$$

if $(x,y) \neq (0,0)$.

(a) Prove, for all $(x,y) \in \mathbb{R}^2$, that

$$4x^4 y^2 \leq (x^4 + y^2)^2. \tag{324}$$

Conclude that f is continuous.

(b) For $0 \leq \theta \leq 2\pi$, $-\infty < t < \infty$, define

$$g_\theta(t) = f(t\cos\theta, t\sin\theta). \tag{325}$$

Show that

$$g_\theta(0) = 0, \qquad g_\theta'(0) = 0, \qquad g_\theta''(0) = 2. \tag{326}$$

Each g_θ has therefore a strict local minimum at $t = 0$.
In other words, the restriction of f to each line through $(0,0)$ has a strict local minimum at $(0,0)$.

(c) Show that $(0,0)$ is nevertheless not a local minimum for f, since

$$f(x,x^2) = -x^4. \tag{327}$$

Exercise 1.11 (Rudin 9.16)

Show that the continuity of f' at the point a is needed in the inverse function theorem, when the case $n = 1$: If

$$f(t) = t + 2t^2 \sin\left(\frac{1}{t}\right) \tag{328}$$

for $t \neq 0$, and $f(0) = 0$, then $f'(0) = 1$, f' is bounded in $(-1,1)$, but f is not one-to-one in any neighborhood of 0.

Exercise 1.12 (Rudin 9.17)

Let $f = (f_1, f_2)$ be the mapping of $\mathbb{R}^2$ into $\mathbb{R}^2$ given by

$$f_1(x,y) = e^x \cos y, \qquad f_2(x,y) = e^x \sin y. \tag{329}$$

(a) What is the range of f?

(b) Show that the Jacobian of f is not zero at any point of $\mathbb{R}^2$. Thus every point of $\mathbb{R}^2$ has a neighborhood in which f is one-to-one. Nevertheless, f is not one-to-one on $\mathbb{R}^2$.

(c) Put

$$a = \left(0, \frac{\pi}{3}\right), \qquad b = f(a), \tag{330}$$

let g be the continuous inverse of f, defined in a neighborhood of b, such that $g(b) = a$. Find an explicit formula for g, compute $f'(a)$ and $g'(b)$, and verify the formula (52).

(d) What are the images under f of lines parallel to the coordinate axes?

Exercise 1.13 (Rudin 9.18)

Answer analogous questions for the mapping defined by

$$u = x^2 - y^2, \qquad v = 2xy. \tag{331}$$

Exercise 1.14 (Rudin 9.19)

Show that the system of equations

$$3x + y - z + u^2 = 0, \tag{332}$$

$$x - y + 2z + u = 0, \tag{333}$$

$$2x + 2y - 3z + 2u = 0 \tag{334}$$

can be solved for x, y, u in terms of z; for x, z, u in terms of y; for y, z, u in terms of x; but not for x, y, z in terms of u.

Exercise 1.15 (Rudin 9.20)

Take $n = m = 1$ in the implicit function theorem, and interpret the theorem (as well as its proof) graphically.

Exercise 1.16 (Rudin 9.21)

Define f in $\mathbb{R}^2$ by

$$f(x, y) = 2x^3 - 3x^2 + 2y^3 + 3y^2. \tag{335}$$

(a) Find the four points in $\mathbb{R}^2$ at which the gradient of f is zero. Show that f has exactly one local maximum and one local minimum in $\mathbb{R}^2$.

(b) Let S be the set of all $(x, y) \in \mathbb{R}^2$ at which $f(x, y) = 0$. Find those points of S that have no neighborhoods in which the equation $f(x, y) = 0$ can be solved for y in terms of x (or for x in terms of y). Describe S as precisely as you can.

Exercise 1.17 (Rudin 9.22)

Give a similar discussion for

$$f(x, y) = 2x^3 + 6xy^2 - 3x^2 + 3y^2. \tag{336}$$

Exercise 1.18 (Rudin 9.23)

Define f in $\mathbb{R}^3$ by

$$f(x, y_1, y_2) = x^2 y_1 + e^x + y_2. \tag{337}$$

Show that $f(0, 1, -1) = 0$, $(D_1 f)(0, 1, -1) \neq 0$, and that there exists therefore a differentiable function g in some neighborhood of $(1, -1)$ in $\mathbb{R}^2$, such that $g(1, -1) = 0$ and

$$f(g(y_1, y_2), y_1, y_2) = 0. \tag{338}$$

Find $(D_1 g)(1, -1)$ and $(D_2 g)(1, -1)$.

Exercise 1.19 (Rudin 9.24)

For $(x, y) \neq (0, 0)$, define $f = (f_1, f_2)$ by

$$f_1(x, y) = \frac{x^2 - y^2}{x^2 + y^2}, \qquad f_2(x, y) = \frac{xy}{x^2 + y^2}. \tag{339}$$

Compute the rank of $f'(x, y)$, and find the range of f.

Exercise 1.20 (Rudin 9.25)

Suppose $A \in L(\mathbb{R}^n, \mathbb{R}^m)$, and let r be the rank of A.

(a) Define S as in the proof of Theorem 9.32. Show that SA is a projection in $\mathbb{R}^n$ whose null space is $\mathscr{N}(A)$ and whose range is $\mathscr{R}(S)$.
Hint: By (68), $SASA = SA$.

(b) Use (a) to show that

$$\dim \mathscr{N}(A) + \dim \mathscr{R}(A) = n. \tag{340}$$

Exercise 1.21 (Rudin 9.26)

Show that the existence (and even the continuity) of $D_{12}f$ does not imply the existence of $D_1 f$. For example, let $f(x, y) = g(x)$, where g is nowhere differentiable.

Exercise 1.22 (Rudin 9.27)

Put $f(0, 0) = 0$, and

$$f(x, y) = \frac{xy(x^2 - y^2)}{x^2 + y^2} \tag{341}$$

if $(x, y) \neq (0, 0)$. Prove that

(a) f, $D_1 f$, and $D_2 f$ are continuous in $\mathbb{R}^2$;

(b) $D_{12}f$ and $D_{21}f$ exist at every point of $\mathbb{R}^2$, and are continuous except at $(0, 0)$;

(c)

$$(D_{12}f)(0, 0) = 1, \qquad (D_{21}f)(0, 0) = -1. \tag{342}$$

Exercise 1.23 (Rudin 9.28)

For $t \geq 0$, put

$$\varphi(x, t) = \begin{cases} x, & 0 \leq x \leq \sqrt{t}, \\ -x + 2\sqrt{t}, & \sqrt{t} \leq x \leq 2\sqrt{t}, \\ 0, & \text{otherwise}, \end{cases} \tag{343}$$

and put $\varphi(x, t) = -\varphi(x, |t|)$ if $t < 0$.
Show that φ is continuous on $\mathbb{R}^2$, and

$$(D_2\varphi)(x, 0) = 0 \tag{344}$$

for all x. Define

$$f(t) = \int_{-1}^{1} \varphi(x, t)\, dx. \tag{345}$$

Show that $f(t) = t$ if $|t| < \frac{1}{4}$. Hence

$$f'(0) \neq \int_{-1}^{1} (D_2\varphi)(x, 0)\, dx. \tag{346}$$

Exercise 1.24 (Rudin 9.29)

Let E be an open set in $\mathbb{R}^n$. The classes $\mathscr{C}^{(k)}(E)$ and $\mathscr{C}^{(\infty)}(E)$ are defined in the text. By induction, $\mathscr{C}^{(k)}(E)$ can be defined as follows, for all positive integers k: To say that $f \in \mathscr{C}^{(k)}(E)$ means that the partial derivatives $D_1 f, \ldots, D_n f$ belong to $\mathscr{C}^{(k-1)}(E)$.
Assume $f \in \mathscr{C}^{(k)}(E)$, and show (by repeated application of Theorem 9.41) that the kth-order derivative

$$D_{i_1 i_2 \cdots i_k} f = D_{i_1} D_{i_2} \cdots D_{i_k} f \tag{347}$$

is unchanged if the subscripts $i_1, \ldots, i_k$ are permuted.
For instance, if $n \geq 3$, then

$$D_{1213} f = D_{3112} f \tag{348}$$

for every $f \in \mathscr{C}^{(4)}$.

Exercise 1.25 (Rudin 9.30)

Let $f \in \mathscr{C}^{(m)}(E)$, where E is an open subset of $\mathbb{R}^n$. Fix $a \in E$, and suppose $x \in \mathbb{R}^n$ is so close to 0 that the points

$$p(t) = a + tx \tag{349}$$

lie in E whenever $0 \leq t \leq 1$. Define

$$h(t) = f(p(t)) \tag{350}$$

for all $t \in \mathbb{R}^1$ for which $p(t) \in E$.

(a) For $1 \leq k \leq m$, show (by repeated application of the chain rule) that

$$h^{(k)}(t) = \sum (D_{i_1 \cdots i_k} f)(p(t)) x_{i_1} \cdots x_{i_k}. \tag{351}$$

The sum extends over all ordered k-tuples $(i_1, \ldots, i_k)$ in which each i_j is one of the integers $1, \ldots, n$.

(b) By Taylor's theorem (5.15),

$$h(1) = \sum_{k=0}^{m-1} \frac{h^{(k)}(0)}{k!} + \frac{h^{(m)}(t)}{m!} \tag{352}$$

for some $t \in (0, 1)$. Use this to prove Taylor's theorem in n variables by showing that the formula

$$f(a + x) = \sum_{k=0}^{m-1} \frac{1}{k!} \sum (D_{i_1 \cdots i_k} f)(a) x_{i_1} \cdots x_{i_k} + r(x) \tag{353}$$

represents $f(a + x)$ as the sum of its so-called "Taylor polynomial of degree $m - 1$," plus a remainder that satisfies

$$\lim_{x \to 0} \frac{r(x)}{|x|^{m-1}} = 0. \tag{354}$$

Each of the inner sums extends over all ordered k-tuples $(i_1, \ldots, i_k)$, as in part (a); as usual, the zero-order derivative of f is simply f, so that the constant term of the Taylor polynomial of f at a is $f(a)$.

(c) Exercise 29 shows that repetition occurs in the Taylor polynomial written in part (b). For instance, D_{113} occurs three times, as D_{113}, D_{131}, and D_{311}. The sum of the corresponding three terms can be written in the form

$$3(D_1^2 D_3 f)(a) x_1^2 x_3. \tag{355}$$

Prove, by calculating how often each derivative occurs, that the Taylor polynomial in (b) can be written in the form

$$\sum \frac{(D_1^{s_1} \cdots D_n^{s_n} f)(a)}{s_1! \cdots s_n!} x_1^{s_1} \cdots x_n^{s_n}. \tag{356}$$

Here the summation extends over all ordered n-tuples $(s_1, \ldots, s_n)$ such that each s_i is a nonnegative integer, and

$$s_1 + \cdots + s_n \leq m - 1. \tag{357}$$

Exercise 1.26 (Rudin 9.31)

Suppose $f \in \mathscr{C}^{(3)}$ in some neighborhood of a point $a \in \mathbb{R}^2$, the gradient of f is 0 at a, but not all second-order derivatives of f are 0 at a. Show how one can then determine from the Taylor polynomial of f at a (of degree 2) whether f has a local maximum, or a local minimum, or neither, at the point a. Extend this to $\mathbb{R}^n$ in place of $\mathbb{R}^2$.

2 Higher Order Derivatives

So far, we have talked about derivatives as linear maps. However, this language is a bit too restrictive because it doesn't allow us to represent the derivatives of higher order using matrices. It turns out that these higher order derivatives are tensors.

2.1 Second Order Derivatives

Definition 2.1 (Second Derivative)

Let $f : E \subset \mathbb{R}^n \to \mathbb{R}$ be a function that is differentiable at a. Then, $Df_a : \mathbb{R}^n \to \mathbb{R}$.
Then, the **second total derivative** is a bilinear map

Definition 2.2 (Hessian)

Given $f : E \subset V \to \mathbb{R}$ twice-differentiable at x, the **Hessian** of f at x is the matrix realization of the total derivative, denoted Hf_x. The entries of the Hessian matrix are precisely the partials.

$$Hf_x := \begin{pmatrix} \partial_{x_1 x_1}(a) & \dots & \partial_{x_1 x_n}(a) \\ \vdots & \ddots & \vdots \\ \partial_{x_n x_1}(a) & \dots & \partial_{x_n x_n}(a) \end{pmatrix} \text{ and } Hf := \begin{pmatrix} \partial_{x_1 x_1} & \dots & \partial_{x_1 x_n} \\ \vdots & \ddots & \vdots \\ \partial_{x_n x_1} & \dots & \partial_{x_n x_n} \end{pmatrix} \tag{358}$$

Proof.

So, we can see that the Hessian matrix must be symmetric for a twice-differentiable function.

Theorem 2.1 (Clairut's Theorem)

Given $f \in C^2$ at point a, its second iterated partials are equal.

$$\partial_{x_i x_j} f(a) = \partial_{x_j x_i} f(a) \text{ for } i, j = 1, 2, \dots, n \tag{359}$$

Proof. For clarity, denote x_i, x_j as x, y and ignore the rest of the variables. Then, the partial derivatives $\partial_{xy} f$ and $\partial_{yx} f$ at a point (x_0, y_0) can be expressed as double limits:

$$\partial_{xy} f(x_0, y_0) = \lim_{y \to y_0} \frac{\partial_x f(x_0, y) - \partial_x f(x_0, y_0)}{y - y_0} \tag{360}$$

where $\partial_x f : D \subset \mathbb{R}^n \longrightarrow \mathbb{R}$. We can use the two limit definitions of partial derivatives

$$\partial_x f(x_0, y) = \lim_{x \to x_0} \frac{f(x, y) - f(x_0, y)}{x - x_0} \text{ and } \partial_x f(x_0, y_0) = \lim_{x \to x_0} \frac{f(x, y_0) - f(x_0, y_0)}{x - x_0} \tag{361}$$

and substitute them to get the two partials

$$\begin{aligned} \partial_{xy} f(x_0, y_0) &= \lim_{y \to y_0} \frac{\lim_{x \to x_0} \frac{f(x,y) - f(x_0,y)}{x - x_0} - \lim_{y \to y_0} \frac{f(x,y_0) - f(x_0,y_0)}{x - x_0}}{y - y_0} && (362) \\ &= \lim_{y \to y_0} \lim_{x \to x_0} \bigg(\frac{f(x,y) - f(x_0,y) - f(x,y_0) + f(x_0,y_0)}{(x - x_0)(y - y_0)} \bigg) && (363) \\ \partial_{yx} f(x_0, y_0) &= \lim_{x \to x_0} \frac{\lim_{y \to y_0} \frac{f(x,y) - f(x,y_0)}{y - y_0} - \lim_{y \to y_0} \frac{f(x_0,y) - f(x_0,y_0)}{y - y_0}}{x - x_0} && (364) \\ &= \lim_{x \to x_0} \lim_{y \to y_0} \bigg(\frac{f(x,y) - f(x,y_0) - f(x_0,y) + f(x_0,y_0)}{(y - y_0)(x - x_0)} \bigg) && (365) \end{aligned}$$

Now invoking our assumption that f is C^2, the two limits, which approach (x_0, y_0) along different paths, both exist and are equal to

$$\lim_{(x,y)\to(x_0,y_0)} \frac{f(x,y) - f(x_0,y) - f(x,y_0) + f(x_0,y_0)}{(x-x_0)(y-y_0)} \tag{366}$$

and therefore $\partial_{xy} f = \partial_{yx} f$.

Once we have computed the Hessian, let's take the eigendecomposition of it. Since Hf_a is a real symmetric matrix, by the spectral theorem, it will have n real eigenvalues $\lambda_1, \ldots, \lambda_n$ (in descending values) and corresponding orthonormal eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$. Now given the gradient $\nabla f(a)$ at a, we can approximate $\nabla f(a + \mathbf{h})$ at $a + \mathbf{h}$ using its total derivative as

$$\nabla f(a + \mathbf{h}) \approx \nabla f(a) + D\nabla f_a \mathbf{h} = \nabla f(a) + Hf(a)\mathbf{h} \tag{367}$$

The eigenvalues of $Hf(a)$ will tell us how "fast" the gradient changes at a. That is, given a small displacement vector $\mathbf{h}$, we can take an orthonormal decomposition of it in the form

$$\mathbf{h} = \sum_i h_i \mathbf{v}_i \tag{368}$$

and now the approximate gradient can be written as

$$\nabla f(a + \mathbf{h}) = \nabla f(a) + \sum_i h_i \lambda_i \mathbf{v}_i \tag{369}$$

Therefore, bigger λ_i's will contribute to a greater change in $f(a)$, and smaller ones will contribute less. We can use this information to speed up convergence by scaling along different axes of $\mathbf{h}$ when sampling.

2.2 Convexity

Recall what a convex function is. For smooth functions $f : \mathbb{R}^n \to \mathbb{R}$, we have an equivalent formulation of convexity.

Theorem 2.2 (Convexity of C^1 Functions)

Let D be a convex set and $f : E \subset \mathbb{R}^n \longrightarrow \mathbb{R}$ be C^1. Then, f is convex over E if and only if

$$f(a) + Df_a(b - a) \leq f(b) \tag{370}$$

for all $a, b \in E$.

This theorem for C^1 functions is quite intuitive, since for a convex function, the tangent plane on its graph must never be "above" the graph. In other words, the first order approximation must be a global underestimate of f.

Theorem 2.3 (Convexity of C^2 Functions)

Let $f : E \subset \mathbb{R}^n \to \mathbb{R}$ be a C^2-function over E. Then, f is convex over E if and only if Hf is positive semidefinite over all interior points of E.

In order to determine whether a critical point a is a relative maximum, minimum, or neither, we use the second derivative test.

Theorem 2.4 (2nd Derivative Test)

Let a be a critical point of C^2 function $f : \mathbb{R}^n \longrightarrow \mathbb{R}$. That is, $Df_a = \mathbf{0}$. Then,

1. a is a local minimum if Hf_a is positive definite.
2. a is a local maximum if Hf_a is negative definite.
3. a is a *saddle point* if Hf_a is not positive definite nor negative definite.

Visually, this makes sense since given a critical point a, the derivative matrix would be 0, meaning that the 2nd degree Taylor expansion of f near a would be in form

$$f(x) \approx f(a) + \frac{1}{2}(x-a)^T Hf_a(x-a) \tag{371}$$

If Hf_a is positive definite, then by definition $\frac{1}{2}(x-a)^T Hf_a(x-a) > 0$ for all x near a, and so f would increase in every direction within the neighborhood of a. The logic follows similarly for negative definite matrix Hf_a. If Hf_a is not positive nor negative definite, then $\frac{1}{2}(x-a)^T Hf_a(x-a)$ could be positive or negative, depending on which direction vector $\mathbf{h} = x - a$ we choose for computing the directional derivative. Therefore, f will increase for certain $\mathbf{h}$ and decrease for other $\mathbf{h}$.

2.3 k-Times Continuously Differentiable Functions

Recall from linear algebra that a linear map $f : V \to W$ is an element of the tensor product space $V^\dagger \otimes W$, where $V^\dagger$ represents the dual space of V. That is,

$$\text{Hom}(V, W) \simeq V^\dagger \otimes W \tag{372}$$

Definition 2.3 (Frechet Derivative)

A function $f : V \longrightarrow W$ is **differentiable** at $x \in V$ if there exists a unique tensor $Df_x \in V^\dagger \otimes W$—called the **total derivative** or **Frechet derivative**—such that

$$\lim_{h \to 0} \frac{\|f(x+h) - f(x) - Df_x h\|}{\|h\|} = 0 \tag{373}$$

It's a simple—almost trivial—change, but extremely powerful. Note that if f is differentiable over V, then the transformation $Df : V \to (V^\dagger \otimes W)$ that maps x_1 to its Frechet derivative Df_{x_1} can also be analyzed. Supposing that it is differentiable, the derivative of it at x is

$$D^2 f_x := D(Df)_x \in V^\dagger \otimes (V^\dagger \otimes W) \simeq (V^\dagger)^{\otimes 2} \otimes W \tag{374}$$

by the definition above. Intuitively this also make sense: if we want to take the second derivative of f, we must first specify the point in the domain of Df (our velocity) plus the point in the domain of f itself (our position) to compute it (the acceleration). Continuing this pattern, we get the following recursive definition.

Definition 2.4 (kth Frechet Derivative)

For $k > 1$, a function $f : V \longrightarrow W$ is **k-times differentiable** at $x \in V$ if

1. f is $k-1$ times differentiable at $x \in V$, and
2. there exists a unique rank $(k, 1)$ tensor $D^k f_x \in (V^\dagger)^{\otimes k} \otimes W$—called the **$k$th total derivative** or **kth Frechet derivative**—such that

$$\lim_{h \to 0} \frac{\|D^{k-1} f_x(x+h) - D^{k-1} f_x(x) - D^k f_x h\|}{\|h\|} = 0 \tag{375}$$

where $D^{k-1} f_x$ is the $(k-1)$th derivative of f at x.

Generally, multivariate calculus courses do not go into tensor algebra, so these higher order derivatives are omitted; you only work with the Hessian for scalar-valued functions. But we won't be scared off and will present this in full generality, and this gives us two major benefits:

1. We can compute higher order derivative as we have seen.
2. We can compute functions that also input or output tensors, such as matrix-valued functions. This knowledge becomes very useful for optimization, e.g. when you optimize a neural network with respect to matrices (in fully-connected layers) and 3-tensors (e.g. convolutions) and is a must-know for building autograd libraries.

Definition 2.5 (kth Partial Derivative)

Directional derivatives lose meaning in these higher order regimes, and we just work with iterated partials.

Definition 2.6 (C^k Functions)

A function $f : D \subset \mathbb{R}^n \longrightarrow \mathbb{R}$ is said to be a C^k function if all k-times iterated partial derivatives

$$\partial_{x_{i_1} x_{i_2} \ldots x_{i_k}} f \tag{376}$$

exist and are continuous. The vector space of all C^k functions is denoted $C^k(D; \mathbb{R})$, or $C^k(D)$.

Whenever we want to get the kth iterated partial derivative of f, we will assume that $f \in C^k$. Again, this is overkill, but it is conventional since we don't really work with the set of functions with existing partial derivatives. Note that mathematicians throw around the word "smooth" a lot. Usually, it means one of three things: it is of class C^1, C^∞, or C^k where k is however high it needs to be to satisfy our assumptions. For example, if I say let us differentiate smooth f two times, then I am assuming that $f \in C^2(\mathbb{R}^n)$.

Visualizing C^k-functions is easy for low orders. A C^0 function produces a graph that isn't "ripped" or "punctured," since this is exactly what a discontinuity would look like. A C^1 function requires the surface to be smooth in such a way that there is a well defined affine tangent subspace at every point. This means that there cannot be any sharp "points" or "edges" on the graph since a tangent subspace cannot be well defined.

Corollary 2.5 (Symmetry in kth Iterated Partials)

Given $f \in C^k$, its kth iterated partials are equal. That is, given any permutation σ,

$$\partial_{x_{i_1} x_{i_2} \ldots x_{i_k}} f = \partial_{x_{\sigma(i_1)} x_{\sigma(i_2)} \ldots x_{\sigma(i_k)}} f \text{ for } i_1, \ldots, i_k = 1, \ldots, n \tag{377}$$

Proof. By induction.

2.4 Taylor Series

To talk about convergence, the big-O notation is very useful.

Definition 2.7 (Classes of Infinitesimal Functions)

A function $\alpha : \mathbb{R}^n \longrightarrow \mathbb{R}$ is **infinitesimal** if $\alpha \rightarrow 0$ as $x \rightarrow a$. There are multiple "levels" of infinitesimal functions, i.e. how fast they converge to 0. We can classify them by comparing their limits to polynomials.

1. α is of class $O(1)$ if
$$\lim_{x \to a} \frac{\alpha(x)}{1} = 0$$
This means that $\alpha(x)$ tends to 0 infinitely faster than 1 (which just means that it tends to 0).
2. α is of class $O(h)$ if
$$\lim_{x \to a} \frac{\alpha(x)}{||x-a||} = 0$$
This means that $\alpha(x)$ tends to 0 infinitely faster than the linear $||\mathbf{h}||$, where $\mathbf{h} = x - a$.
3. α is of class $O(h^2)$ if
$$\lim_{x \to a} \frac{\alpha(x)}{||x-a||^2} = 0$$
This means that $\alpha(x)$ tends to 0 infinitely faster than the quadratic $||\mathbf{h}||^2$, where $\mathbf{h} = x - a$.
4. α is of class $O(h^k)$ if
$$\lim_{x \to a} \frac{\alpha(x)}{||x-a||^k} = 0$$
This means that $\alpha(x)$ tends to 0 infinitely faster than the kth-order $||\mathbf{h}||^k$, where $\mathbf{h} = x - a$.

Clearly, $O(h^k) \supset O(h^{k+1})$.

Now given a $f : D \subset \mathbb{R}^n \longrightarrow \mathbb{R}$, we present some polynomial approximations of f at $a \in D$:

1. If $f \in C^0$, the zeroth (constant) approximation is just
$$P_0(x) = f(a)$$
This is not interesting at all, since it is just constant. Furthermore, the error term $\epsilon_0(x) = f(x) - P_0(x)$ is an infinitesimal function as $x \rightarrow a$ and is of class $O(1)$, since
$$\lim_{x \to \mathbf{x_0}} \frac{f(x) - P_0(x)}{1} = 0$$
2. If $f \in C^1$, the first (linear) approximation requires us to use our total derivative:
$$P_1(x) = f(a) + Df_a(x-a)$$
and we know that the error $\epsilon_1(x) = f(x) - P_1(x)$ is infinitesimal as $x \rightarrow a$ and is of class $O(h)$, since
$$\lim_{x \to a} \frac{f(x) - P_1(x)}{||x-a||} = 0$$
3. If $f \in C^2$, the second (quadratic) approximation requires us to use a quadratic term (i.e. a bilinear form of $\mathbf{h} = x - a$) centered at a. Call it $H_a : \mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathbb{R}$, and our estimation is
$$P_2(x) = f(a) + Df_a(x-a) + \frac{1}{2} H(x-a, x-a)$$
which we would like the error term $\epsilon_2(x) = f(x) - P_2(x)$ to be $O(h^2)$, or in limit terms, P_2 must satisfy
$$\lim_{x \to a} \frac{f(x) - P_2(x)}{||x-a||^2} = 0$$
We show that this form H is precisely the Hessian matrix.

Theorem 2.6 (2nd Order Taylor Polynomial)

The second order approximation of a C^2-differentiable function f about a point a is

$$f(x) = f(a) + Df_a(x-a) + \frac{1}{2}(x-a)^T Hf_a(x-a) + O(h^2)$$

where Df_a is the total derivative at a and Hf_a is the Hessian matrix at a.

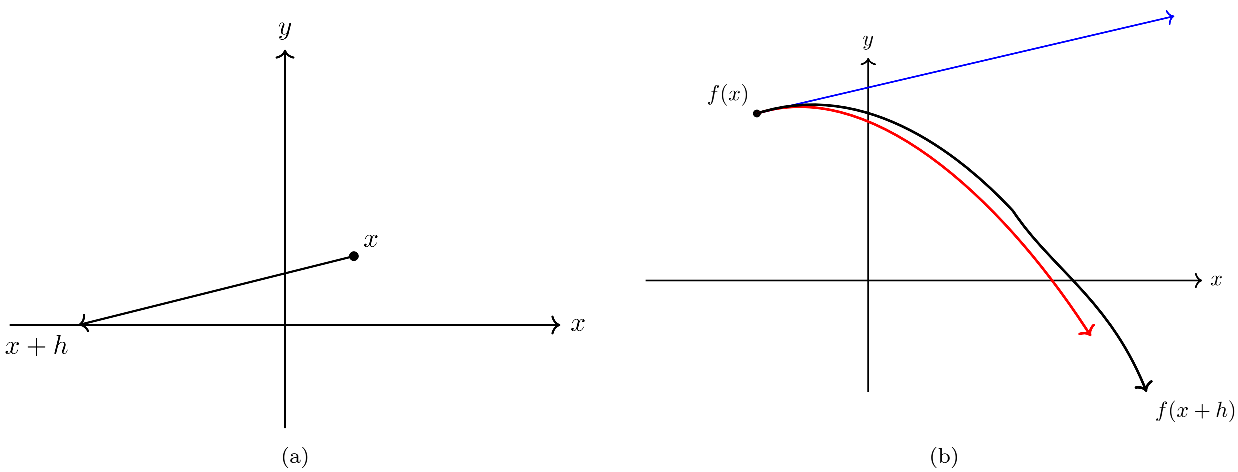


Figure 13: As we change $x \to x+h$, the actual function deviates from $f(x)$ to $f(x+h)$. The first order approximation (blue) and second order approximation (red) are shown.

2.5 Matrix Calculus

Now we will take a look at functions that have either an input or output as matrices. Essentially, matrices are also vectors, so there is nothing new here to learn, but having a concrete set of notation is useful. First, note that when we talk about a total derivative $D\mathbf{f}_a$, we can interpret this as a linear map that takes in some small perturbation $\mathbf{h}$ and gives us the result $D\mathbf{f}_a(\mathbf{h})$. In our column-vector setting, this just corresponded to left matrix multiplication:

$$D\mathbf{f}_a(\mathbf{h}) = D\mathbf{f}_a\mathbf{h} \tag{378}$$

This is not the case in the matrix setting. Let us compare the following:

1. The derivative of $f : \mathbb{R} \to \mathbb{R}$ at a is a linear function $Df_a : \mathbb{R} \longrightarrow \mathbb{R}$ satisfying $f(a+h) \approx f(a) + Df_a(h) + O(h^2)$. But linearity reduces Df_a to simply a scalar, and so our condition reduces to

$$f(a+h) \approx f(a) + Df_a h + O(h^2)$$

2. The derivative of a path function $\mathbf{f} : \mathbb{R} \to \mathbb{R}^m$ is a linear function $D\mathbf{f}_a : \mathbb{R} \longrightarrow \mathbb{R}^m$ satisfying $\mathbf{f}(a+h) = \mathbf{f}(a) + D\mathbf{f}_a(h) + O(h^2)$. Linearity implies that $D\mathbf{f}_a$ is a rank-1 linear map, which reduces to it being a column vector, and our condition reduces to

$$\underbrace{\mathbf{f}(a+h)}_{m\times 1} \approx \underbrace{\mathbf{f}(a)}_{m\times 1} + \underbrace{D\mathbf{f}_a}_{m\times 1} \underbrace{h}_{1\times 1}$$

3. The derivative of a matrix function $\mathbf{F} : \mathbb{R} \to \mathbb{R}^{m\times n}$ is a linear map $D\mathbf{F}_a : \mathbb{R} \longrightarrow \mathbb{R}^{m\times n}$ satisfying $\mathbf{F}(a+h) \approx \mathbf{F}(a) + D\mathbf{F}_a(h)$. This time, linearity does not reduce it to simple left-hand matrix multiplication.

We could just say that this is a left-hand scalar multiplication, but this doesn't generalize well, so we are stuck with just saying that $D\mathbf{F}_a$ is a linear map.

$$\underbrace{\mathbf{F}(a+h)}_{m\times n} \approx \underbrace{\mathbf{F}(a)}_{m\times n} + \underbrace{D\mathbf{F}_a(h)}_{m\times n}$$

Now let us take a look at when we have matrix inputs.

1. The derivative of $f : \mathbb{R}^{m\times n} \longrightarrow \mathbb{R}$ is a linear function $Df_{\mathbf{A}} : \mathbb{R}^{m\times n} \longrightarrow \mathbb{R}$ satisfying $f(\mathbf{A}+\mathbf{H}) \approx f(\mathbf{A}) + Df_{\mathbf{A}}(\mathbf{H})$. We could let $Df_{\mathbf{A}}$ be some linear map like $\mathbf{M} \mapsto \mathbf{v}^T\mathbf{M}\mathbf{u}$, where $\mathbf{v}, \mathbf{u}$ is fixed. But in generality, we just have the condition

$$\underbrace{f(\mathbf{A}+\mathbf{H})}_{1\times 1} \approx \underbrace{f(\mathbf{A})}_{1\times 1} + \underbrace{Df_{\mathbf{A}}(\mathbf{H})}_{1\times 1}$$

2. The derivative of $\mathbf{f} : \mathbf{R}^{m\times n} \longrightarrow \mathbb{R}^d$ is some linear map $D\mathbf{f}_{\mathbf{A}} : \mathbf{R}^{m\times n} \longrightarrow \mathbb{R}^d$ satisfying $\mathbf{f}(\mathbf{A}+\mathbf{H}) \approx \mathbf{f}(\mathbf{A}) + D\mathbf{f}_{\mathbf{A}}(\mathbf{H})$. Again, we could construct some form that would give us a linear map in terms of some matrix multiplication, but in generality, we have the condition

$$\underbrace{\mathbf{f}(\mathbf{A}+\mathbf{H})}_{d\times 1} \approx \underbrace{\mathbf{f}(\mathbf{A})}_{d\times 1} + \underbrace{D\mathbf{f}_{\mathbf{A}}(\mathbf{H})}_{d\times 1}$$

Now we present some theorems on basic differentiation. Proving these just requires us to expand the function and compute the derivatives component-wise.

Theorem 2.7 (Derivative of Affine Map)

Given $f : \mathbb{R}^n \longrightarrow \mathbb{R}^m$ defined $f(x) = \mathbf{A}x + \mathbf{b}$ (where $\mathbf{A}, \mathbf{b}$ is not dependent on x), its derivative is

$$D\mathbf{f} = \mathbf{A}$$

Theorem 2.8

Given the scalar α defined by

$$\alpha = \mathbf{y}^T\mathbf{A}x$$

where $\mathbf{y} \in \mathbb{R}^{m\times 1}, \mathbf{A} \in \mathbb{R}^{m\times n}$, and $x \in \mathbb{R}^{n\times 1}$, then

$$\frac{\partial \alpha}{\partial x} = \mathbf{y}^T\mathbf{A} : \mathbb{R}^n \longrightarrow \mathbb{R}$$

and

$$\frac{\partial \alpha}{\partial \mathbf{y}} = x^T\mathbf{A}^T : \mathbb{R}^m \longrightarrow \mathbb{R}$$

Rewritten in the total derivative notation, we can interpret α as a function of both x and $\mathbf{y}$ and write

$$D\alpha_{(x,\mathbf{y})} = \begin{pmatrix} \mathbf{y}^T\mathbf{A} \\ x^T\mathbf{A}^T \end{pmatrix} : \mathbb{R}^{n+m} \longrightarrow \mathbb{R}$$

Theorem 2.9

Given the scalar α defined by the quadratic form

$$\alpha = x^T\mathbf{A}x \tag{379}$$

where $x \in \mathbb{R}^{n \times 1}$ and $\mathbf{A} \in \mathbb{R}^{n \times n}$, then

$$\frac{\partial \alpha}{\partial x} = x^T \big(\mathbf{A} + \mathbf{A}^T\big) \tag{380}$$

or in the total derivative notation,

$$D\alpha_{\mathbf{a}} = \mathbf{a}^T \big(\mathbf{A} + \mathbf{A}^T\big) : \mathbb{R}^n \longrightarrow \mathbb{R} \tag{381}$$

3 Riemann Integration

We have done integration over closed intervals $[a, b]$. The natural extension is to define integration over *boxes* $B = \prod_i [a_i, b_i]$. Essentially the construction is exactly the same.

Definition 3.1 (Partition/Mesh)

Let $B = \prod_i [a_i, b_i] \subset \mathbb{R}^n$ be a *box*. Then, a **partition**—or **mesh**—of B is a finite set of points $P = \{x_{ij}\}_{1 \leq i \leq n, 0 \leq j \leq m_i}$ [a] s.t.

$$a_i = x_{i,0} \leq x_{i,1} \leq x_{i,2} \leq \ldots \leq x_{i,m_i - 1} \leq x_{i,m_i} = b \quad \forall i = 1, \ldots, n \tag{382}$$

We denote each **box** within the partition as

$$\Delta_j = \Delta_{j_1, j_2, \ldots, j_n} = \prod_i [x_{i,j_k - 1}, x_{i,j_k}] \quad \forall j_i = 1, \ldots, m_i \tag{383}$$

of volume $|\Delta_j| \coloneqq \prod_i (x_{i,j_k} - x_{i,j_{k-1}})$.

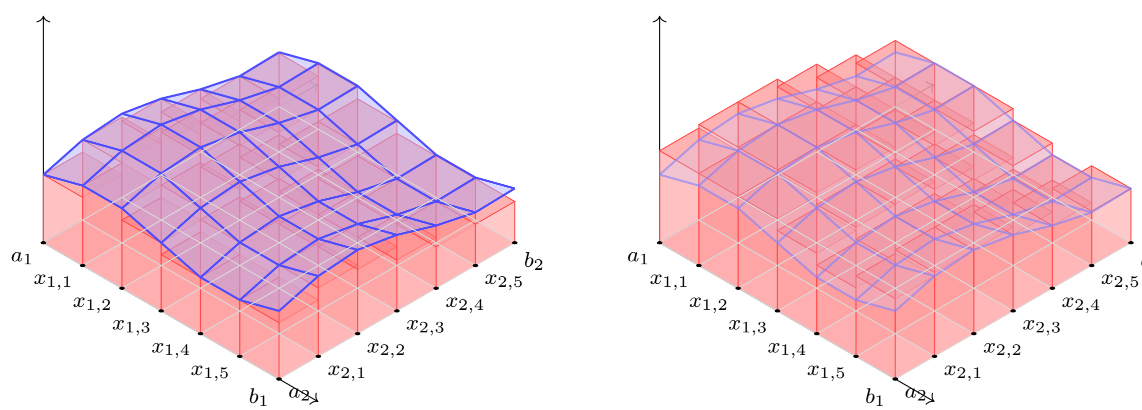
[a] Therefore, for each dimension i, we want to take the interval $[a_i, b_i]$ and subdivide it into $m_i + 1$ subintervals.

This nearly[2] partitions B into a grid of $\prod_i m_i$ smaller boxes, and can be seen as a discretization of the integral which we will construcct.

Definition 3.2 (Darboux Sums with Respect to Partition)

Let $f : E \subset \mathbb{R}^n \to \mathbb{R}$ be a bounded function, let $B \subset E$ be a box, and P be a partition of B. Then, the **upper and lower Darboux integrals** of f with respect to P are defined

$$U(P, f) \coloneqq \sum_J \sup_{x \in \Delta_J} f(x) |\Delta_J|, \qquad L(P, f) \coloneqq \sum_J \inf_{x \in \Delta_J} f(x) |\Delta_J| \tag{384}$$



(a) Lower Riemann sum.

(b) Upper Riemann sum.

Figure 14

[2] since boundaries overlap

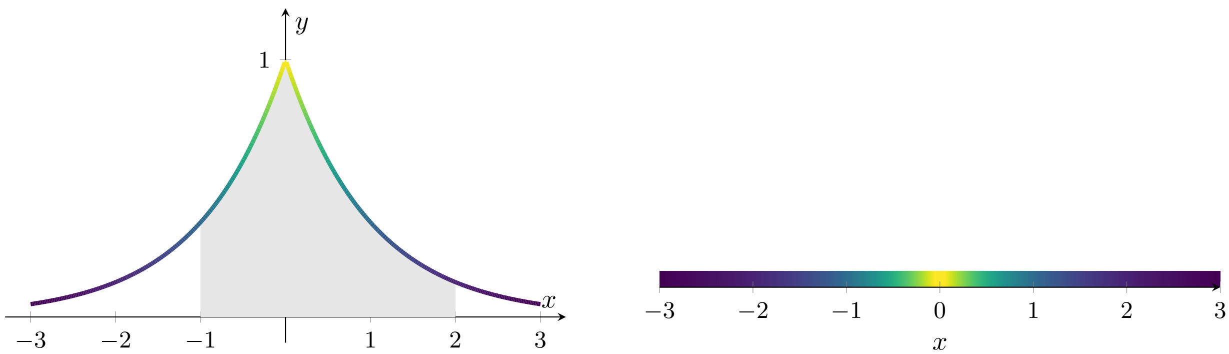
Definition 3.3 (Darboux Integral)

Let $f : E \subset \mathbb{R}^n \to \mathbb{R}$ be a bounded function, and let $B \subset E$ be a box. Then, the **upper and lower Darboux integrals** of f are defined

$$\overline{\int_B} f(x)\,dx := \inf_P U(P,f), \qquad \underline{\int_B} f(x)\,dx := \sup_P L(P,f) \tag{385}$$

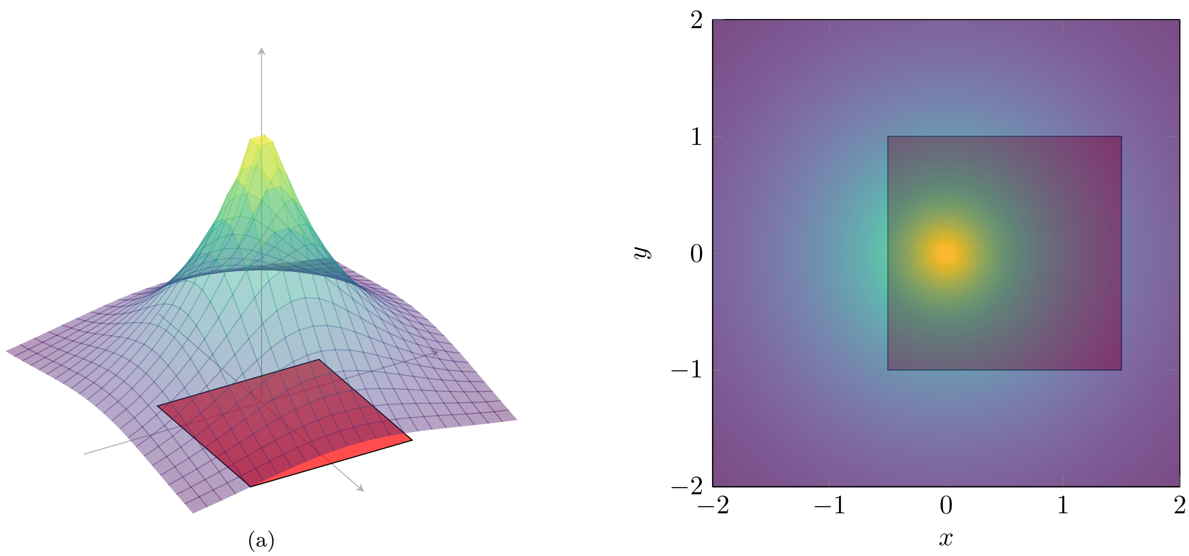
If the upper and lower Riemann integrals are equal then f is said to be **Darboux integrable** over B and the **Darboux integral** of f is

$$\int_B f(x)\,dx \tag{386}$$



(a) (b)

Figure 15: Two interpretations of the integral $\int_{-1}^{2} e^{-|x|}\,dx$. On the left, we view it as the area under a curve. On the right, we view it as the mass of a 1-dimensinoal rod lying on $[-1,2]$ of varying density defined by function $f(x)$.



(a)

Figure 16: Two interpretations of double integral $\iint_B e^{-\sqrt{x^2+y^2}}\,dx$. On the left, we can view it as the area under a surface. On the right, we view it as the mass of a 2-dimensional sheet lying on $[-0.5,1.5] \times [-1,1]$ of varying density defined by function $f(x,y)$.

3.1 Conditions for Integrability

Theorem 3.1 (Continuous Implies Integrable)

$f : E \subset \mathbb{R}^n \to \mathbb{R}$ continuous means f is integrable over B.

Proof.

However, there are some functions with discontinuities that are in fact integrable.

(a) A removable hole on an otherwise smooth surface.

(b) A jump discontinuity across a curve.

Figure 17: Two typical discontinuities that can still occur in integrable functions: a removable hole and a jump discontinuity.

1. Given that there is a subset N in B with volume 0 over which f is not defined, we can integrate over $B \setminus N$. In the one and two dimensional cases,
$$\int_{B \setminus N} f(x) dx \text{ and } \iint_{B \setminus N} f(x) dA$$
are well-defined. Visually, this corresponds to a surface with a puncture over a set of measure zero.
2. The function is defined for all values in the region, but there is a jump in the value of the function. Here the domain remains intact, but the graph splits into two sheets separated by a finite jump.

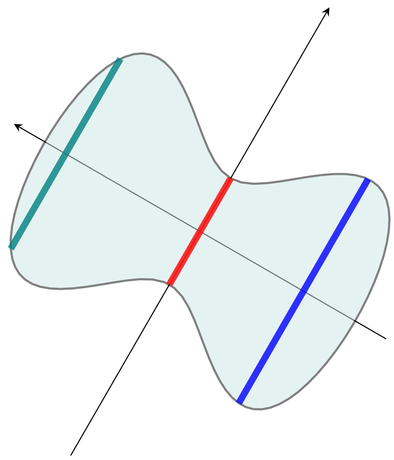
Informally, if we can visualize the Riemann sum converging to a well-defined area as the rectangles get thinner and thinner, then a discontinuous function is integrable. Indeed, all continuous functions (over a bounded set) are integrable since their Riemann sums are well defined.

3.2 Iterated Integrals

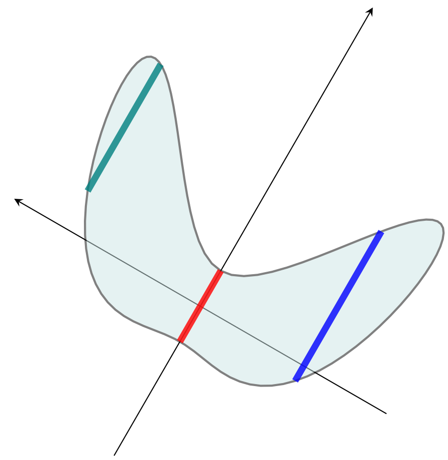
The construction of the integral is one step, but we should know how to practically compute such an integral. We can do this by recursively "reducing" the dimension of the region of the integral until we can work with one dimension at which point we can use the univariate Riemann integral. This is informally known as *Cavalieri's principle*, which states the following.

Theorem 3.2 (Cavalieri's Principle)

1. Suppose two shapes in $\mathbb{R}^2$ are included between two parallel lines in that plane. If every line parallel to these two lines intersects both regions in line segments of equal length, then the two regions have equal areas.



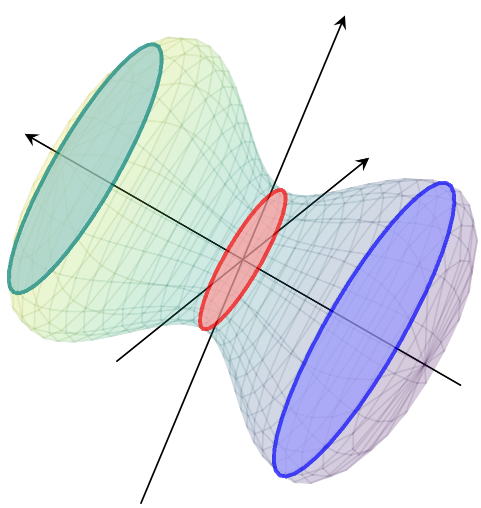
(a) Original region in $\mathbb{R}^2$



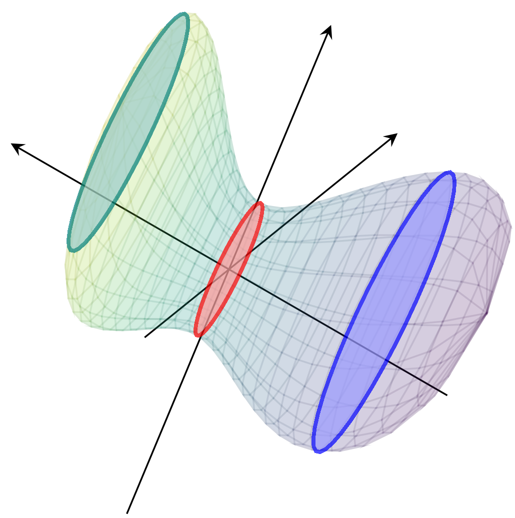
(b) Sheared region preserving 1D cross-sections

Figure 18: By Cavalieri's Principle in 2D, the area of the region is preserved because every 1D horizontal cross-section remains identical in length, despite the parabolic horizontal shear.

2. Suppose two bodies in $\mathbb{R}^3$ are included between two parallel planes. If every plane parallel to these two planes intersects both regions in cross-sections of equal area, then the two regions have equal volumes.



(a)



(b)

Figure 19: Even though the two are different solids in $\mathbb{R}^3$, the cross sections are the same. Therefore, Cavalieri's principle states that the volumes of the two solids must be equal.

In the case when S is a box in $\mathbb{R}^n$, Fubini's theorem states that whether we cut S up along the x_1-axis, x_2-axis, ..., or the x_n-axis, the symmetry in volume is always preserved. This theorem is really just a specific case of this general symmetry in volume.

Theorem 3.3 (Euler-Fubini Theorem)

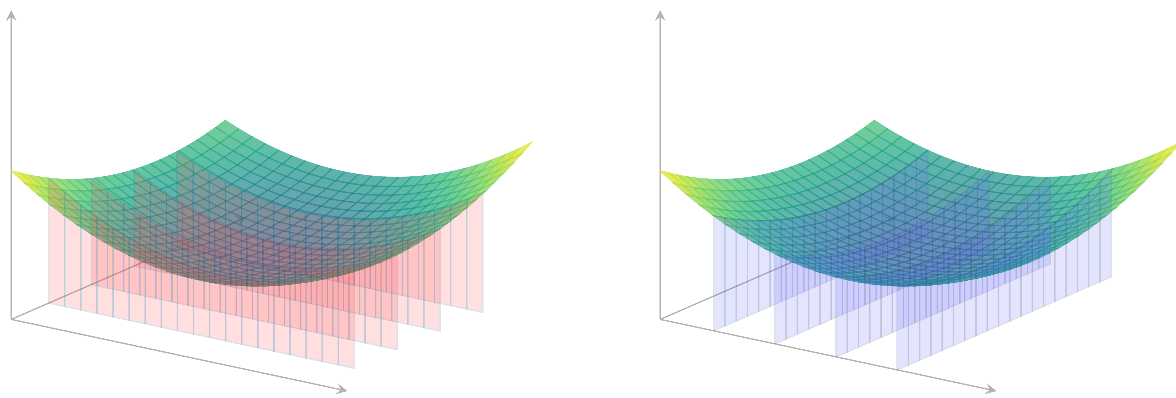
Given a function $f : \mathbb{R}^n \longrightarrow \mathbb{R}$, let $B \coloneqq \prod_{i=1}^n [\alpha_i, \beta_i]$ and let p be any permutation of the elements $\{x_1, x_2, \ldots, x_n\}$. Then

$$\int_B f\, dV = \int_{\alpha_n}^{\beta_n} \ldots \int_{\alpha_1}^{\beta_1} f\, dx_1 \ldots dx_n = \int_{p(\alpha_n)}^{p(\beta_n)} \ldots \int_{p(\alpha_1)}^{p(\beta_1)} f\, dp(x_1) \ldots dp(x_n) \tag{387}$$

Proof.

In the two dimensional case, we have

$$\iint_B f\, dA = \int_c^d \int_a^b f(x,y)\, dx\, dy = \int_a^b \int_c^d f(x,y)\, dy\, dx \tag{388}$$



(a) Slicing at fixed values of y.

(b) Slicing at fixed values of x.

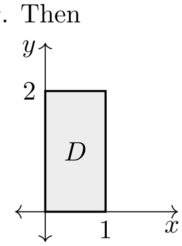
and in the three dimensional case, we can use any of the 6 permutations. Computation of these integrals is simple. You do the innermost integral first with respect to the corresponding variable, while treating the rest of the variables constant. Evaluating each integral outputs a formula for a higher dimensional cross section of the solid S. It is clear that computing iterated integrals is really just doing Cavalieri's principle repeatedly.

Example 3.1 (Double Integral over a Rectangle: Easy)

Let

$$D = [0,1] \times [0,2]$$

and consider the function $f(x,y) = x + 2y$. Then



Since D is a rectangle, we may write

$$\begin{aligned}\iint_D (x+2y)\,dA &= \int_0^2 \int_0^1 (x+2y)\,dx\,dy \\ &= \int_0^2 \left[\frac{x^2}{2} + 2yx\right]_{x=0}^{x=1} dy \\ &= \int_0^2 \left(\frac{1}{2} + 2y\right) dy \\ &= \left[\frac{y}{2} + y^2\right]_{y=0}^{y=2} \\ &= 1 + 4 = 5.\end{aligned}$$

Therefore,

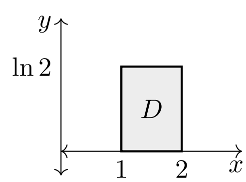
$$\iint_D (x+2y)\,dA = 5.$$

Example 3.2 (Double Integral over a Rectangle: Hard)

Let

$$D = [1,2] \times [0, \ln 2]$$

and let $f(x,y) = xe^y$. Then



Again, the region is rectangular, so

$$\begin{aligned}\iint_D xe^y\,dA &= \int_0^{\ln 2} \int_1^2 xe^y\,dx\,dy \\ &= \int_0^{\ln 2} e^y \left[\frac{x^2}{2}\right]_{x=1}^{x=2} dy \\ &= \int_0^{\ln 2} \frac{3}{2} e^y\,dy \\ &= \frac{3}{2}\left[e^y\right]_{y=0}^{y=\ln 2} \\ &= \frac{3}{2}(2-1) = \frac{3}{2}.\end{aligned}$$

Hence,

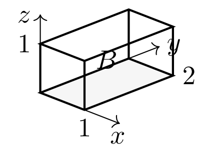
$$\iint_D xe^y\,dA = \frac{3}{2}.$$

Example 3.3 (Triple Integral over a Box: Easy)

Let

$$B = [0,1] \times [0,2] \times [0,1]$$

and let $f(x,y,z) = x + y + z$. Then



Using the order $dz\,dy\,dx$, we get

$$\begin{aligned}
\iiint_B (x+y+z)\,dV &= \int_0^1 \int_0^2 \int_0^1 (x+y+z)\,dz\,dy\,dx \\
&= \int_0^1 \int_0^2 \left[(x+y)z + \frac{z^2}{2}\right]_{z=0}^{z=1} dy\,dx \\
&= \int_0^1 \int_0^2 \left(x+y+\frac{1}{2}\right) dy\,dx \\
&= \int_0^1 \left[xy + \frac{y^2}{2} + \frac{y}{2}\right]_{y=0}^{y=2} dx \\
&= \int_0^1 (2x+3)\,dx \\
&= \left[x^2+3x\right]_{x=0}^{x=1} \\
&= 1+3 = 4.
\end{aligned}$$

Therefore,

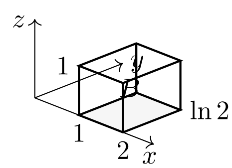
$$\iiint_B (x+y+z)\,dV = 4.$$

Example 3.4 (Triple Integral over a Box: Hard)

Let

$$B = [1,2] \times [0, \ln 2] \times [0,1]$$

and let $f(x,y,z) = xe^y(1+z^2)$. Then



With the order $dz\,dy\,dx$, we obtain

$$\begin{aligned}
\iiint_B xe^y(1+z^2)\,dV &= \int_1^2 \int_0^{\ln 2} \int_0^1 xe^y(1+z^2)\,dz\,dy\,dx \\
&= \int_1^2 \int_0^{\ln 2} xe^y \left[z + \frac{z^3}{3}\right]_{z=0}^{z=1} dy\,dx \\
&= \int_1^2 \int_0^{\ln 2} \frac{4}{3}xe^y\,dy\,dx \\
&= \int_1^2 \frac{4}{3}x\,[e^y]_{y=0}^{y=\ln 2}\,dx \\
&= \int_1^2 \frac{4}{3}x\,dx \\
&= \frac{4}{3}\left[\frac{x^2}{2}\right]_{x=1}^{x=2} \\
&= \frac{4}{3}\cdot\frac{3}{2} = 2.
\end{aligned}$$

Hence,

$$\iiint_B xe^y(1+z^2)\,dV = 2.$$

3.3 Integration over Regions between Curves

Definition 3.4 (Simple Regions w.r.t. a Variable)

A bounded region D in $\mathbb{R}^n$ is said to be x_i-simple if it is bounded by the graphs of two continuous functions $u_1, u_2 : \mathbb{R}^{n-1} \longrightarrow \mathbb{R}$ of the variables

$$x_1, x_2, ..., x_{i-1}, x_{i+1}, ..., x_n$$

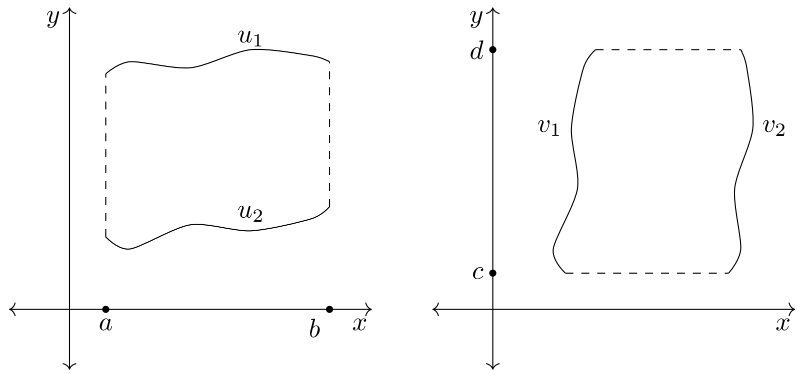
That is, D can be expressed in the form

$$\{x \in \mathbb{R}^n \mid u_1(x_1, ..., x_{i-1}, x_{i+1}, ..., x_n) \leq x_i \leq u_2(x_1, ..., x_{i-1}, x_{i+1}, ..., x_n)\}$$

If a region is simple in all of its variables, it is simply called *simple*. Note that n-dimensional boxes are simple regions.

Example 3.5

In $\mathbb{R}^2$, the region on the left graph is an y-simple region and the region on the right is a x-simple region.



We now describe the method of calculating double integrals over elementary regions.

Theorem 3.4 (Iterated Integrals over Simple Regions)

The double integral over a y-simple region D bounded by functions u_1 and u_2 in $\mathbb{R}^2$ and the x-values a and b (as shown in the left graph of example 2.1) is

$$\iint_D f(x,y) = \int_a^b \int_{u_2(x)}^{u_1(x)} f(x,y)\,dy\,dx \tag{389}$$

The double integral over an x-simple region D bounded by functions v_1 and v_2 in $\mathbb{R}^2$ and the y-values c and d (shown in the right of graph of example 2.1) is

$$\iint_D f(x,y) = \int_c^d \int_{v_2(y)}^{v_1(y)} f(x,y)\,dx\,dy \tag{390}$$

Example 3.6 (Unit Disk)

Integrating $f(x,y)$ over the unit disk would have the form

$$\int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} f(x,y)\,dy\,dx \text{ or } \int_{-1}^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} f(x,y)\,dx\,dy \tag{391}$$

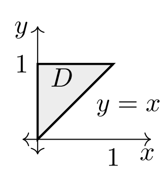
Note that the unit disk is both x and y simple.

Example 3.7 (Double Integral over a Non-Rectangular Region: Easy)

Consider the triangular region

$$D = \{(x,y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1,\ x \leq y \leq 1\}$$

and let $f(x,y) = x + y$. Then



Since D is y-simple,

$$\begin{aligned}\iint_D (x+y)\,dA &= \int_0^1 \int_x^1 (x+y)\,dy\,dx \\ &= \int_0^1 \left[xy + \frac{y^2}{2}\right]_{y=x}^{y=1} dx \\ &= \int_0^1 \left(x + \frac{1}{2} - x^2 - \frac{x^2}{2}\right) dx \\ &= \int_0^1 \left(x + \frac{1}{2} - \frac{3x^2}{2}\right) dx \\ &= \left[\frac{x^2}{2} + \frac{x}{2} - \frac{x^3}{2}\right]_{x=0}^{x=1} \\ &= \frac{1}{2}.\end{aligned}$$

Therefore,

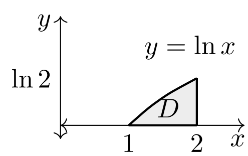
$$\iint_D (x+y)\,dA = \frac{1}{2}.$$

Example 3.8 (Double Integral over a Non-Rectangular Region: Hard)

Let

$$D = \{(x,y) \in \mathbb{R}^2 \mid 1 \le x \le 2,\ 0 \le y \le \ln x\}$$

and let $f(x,y) = ye^y$. Then



This region is not a rectangle because its upper boundary depends on x. We compute

$$\begin{aligned}\iint_D ye^y\,dA &= \int_1^2 \int_0^{\ln x} ye^y\,dy\,dx \\ &= \int_1^2 [(y-1)e^y]_{y=0}^{y=\ln x}\,dx \\ &= \int_1^2 (x\ln x - x + 1)\,dx \\ &= \left[\frac{x^2}{2}\ln x - \frac{x^2}{4} - \frac{x^2}{2} + x\right]_{x=1}^{x=2} \\ &= (2\ln 2 - 1) - \frac{1}{4} \\ &= 2\ln 2 - \frac{5}{4}.\end{aligned}$$

Thus,

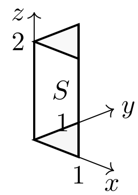
$$\iint_D ye^y\,dA = 2\ln 2 - \frac{5}{4}.$$

Example 3.9 (Triple Integral over a Non-Rectangular Region: Easy)

Consider the solid

$$S = \{(x, y, z) \in \mathbb{R}^3 \mid 0 \leq x \leq 1,\ 0 \leq y \leq 1 - x,\ 0 \leq z \leq 2\}$$

and let $f(x, y, z) = x + y + z$. Then



The base in the xy-plane is triangular, so the bounds are not independent. We compute

$$\begin{aligned}
\iiint_S (x + y + z)\, dV &= \int_0^1 \int_0^{1-x} \int_0^2 (x + y + z)\, dz\, dy\, dx \\
&= \int_0^1 \int_0^{1-x} \left[(x + y)z + \frac{z^2}{2} \right]_{z=0}^{z=2} dy\, dx \\
&= \int_0^1 \int_0^{1-x} (2x + 2y + 2)\, dy\, dx \\
&= \int_0^1 \left[2xy + y^2 + 2y \right]_{y=0}^{y=1-x} dx \\
&= \int_0^1 \left(3 - 2x - x^2 \right) dx \\
&= \left[3x - x^2 - \frac{x^3}{3} \right]_{x=0}^{x=1} \\
&= 3 - 1 - \frac{1}{3} = \frac{5}{3}.
\end{aligned}$$

Hence,

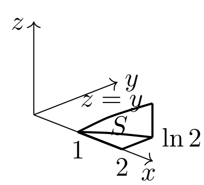
$$\iiint_S (x + y + z)\, dV = \frac{5}{3}.$$

Example 3.10 (Triple Integral over a Non-Rectangular Region: Hard)

Let

$$S = \{(x, y, z) \in \mathbb{R}^3 \mid 1 \leq x \leq 2,\ 0 \leq y \leq \ln x,\ 0 \leq z \leq y\}$$

and let $f(x, y, z) = e^y(1 + z)$. Then



Here both the y- and z-bounds depend on earlier variables. Using the order $dz\,dy\,dx$,

$$\begin{aligned}
\iiint_S e^y(1+z)\,dV &= \int_1^2 \int_0^{\ln x} \int_0^y e^y(1+z)\,dz\,dy\,dx \\
&= \int_1^2 \int_0^{\ln x} e^y \left[z + \frac{z^2}{2}\right]_{z=0}^{z=y} dy\,dx \\
&= \int_1^2 \int_0^{\ln x} e^y \left(y + \frac{y^2}{2}\right) dy\,dx \\
&= \int_1^2 \left[\frac{y^2}{2} e^y\right]_{y=0}^{y=\ln x} dx \\
&= \int_1^2 \frac{x(\ln x)^2}{2}\,dx \\
&= \left[\frac{x^2}{4}(\ln x)^2 - \frac{x^2}{4}\ln x + \frac{x^2}{8}\right]_{x=1}^{x=2} \\
&= \left((\ln 2)^2 - \ln 2 + \frac{1}{2}\right) - \frac{1}{8} \\
&= (\ln 2)^2 - \ln 2 + \frac{3}{8}.
\end{aligned}$$

Therefore,

$$\iiint_S e^y(1+z)\,dV = (\ln 2)^2 - \ln 2 + \frac{3}{8}.$$

3.4 Change of Basis

Sometimes, integrating a region over a different basis would make the integral computation much more simpler. In this case, we may be able to transform more complicated regions into elementary regions. We first introduce a change of basis in 2 dimensions and then generalize it into higher dimensions.

Let $\mathbb{R}^2$ have the standard orthonomal basis e_1, e_2, commonly known as the x, y basis. Now, let us construct new basis vectors of $\mathbb{R}^2$, denoted f_1, f_2 such that f_1, f_2 are functions of e_1, e_2. Since they are both bases that span $\mathbb{R}^2$, we can equally represent e_1, e_2 as functions of f_1, f_2.

$$\begin{aligned}
e_1 &= g(f_1, f_2) \\
e_2 &= h(f_1, f_2)
\end{aligned}$$

Note that this change of basis does not necessarily have to be linear, as in the context of passive transformation in linear algebra. Then, every point (x, y) in the (e_1, e_2)-basis can be rewritten as

$$\begin{aligned}
(x, y) &= xe_1 + ye_2 \\
&= x\,g(f_1, f_2) + y\,h(f_1, f_2) \\
&= uf_1 + vf_2
\end{aligned}$$

Note that it is customary to denote x, y as the coefficients in the e_1, e_2 basis and u, v as the coefficients in the new f_1, f_2 basis. This way, we can not only write e_1 and e_2 as functions of f_1 and f_2, but we can also write the coefficents x, y as functions of the coeffiecents u, v! That is,

$$\begin{aligned}
x &= x(u, v) \\
y &= y(u, v)
\end{aligned}$$

which is really just a function

$$B : \mathbb{R}^2 \longrightarrow \mathbb{R}^2, \;\; B(u, v) = \begin{pmatrix} x(u, v) \\ y(u, v) \end{pmatrix}$$

Notice that B changes the u, v coordinates to the x, y coordinates, and B^{-1} changes the x, y coordinates to the u, v coordinates.

$$B^{-1} : \mathbb{R}^2 \longrightarrow \mathbb{R}^2, \;\; B^{-1}(x, y) = \begin{pmatrix} u(x, y) \\ v(x, y) \end{pmatrix}$$

Note that these coefficients actually change *contravariantly*, that is, they change inversely with respect to how the basis vectors are changed. In vector calculus, it is conventional to represent a change of basis with functions that relate the coefficients x, y with u, v, rather than the bases f_1, f_2 with e_1, e_2.

Theorem 3.5 (Integration over Change of Bases in $\mathbb{R}^2$)

Let $\mathbb{R}^2$ have the standard orthonomal basis e_1, e_2. Now, let us construct new basis vectors of $\mathbb{R}^2$, denoted f_1, f_2 such that the coefficients of the vectors in $\mathbb{R}^2$ are related by the change of basis function

$$B = \begin{pmatrix} x \\ y \end{pmatrix} \implies B(u, v) = \begin{pmatrix} x(u, v) \\ y(u, v) \end{pmatrix}$$

Given region $D \subset \mathbb{R}^2$ and $S = B(D)$ is the region transformed by B, the integral of function $f(x, y)$ over region D can be expressed as

$$\iint_D f(x, y)\, dA = \iint_S f\big(x(u, v), y(u, v)\big) \, \big|JB(u, v)\big| \, d\bar{A}$$

where $\big|JB(u, v)\big|$ is the determinant of the Jacobian matrix of B. Expanding the Facobian determinant gives

$$\big|JB(u, v)\big| = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u}$$

Theorem 3.6 (Integration over Change of Bases in $\mathbb{R}^3$)

Given that we have the change of basis function

$$B : \mathbb{R}^3 \longrightarrow \mathbb{R}^3, \quad B(u, v, w) = \begin{pmatrix} x(u, v, w) \\ y(u, v, w) \\ z(u, v, w) \end{pmatrix}$$

a region $D \in \mathbb{R}^3$ and $S = B(D)$, the region transformed by B, the integral of $f(x, y, z)$ over region D can be expressed as

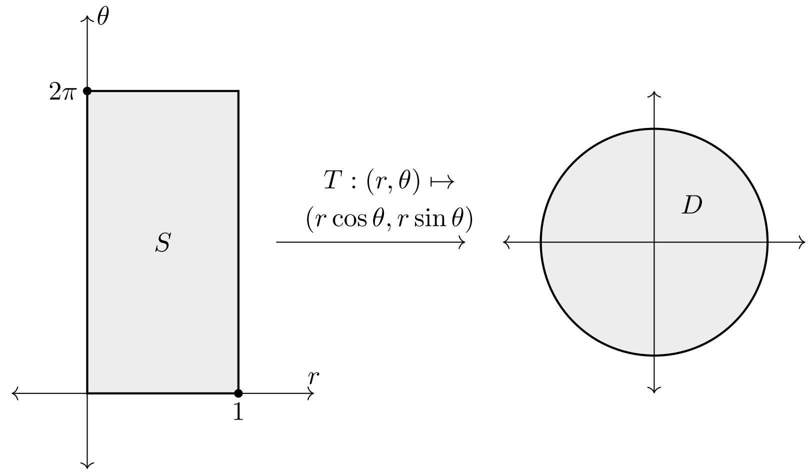
$$\iiint_D f(x, y, z)\, dV = \iiint_S f\big(x(u, v, w), y(u, v, w), z(u, v, w)\big) \big|JB(u, v, w)\big| \, d\bar{V}$$

where $\big|JB(u, v, w)\big|$ is the Jacobian determinant of B.

Example 3.11

Given a real-valued function f defined over the region $D \subset \mathbb{R}^2$, we can perform a change of basis of the x, y coordinates into polar ones within a new region S. The change of basis

$$\begin{aligned} x &= r \cos\theta \\ y &= r \sin\theta \end{aligned}$$



Theorem 3.7 (Integration over Change of Bases in $\mathbb{R}^n$)

Let $\mathbb{R}^n$ have the standard orthonormal basis $e_1, e_2, ..., e_n$, and let us construct a new basis $f_1, f_2, ..., f_n$ such that the coefficients of the vectors in $\mathbb{R}^n$ are related with the functions

$$B : \mathbb{R}^n \longrightarrow \mathbb{R}^n, \quad B(u_1, u_2, \ldots, u_n) = \begin{pmatrix} x_1(u_1, \ldots, u_n) \\ x_2(u_1, \ldots, u_n) \\ \vdots \\ x_n(u_1, \ldots, u_n) \end{pmatrix}$$

Given that the region $D \subset \mathbb{R}^n$ is transformed into a new region $S = B(D) \subset \mathbb{R}^n$ under this basis transformation, the integral of function $f(x_1, \ldots, x_n)$ over region D can be expressed as

$$\int_D f(x)\, dH = \int_S f\big(x_1(u), x_2(u), ..., x_n(u)\big) \big|JB(u_1, \ldots, u_n)\big|\, d\bar{H}$$

where the integral on both the left and right hand side represents integration over an n-dimensional region, x represents the n-tuple $(x_1, \ldots, x_n)$, u represents the n-tuple $(u_1, \ldots, u_n)$, and $\big|JB(u_1, \ldots, u_n)\big|$ represents the Jacobian determinant of function B.

We now describe some common change of basis formulas for polar, cylindrical, and spherical coordinates.

Theorem 3.8 (Integration in Polar Coordinates)

$$\iint_D f(x, y)\, dx\, dy = \iint_S f(r\cos\theta, r\sin\theta) r\, dr\, d\theta \tag{392}$$

Definition 3.5 (Cylindrical, Spherical Coordinates)

In $\mathbb{R}^3$, *cylindrical coordinates* have the following relation to rectangular coordinates.

$$x = r\cos\theta \tag{393}$$
$$y = r\sin\theta \tag{394}$$
$$z = z \tag{395}$$

In $\mathbb{R}^3$, *spherical coordinates* have the following relation to rectangular coordinates.

$$\begin{aligned} x &= \rho \sin\phi \cos\theta \\ y &= \rho \sin\phi \sin\theta \\ z &= \rho \cos\phi \end{aligned}$$

Corollary 3.9 (Integration in Cylindrical Coordinates)

$$\iiint_D f(x,y,z)\, dx\, dy\, dz = \iiint_S f(r\cos\theta, r\sin\theta, z) r\, dr\, d\theta\, dz$$

Corollary 3.10 (Integration in Spherical Coordinates)

$$\iiint_D f(x,y,z)\, dx\, dy\, dz = \iiint_S f(\rho\sin\phi\cos\theta, \rho\sin\phi\sin\theta, \rho\cos\phi)\rho^2 \sin\theta\, d\rho\, d\theta\, d\phi$$

Example 3.12 (Gaussian Integral)

The following is the (un-normalized) probability distribution function of the Gaussian distribution.

$$\int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi}$$

3.5 Improper Integrals

There are generally two types of improper integrals.

1. The region D integrated over is unbounded.
2. The function f that is integrated is unbounded within the region D.

These types of improper integrals are usually evaluated using a limiting process. When the interval I is unbounded, say $(1,\infty)$, the integral can be evaluated as

$$\int_1^\infty \frac{1}{x^2}\, dx = \lim_{b\to\infty} \int_1^b \frac{1}{x^2}\, dx = \lim_{b\to\infty} \bigg(1 - \frac{1}{b}\bigg) = 1$$

In case 2, we can add a limit at the point where the function f diverges as such.

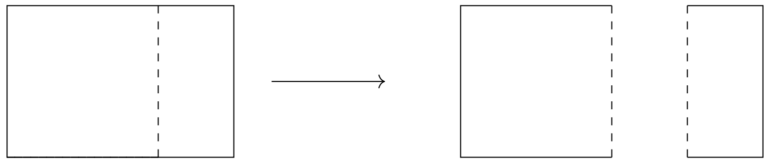
$$\int_0^1 \frac{1}{\sqrt{x}}\, dx = \lim_{a\to 0} \int_a^1 \frac{1}{\sqrt{x}}\, dx = \lim_{a\to 0}(2 - 2\sqrt{a}) = 2$$

We now describe how to integrate over a certain path p embedded in a higher dimensional space $\mathbb{R}^n$, possibly with a scalar or vector field f. We must first go over oriented paths.

Extending the previous case, we use a multivariate limiting process in $\mathbb{R}^2$. We will first work with case 2, when f is unbounded within the region D. Let us define an elementary region D in $\mathbb{R}^2$; without loss of generality, we will make it y-simple, meaning that D can be expressed as

$$D \equiv \{(x,y) \in \mathbb{R}^2 \mid a \leq x \leq b,\ \phi_1(x) \leq y \leq \phi_2(x)\}$$

We can actually assume that the region in which f is unbounded lies in the boundary ∂D. This is because if it lied in the interior of D, we could split D into pieces across a path that intersects this region with divergent values, evaluate the integrals over the pieces separately, and then sum the integrals. For example, in the rectangular region below, let the dashed line represent the values where the function f diverges. Then, we can split the region into two rectangular regions shown in the right.



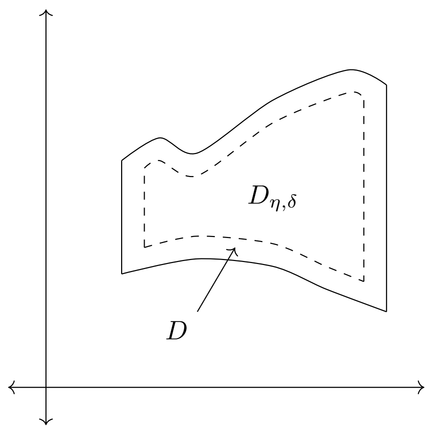
Therefore, assuming that f is unbounded in ∂D, we can construct a new region

$$D_{\eta,\delta} \equiv \{(x,y) \in \mathbb{R}^2 \mid a+\eta \leq x \leq b-\eta,\ \phi_1(x)+\delta \leq y \leq \phi_2(x)-\delta\}$$

for some arbitrarily small numbers $\eta, \delta > 0$, meaning that the integral (reduced to iterated integrals using Fubini's theorem)

$$F(\eta,\delta) \equiv \iint_{D_{\eta,\delta}} f(x,y)\,dA = \int_{a+\eta}^{b-\eta} \int_{\phi_1(x)+\delta}^{\phi_2(x)-\delta} f(x,y)\,dy\,dx$$

is well defined.



Clearly, the function $F(\eta,\delta)$ is a function of two variables η and δ. So, if the limit

$$\lim_{(\eta,\delta)\to(0,0)} F(\eta,\delta)$$

is well defined, then so is the improper integral. For it to exist, the iterated limits must both equal to a well-defined real number $\mathcal{L}$ (and to each other). That is,

$$\lim_{\eta\to 0}\lim_{\delta\to 0} F(\eta,\delta) = \lim_{\delta\to 0}\lim_{\eta\to 0} F(\eta,\delta) = \mathcal{L} \implies \lim_{(\eta,\delta)\to(0,0)} F(\eta,\delta) = \mathcal{L}$$

It is also worthwhile to note that functions unbounded at isolated points can be evaluated using the methods above using a change of basis. Consider the example below.

Example 3.13

In the unit disk $D \subset \mathbb{R}^2$, let the function f be defined as

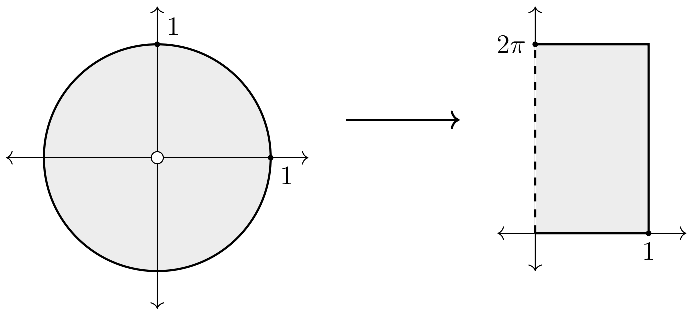
$$f(x,y) \equiv \frac{1}{\sqrt{x^2+y^2}}$$

Clearly, f is continuous at every point except $0 = (0,0)$, meaning that

$$\iint_{D\setminus\{0\}} f(x,y)\, dA$$

is well-defined. In order to solve the integral over the entire disk, we convert to polar coordinates and evaluate the limit

$$\iint_{D\setminus\{0\}} f(x,y)\, dA = \lim_{\delta\to 0} \int_{\delta}^{1} \int_{0}^{2\pi} r\, f(r\cos\theta, r\sin\theta)\, d\theta\, dr$$



If we are given an unbounded region $D \subset \mathbb{R}^2$, we can first create a bounded region and expand that region using a limit to cover all of D.

4 Line and Surface Integrals

We will introduce two objects: *vector fields* as the thing that we are trying to integrate, and *parameterized sets* as the thing we are integrating over.

4.1 Vector Fields

Definition 4.1 (Gradient)

The **gradient** of differentiable $f : E \subset \mathbb{R}^n \to \mathbb{R}$ is the vector field defined in the equivalent ways:

1. *Vector of Partials.* The gradient is
$$\nabla f : E \to \mathbb{R}^n, \qquad \nabla f(x) \coloneqq \begin{pmatrix} \frac{\partial f}{\partial x_1}(x) \\ \vdots \\ \frac{\partial f}{\partial x_n}(x) \end{pmatrix} \tag{396}$$
2. *Coordinate-Free.* It is the unique vector field whose dot product with any vector v at each point is the directional derivative of f along v. That is,
$$\nabla f_a \cdot v = Df_a v \text{ for all } a \in D \tag{397}$$

Proof.

Note that the gradient is a vector field (a bundle of vectors), while the total derivative is a covector field (a bundle of covectors). Since $\mathbb{R}^n$ is an inner product space, we can invoke Riesz Representation theorem and see that they are related in the way that
$$Df_a v = \nabla f(a) \cdot v \tag{398}$$
where $\cdot$ represents the dot product. At this point, it's a bit hard to see the difference between these two, but in more abstract spaces the total derivative generalizes much better than the gradient, which exists for inner product spaces. From this, we can write a coordinate independent definition of the gradient.

Lemma 4.1 (Gradient as Direction of Fastest Increase)

Let f be a real-valued function such that $\nabla f(x) \neq 0$. Then, at the point x,

1. $\nabla f(x)$ points in the direction along which f is increasing the fastest.
2. $-\nabla f(x)$ points in the direction along which f is decreasing the fastest.

Proof. Note that this is a coordinate-independent proof. Given a directional vector v, we can normalize it since we are only interested in direction. Evaluating it with the total derivative at x gives us $D_a f v$. But by definition,
$$\nabla f_a \cdot v = Df_a v \tag{399}$$
which means that
$$\begin{aligned} \sup_{\|v\|=1} \{Df_a v\} &= \sup_{\|v\|=1} \{\nabla f_a \cdot v\} && (400) \\ &= \sup_{\|v\|=1} \{\|\nabla f_a\| \, \|v\| \cos(\theta)\} && (401) \\ &= \sup\{\|\nabla f_a\| \cos(\theta)\} && (402) \\ &= \|\nabla f_a\| \text{ when } \theta = 0 && (403) \end{aligned}$$

Therefore, v must point in the direction of ∇f_a.

Therefore, we can interpret the gradient evaluated at a point as the tangent vector that points in the direction of fastest increase. We can also interpret the gradient ∇f itself as the vector field that determines some sort of "flow" in the domain $\mathbb{R}^n$. Therefore, if we drop a point in this field, the point will flow through $\mathbb{R}^n$ through a current determined by ∇f and will eventually end up at a local maximum.

Definition 4.2 (Del Operator)

The **del operator** is notational shorthand used to represent the following operations.

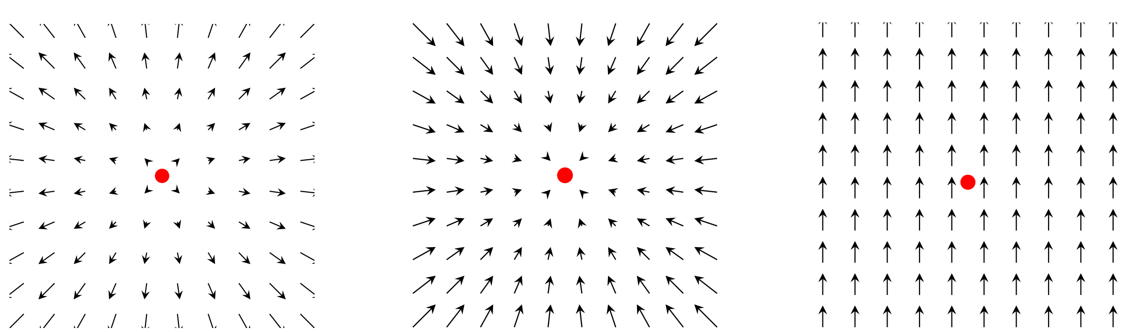
1. *Gradient.* Given $f : \mathbb{R}^n \to \mathbb{R}$, ∇f represents the gradient.
2. *Dot Product.* Given $F : \mathbb{R}^n \to \mathbb{R}^n$, by abuse of notation from treating ∇ as a vector, we can define
$$\nabla \cdot F \coloneqq \begin{pmatrix} \frac{\partial}{\partial x_1} \\ \vdots \\ \frac{\partial}{\partial x_n} \end{pmatrix} \cdot \begin{pmatrix} F_1 \\ \vdots \\ F_n \end{pmatrix} = \sum_i \frac{\partial F_i}{\partial x_i} \tag{404}$$
3. *Cross Product.* Given $F : \mathbb{R}^3 \to \mathbb{R}^3$, by abuse of notation from treating ∇ as a vector, we can define
$$\nabla \times F \coloneqq \begin{pmatrix} \frac{\partial}{\partial x_1} \\ \frac{\partial}{\partial x_2} \\ \frac{\partial}{\partial x_3} \end{pmatrix} \cdot \begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} = \tag{405}$$

For convenience, we use the del operator to denote the gradient. The **del operator** $\nabla : f \mapsto \nabla f$ takes in a differentiable function and outputs the gradient of it.

We would like to observe some nice properties of vector fields. Two natural properties can be analyzed through the divergence and curl. Colloquially, the divergence is an operator div that operates on a vector field and produces a scalar field which provides the quantity of the vector field's source at each point. Technically, the divergence represents the volume density of the outward flux of a vector field from an infinitesimal volume around a given point.

Definition 4.3 (Divergence)

The **divergence** of a vector field $F : \mathbb{R}^n \to \mathbb{R}^n$ is a scalar field defined
$$\operatorname{div} F \coloneqq \nabla \cdot F \tag{406}$$



(a) $\nabla \cdot F > 0$ (Source) (b) $\nabla \cdot F < 0$ (Sink) (c) $\nabla \cdot F = 0$ (Constant)

Figure 21: Imagine that the vector field F represents fluid flow in $\mathbb{R}^n$. Divergence is then the measure of the net amount of fluid flowing in and out of an infinitesimally small region, labeled at each point. If the net fluid flow is positive (i.e. more fluid is flowing in than out) at point x_0, then $\operatorname{div} F(x_0) > 0$. If the net fluid flow is negative (i.e. more fluid is flowing out than in) at point x_0, then $\operatorname{div} F(x_0) < 0$. Therefore, each point either acts as a "source" of fluid emanating from it or as a "sink" that sucks in more fluid than it puts out.

When $n = 1$, F reduces to a regular function and $\operatorname{div} F$ reduces to the ordinary derivative.

Lemma 4.2 (Linearity of Divergence)

By linearity of partials, div is also a linear operator. That is, given two vector fields F, G and two scalars α, β,

$$\operatorname{div}(\alpha F + \beta G) = \alpha \operatorname{div} F + \beta \operatorname{div} G \tag{407}$$

Proof.

Lemma 4.3 (Product Rule)

Divergence satisfies the product rule: Given a vector field $F : \mathbb{R}^n \to \mathbb{R}^n$ and a scalar function $\varphi : \mathbb{R}^n \to \mathbb{R}$.

$$\nabla \cdot (\varphi F) = \nabla \varphi \cdot F + \varphi (\nabla \cdot F) \tag{408}$$

Proof.

Lemma 4.4 (Divergence in Cylindrical Coordinates)

For vector field $F : \mathbb{R}^3 \to \mathbb{R}^3$ expressed in cylindrical coordinates as

$$F = \begin{pmatrix} F_r \\ F_\theta \\ F_z \end{pmatrix} \tag{409}$$

the divergence is

$$\operatorname{div} F = \nabla \cdot F = \frac{1}{r} \frac{\partial}{\partial r} \big(r F_r \big) + \frac{1}{r} \frac{\partial F_\theta}{\partial \theta} + \frac{\partial F_z}{\partial z} \tag{410}$$

Note that the condition of locality is important, since in general a global cylindrical coordinate system would be inconsistent.

Proof.

Lemma 4.5 (Divergence in Spherical Coordinates)

For vector field $F : \mathbb{R}^3 \to \mathbb{R}^3$ expressed in spherical coordinates (r, θ, ϕ), the divergence is

$$\operatorname{div} F = \nabla \cdot F = \frac{1}{r^2} \frac{\partial}{\partial r} \big(r^2 F_r \big) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} \big(\sin \theta F_\theta \big) + \frac{1}{r \sin \theta} \frac{\partial F_\phi}{\partial \phi} \tag{411}$$

Proof.

Example 4.1 (Solenoidal Vector Fields)

A vector field $F : \mathbb{R}^n \to \mathbb{R}^n$ is **solenoidal**, or **incompressible**, if $\operatorname{div} F = 0$ for all $x \in \mathbb{R}^n$.[a]

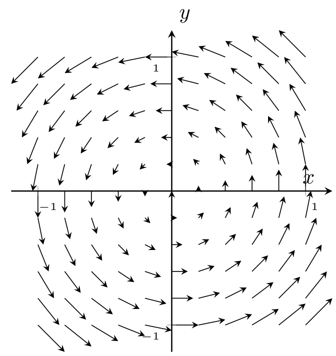


Figure 22: The vector field $F : (x, y) \mapsto (y, -x)$ is solenoidal.

[a] So, F has no sources or sinks.

Colloquially, the curl is a vector operator that describes the infinitesimal circulation of a vector field in 3-dimensional Euclidean space, where the curl at each point is represented by a vector whose length and direction denote the magnitude and axis as the maximum circulation. That is, if one drops a twig or a ball with its center of mass at a certain point, the curl measures how much it will spin. In physics, the rotation of a rigid body in 3-dimensions can be described by a vector ω along the axis of rotation. ω is called the *angular velocity vector*, with $\|\omega\|$ denoting the angular speed of the body. The curl of this vector field measured at the center of mass of the body is measured as 2ω. That is, the curl outputs *twice* the angular velocity vector of any rigid body. Note that unlike the gradient and divergence operators, curl does not generalize as simply to other dimensions.

Definition 4.4 (Curl)

The **curl** of a differentiable vector field $F : \mathbb{R}^3 \to \mathbb{R}^3$ is an operator

$$\operatorname{curl} F := \nabla \times F \tag{412}$$

Example 4.2 (Irrotational Vector Fields)

A vector field F is **irrotational** if $\operatorname{curl} F = 0$.[a]

[a] Visually, this indicates that there are no "whirlpools" everywhere, meaning that any rigid body placed anywhere, while it may travel along a path, will not rotate around its own axis.

It has been shown that fluid draining from a tub is usually irrotational except for right at the center, which is surprising since the fluid itself is "rotating" around the drain.

Theorem 4.6 (Curl Fields Have no Divergence)

For any C^2 vector field $F : \mathbb{R}^3 \to \mathbb{R}^3$,

$$\operatorname{div} \operatorname{curl} F = \nabla \cdot (\nabla \times F) = 0 \tag{413}$$

That is, the divergence of any curl is 0.

Proof. Proved by equality of mixed partials.

Definition 4.5 (Laplacian)

The **Laplacian** of a function $f : \mathbb{R}^n \to \mathbb{R}$ is the divergence of the gradient.

$$\nabla^2 f \coloneqq \nabla \cdot (\nabla f) = \sum_{i=1}^n \frac{\partial^2 f}{\partial x_i^2} \tag{414}$$

Definition 4.6 (Conservative Vector Fields)

A vector field $F : U \subset \mathbb{R}^n \to \mathbb{R}^n$ is a **conservative vector field** if and only if there exists a scalar field $f : U \subset \mathbb{R}^n \to \mathbb{R}$ such that

$$F = \nabla f \tag{415}$$

on U.

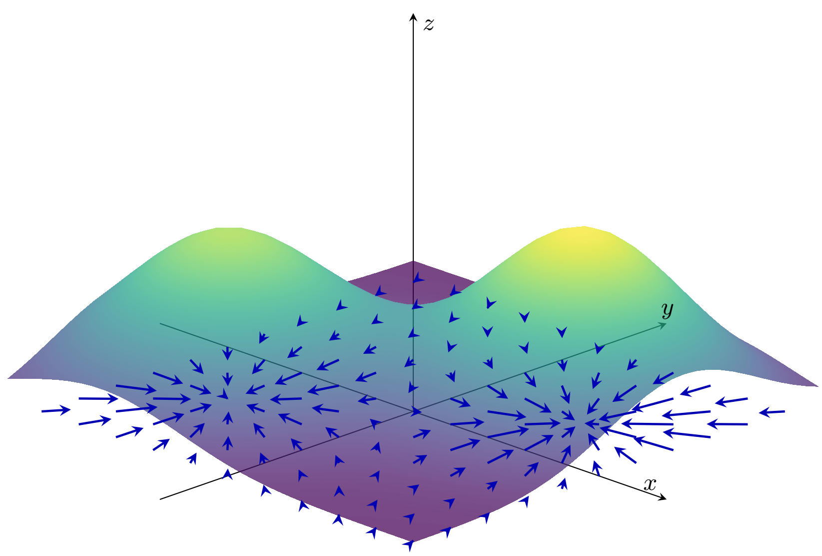


Figure 23: If vector field F is conservative, then there exists a smooth scalar function f such that $\nabla f = F$. For each latitude and longitude on a certain map, we can give it an altitude as a function of those coordinates (picture a map with a bunch of hills and valleys). The gradient and thus the vector field is all the vectors that point in the direction of highest ascent. The vector field is all the vectors that point in the direction of highest ascent.

Conservative vector fields appear naturally in mechanics: they are vector fields representing forces of physical systems in which energy is conserved.

Theorem 4.7 (Gradient Fields Have No Curl)

Given a C^2-function $f : \mathbb{R}^3 \to \mathbb{R}$,

$$\nabla \times (\nabla f) = 0 \tag{416}$$

That is, the curl of any gradient vector field is the zero vector.

Proof. $\nabla \times \nabla f$ can be expanded to

$$\begin{pmatrix} \frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y} \\ \frac{\partial^2 f}{\partial z \partial x} - \frac{\partial^2 f}{\partial x \partial z} \\ \frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \tag{417}$$

by equality of mixed partials.

Theorem 4.8 (Helmholtz Decomposition)

Let $F : \mathbb{R}^3 \to \mathbb{R}^3$ be a C^2 vector field. Then, F can be decomposed into a curl-free component and a divergence-free component. That is, there exists vector fields A and Φ

$$F = -\nabla \cdot \Phi + \nabla \times A \tag{418}$$

4.2 Parameterizations and Orientation

Definition 4.7 (Parameterization)

Given some subset $E \subset \mathbb{R}^n$, a **k-parameterization** of E is a continuous surjective map

$$\varphi : D \subset \mathbb{R}^k \to E \tag{419}$$

where D is a compact subset of $\mathbb{R}^k$.

Definition 4.8 (Reparameterization)

Let $E \subset \mathbb{R}^n$ with parameterization $\varphi : D \subset \mathbb{R}^k \to E$. Given **reparameterization** of E is a continuous bijective map $h : D' \subset \mathbb{R}^k \to D$, where D' is a compact subset of $\mathbb{R}^k$. Therefore,

$$\varphi \circ h : D' \to E \tag{420}$$

is another parameterization of E.

Continuity alone is not enough to guarantee that a parameterized set behaves like a curve in the usual geometric sense. In fact, we will sketch a construction of a continuous surjective map $\gamma : [0, 1] \to [0, 1]^2$, called a *space-filling curve.*

Example 4.3 (Space-Filling Curve in $\mathbb{R}^2$)

Let $C \subset [0, 1]$ be the Cantor set. Every point $x \in C$ has a ternary expansion using only the digits 0 and 2.

$$x = 0.a_1 a_2 a_3 \cdots, \qquad a_i \in \{0, 2\}. \tag{421}$$

Define binary digits $b_i = \frac{a_i}{2} \in \{0, 1\}$. Now split the binary digits into odd and even positions, and define

$$\Gamma(x) = \big(0.b_1 b_3 b_5 \cdots_2 , 0.b_2 b_4 b_6 \cdots_2\big) = \left(\sum_{j=1}^{\infty} \frac{b_{2j-1}}{2^j}, \sum_{j=1}^{\infty} \frac{b_{2j}}{2^j} \right). \tag{422}$$

We claim that $\Gamma : C \to [0, 1]^2$ is continuous and surjective.

1. *Continuous.* The idea is that if two Cantor points x, x' are close, then their tenary expansions will agree for a long time. Fix $\epsilon > 0$. Choose N large enough so that
$$\sum_{j>N/2} \frac{1}{2^j} < \epsilon. \tag{423}$$
If two points $x, x'' \in C$ agree in their first N ternary digits, then the corresponding binary digits b_i, b_i' also agree for $1 \leq i \leq N$. Hence the first $\lfloor N/2 \rfloor$ binary digits of both coordinates of $\Gamma(x)$ and $\Gamma(x')$ agree. Therefore
$$\|\Gamma(x) - \Gamma(x')\|_\infty \leq \sum_{j>N/2} \frac{1}{2^j} < \epsilon. \tag{424}$$
Since agreement of sufficiently many initial ternary digits is guaranteed whenever $x, x' \in C$ are sufficiently close, Γ is continuous.
2. *Surjective.* Take any point $(y, z) \in [0,1]^2$, Write y and z in binary:
$$y = 0.c_1 c_2 c_3 \cdots_2 , \qquad z = 0.d_1 d_2 d_3 \cdots_2 , \tag{425}$$
where $c_i, d_i \in \{0, 1\}$. Now interleave these digits to define
$$b_1 = c_1, \quad b_2 = d_1, \quad b_3 = c_2, \quad b_4 = d_2, \quad \ldots \tag{426}$$
Then define
$$x = 0.(2b_1)(2b_2)(2b_3)(2b_4) \cdots_3 . \tag{427}$$
Since the ternary digits of x are only 0 and 2, we have $x \in C$. By construction,
$$\Gamma(x) = (y, z). \tag{428}$$

Finally, we extend Γ from C to all of $[0, 1]$.
1. On C, define
$$\gamma(t) = \Gamma(t), \qquad t \in C. \tag{429}$$
2. On $[0,1] \setminus C$, note that this complement is a union of disjoint open intervals. On each such gap (a, b), define γ by linearly interpolating between the endpoint values $\Gamma(a)$ and $\Gamma(b)$:
$$\gamma(t) = \frac{b-t}{b-a}\Gamma(a) + \frac{t-a}{b-a}\Gamma(b), \qquad t \in (a, b). \tag{430}$$
Since $[0,1]^2$ is a convex set, γ will land in $[0,1]^2$.

It remains to check that the extension is continuous. On each gap (a, b), the map γ is affine in t, hence continuous. Moreover, as $t \to a^+$,
$$\gamma(t) = \frac{b-t}{b-a}\Gamma(a) + \frac{t-a}{b-a}\Gamma(b) \to \Gamma(a) = \gamma(a), \tag{431}$$
and similarly $\gamma(t) \to \Gamma(b) = \gamma(b)$ as $t \to b^-$.

Finally, let $c \in C$. Since C is compact and Γ is continuous, Γ is uniformly continuous on C. Hence, given $\epsilon > 0$, there exists $\delta > 0$ such that whenever $p, q \in C$ and $|p - q| < \delta$, we have
$$\|\Gamma(p) - \Gamma(q)\| < \epsilon. \tag{432}$$
If t is sufficiently close to c, then either $t \in C$, in which case $\gamma(t) = \Gamma(t)$ is close to $\Gamma(c)$, or t lies in a gap (a, b) whose endpoints $a, b \in C$ are close to c. In the latter case, $\gamma(t)$ is a convex combination of $\Gamma(a)$ and $\Gamma(b)$, both of which are close to $\Gamma(c)$. Therefore $\gamma(t)$ is close to $\Gamma(c) = \gamma(c)$. Thus γ is continuous on all of $[0, 1]$.

This shows that a continuous 1-parameterization can have a 2-dimensional image. Thus continuity alone is not enough for a useful geometric theory of curves and surfaces. We must make sure that the set E is smooth in the sense that (informally) there aren't any wrinkles, points, folds, or self-intersections in such a way that the tangent plane to the surface is not well-defined.

Definition 4.9 (Regular Parameterization)

Let $E \subset \mathbb{R}^n$. A **regular k-parameterization** of E is a continuously differentiable surjective map

$$\varphi : D \subset \mathbb{R}^k \to E, \tag{433}$$

where D is compact and where $D\varphi_p : \mathbb{R}^k \to \mathbb{R}^n$ has rank k for every interior point $u \in D$.

Note that a parameterization is not necessarily a homeomorphism. It may fail to be injective, and its image may have self-intersections or other singular behavior. If φ is a continuous bijection from compact D onto $E \subset \mathbb{R}^n$, then φ is a homeomorphism onto E. In general, however, a parameterized set should not automatically be regarded as a manifold.

There are many special cases of surfaces, most notably 1-dimensional paths in $\mathbb{R}^2$ and $\mathbb{R}^3$, along with 2-dimensional surfaces in $\mathbb{R}^3$.

Definition 4.10 (Curve)

A **parameterized curve** is a C^1 surjective map

$$\varphi : [a, b] \subset \mathbb{R} \to E \subset \mathbb{R}^n \tag{434}$$

A **curve** is the image $\varphi([a, b]) \subset \mathbb{R}^n$.

1. φ is **closed** if $\varphi(a) = \varphi(b)$.
2. φ is **simple** if the restriction of φ to $[a, b)$ is injective.[a]

[a]Visually, the curve doesn't intersect itself except possible at the endpoints.

Definition 4.11 (Surface)

A **parameterized surface** is a C^1 surjective map

$$\varphi : [a_1, b_1] \times [a_2, b_2] \subset \mathbb{R}^2 \to E \subset \mathbb{R}^n \tag{435}$$

A **surface** is the image E.

In lower dimensions, we can find some motivation for the tangent space of a paramaterized set.

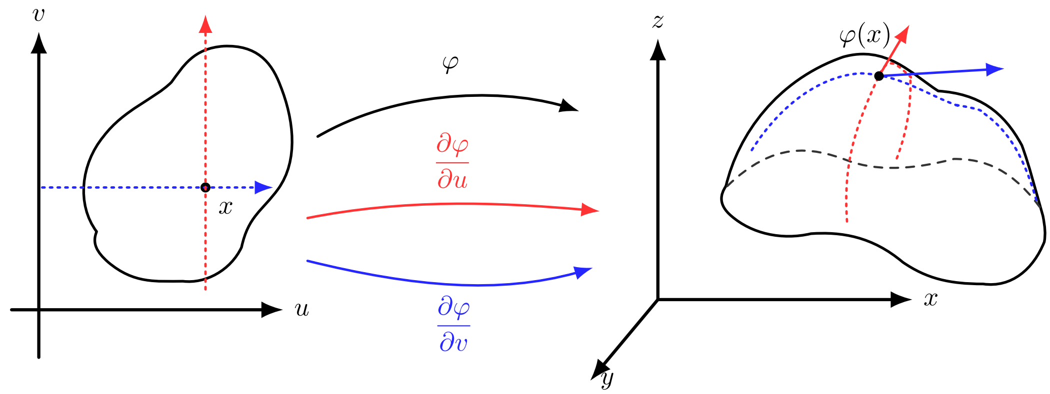


Figure 24: When $\varphi : D \subset \mathbb{R}^2 \to \mathbb{R}^3$, we can see visually that given $(u_0, v_0) \in D$, there should be two linearly independent tangent vectors T_u, T_v sticking out from $\varphi(u_0, v_0)$ on the surface E. So every vector in the "tangent plane" should be a linear combination of T_u and T_v. In fact, these two vectors are precisely $T_u = \frac{\partial \varphi}{\partial u}(u_0, v_0)$ and $T_v = \frac{\partial \varphi}{\partial v}(u_0, v_0)$, so the tangent plane is really just $\varphi(u_0, v_0)$ plus the vectors in the column space of $D\varphi(u_0, v_0)$!

Definition 4.12 (Tangent Space)

Let $\varphi : D \subset \mathbb{R}^k \to \mathbb{R}^n$ be a regular parameterization, and let $u \in D$ be an interior point. The **tangent space** to the parameterized set at $\varphi(u)$ is the k-dimensional subspace

$$T_{\varphi(u)} E \coloneqq \operatorname{Im} D\varphi_u \subset \mathbb{R}^n. \tag{436}$$

The corresponding **affine tangent plane** is

$$\varphi(u) + T_{\varphi(u)} E = \{\varphi(u) + v : v \in T_{\varphi(u)} E\}. \tag{437}$$

If $k = n-1$, then the normal space is one-dimensional. If $N(u)$ is a nonzero vector satisfying $N(u) \perp T_{\varphi(u)} E$, then the affine tangent hyperplane can also be written as

$$\{x \in \mathbb{R}^n : \langle x - \varphi(u), N(u) \rangle = 0\}. \tag{438}$$

Example 4.4 (Tangent Space of Curve in $\mathbb{R}^2$)

Example 4.5 (Tangent Space of Curve in $\mathbb{R}^3$)

Example 4.6 (Tangent Space of Surface in $\mathbb{R}^3$)

Often, we work with 2-parameterizations of some $E \subset \mathbb{R}^3$. Therefore, the definition for a regularity at a point (u_1, u_2) is equivalent to

$$\frac{\partial \varphi}{\partial u_1}(u_1, u_2) \times \frac{\partial \varphi}{\partial u_2}(u_1, u_2) \neq 0 \tag{439}$$

where $\times$ is the Euclidean cross product.

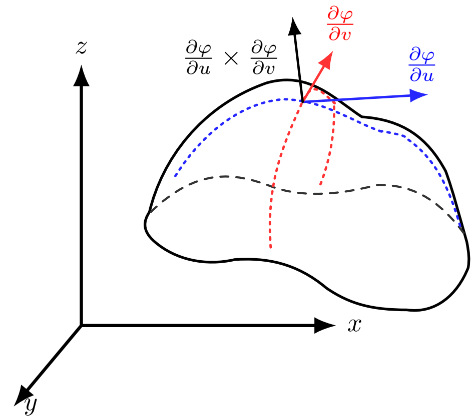


Figure 25: Note that the cross product $\frac{\partial\varphi}{\partial u}(u_0, v_0) \times \frac{\partial\varphi}{\partial v}(u_0, v_0) \neq 0$ if and only if $D\varphi(u_0, v_0)$ is full rank, i.e. the tangent plane exists.

A path has a natural orientation defined by the function. We would like to determine a similar notion of orientation by defining one "side" of a surface to be positive and the other to be negative. Surprisingly, a paramaterization does not have an intrinsic global orientation. Rather, we determine the orientation ourselves by choosing a unit vector that generally points towards the outside of the surface S. Again, this choice is arbitrary, but it is customary to choose a vector that generally points "out."

Definition 4.13 (Orientation of a Vector Space)

Let V be a k-dimensional real vector space.

1. An **ordered basis** is an ordered tuple of basis elements $\mathcal{B} = (v_1, v-2, \ldots, v_k)$ of V. It is encoded in the column matrix
$$B = \begin{pmatrix} | & \ldots & | \\ v_1 & \ldots & v_k \\ | & \ldots & | \end{pmatrix} \tag{440}$$
2. Two ordered bases $\mathcal{B} = (v_1, \ldots, v_k)$ and $\mathcal{C} = (w_1, \ldots, w_k)$ are said to determine the same orientation if the change-of-basis matrix from $\mathcal{B}$ to $\mathcal{C}$ has positive determinant. This defines an equivalence relation on the set of ordered bases of V.
3. An **orientation** of V is one of the two equivalence classes of ordered bases.
4. A **positive orientation** of $\mathbb{R}^n$ is the equivalence class consisting of the standard basis.

Definition 4.14 (Orientation of a Regular Parameterized Set)

Let $E \subset \mathbb{R}^n$ be a regular k-dimensional parameterized set, and let $\varphi : D \subset \mathbb{R}^k \to E$ be a regular parameterization of E.

1. For each $p \in D$, the derivative $D\varphi_p : \mathbb{R}^k \to \mathbb{R}^n$ maps the standard ordered basis $(e_1, \ldots, e_k)$ of $\mathbb{R}^k$ to the ordered tangent basis
$$\big(D\varphi_p(e_1), \ldots, D\varphi_p(e_k)\big) \tag{441}$$
of the tangent space $T_{\varphi(p)}E = \mathrm{Im}(D\varphi_p)$. This ordered basis induces an **orientation of the tangent space** $T_{\varphi(p)}E$.
2. The orientation induced by the (positively oriented) standard basis is known as the **positive orientation** of $T_{\varphi(p)}E$.
3. An **orientation of** E is a choice of orientation of each tangent space T_qE, for every $q \in E$, such that the choice varies continuously with q. Equivalently, an orientation of E is a continuous

choice of positively oriented ordered tangent bases at every point of E.

So, to figure out how the parameterization determines an oorientation to E, we go through the following steps: You first choose an orientation on the parameter domain $\mathbb{R}^k$. Usually, we declare the standard ordered basis $(e_1, \ldots, e_k)$ to be positive. Then the regular parameterization $\varphi : D \subset \mathbb{R}^k \to E \subset \mathbb{R}^n$ pushes that positive basis forward to tangent vectors on E:

$$\big(D\varphi_p(e_1), \ldots, D\varphi_p(e_k)\big). \tag{442}$$

Because φ is regular, these form a basis of the tangent space $T_{\varphi(p)}E$. Then we declare this tangent basis to be **positive** in $T_{\varphi(p)}E$. So the idea is:

$$\text{positive basis in parameter space} \quad \xrightarrow{D\varphi_p} \quad \text{positive basis in tangent space.} \tag{443}$$

If this orientation varies continuously over p, then we can declare this to be the orientation of E. Let's see a few examples of this.

Example 4.7 (Orientation of Curve in $\mathbb{R}^2$)

Let $E = \{(x, y) \in \mathbb{R}^2 \colon x^2 + y^2 = 1\} \subset \mathbb{R}^2$ be the unit circle. Let's look at two parameterizations.

1. *Counterclockwise.* Let $\varphi : [0, 2\pi] \to E,\ \varphi(t) = (\cos t, \sin t)$. We give $\mathbb{R}$ the standard orientation by declaring the ordered basis $(e_1) = (1)$ to be positive. For each t, the derivative $D\varphi_t : \mathbb{R} \to \mathbb{R}^2$ maps the positive basis vector $e_1 \in \mathbb{R}$ to

$$D\varphi_t(e_1) = \varphi'(t) = (-\sin t, \cos t) \tag{444}$$

   Since $\varphi'(t) \neq 0$ for all t, this gives a basis of the tangent space $T_{\varphi(t)}E$ for every point on E. Therefore, this parameterization declares

$$\big(\varphi'(t)\big) \tag{445}$$

   to be a positively oriented basis of $T_{\varphi(t)}E$. Geometrically, this means that the positive direction on the circle is the direction in which $\varphi(t)$ travels as t increases, which is counterclockwise.

2. *Clockwise.* Now let's use the parameterization $\phi : [0, 2\pi] \to E,\ \phi(t) = (\cos t, -\sin t)$. We give $\mathbb{R}$ the standard orientation by declaring the ordered basis $(e_1) = (1)$ to be positive. For each t, the derivative $D\phi_t : \mathbb{R} \to \mathbb{R}^2$ maps the positive basis vector $e_1 \in \mathbb{R}$ to

$$D\phi_t(e_1) = \phi'(t) = (\sin t, -\cos t) \tag{446}$$

   Since $\phi'(t) \neq 0$ for all t, this gives a basis of the tangent space $T_{\phi(t)}E$ for every point on E. Therefore, this parameterization declares

$$\big(\phi'(t)\big) \tag{447}$$

   to be a positively oriented basis of $T_{\phi(t)}E$. Geometrically, this means that the positive direction on the circle is the direction in which $\phi(t)$ travels as t increases, which is clockwise.

Example 4.8 (Orientation of Curve in $\mathbb{R}^3$)

Let $E = \{(\cos t, \sin t, t) \in \mathbb{R}^3 : t \in \mathbb{R}\} \subset \mathbb{R}^3$ be the helix. Let's look at two parameterizations.

1. *Upward.* Let $\varphi : \mathbb{R} \to E,\ \varphi(t) = (\cos t, \sin t, t)$. We give $\mathbb{R}$ the standard orientation by declaring the ordered basis $(e_1) = (1)$ to be positive. For each t, the derivative $D\varphi_t : \mathbb{R} \to \mathbb{R}^3$ maps the

positive basis vector $e_1 \in \mathbb{R}$ to

$$D\varphi_t(e_1) = \varphi'(t) = (-\sin t, \cos t, 1) \tag{448}$$

Since $\varphi'(t) \neq 0$ for all t, this gives a basis of the tangent space $T_{\varphi(t)}E$ for every point on E. Therefore, this parameterization declares

$$\big(\varphi'(t)\big) \tag{449}$$

to be a positively oriented basis of $T_{\varphi(t)}E$. Geometrically, this means that the positive direction on the helix is the direction in which $\varphi(t)$ travels as t increases, which is upward.

2. *Downward.* Now let's use the parameterization $\phi : \mathbb{R} \to E$, $\phi(t) = (\cos t, -\sin t, -t)$. We give $\mathbb{R}$ the standard orientation by declaring the ordered basis $(e_1) = (1)$ to be positive. For each t, the derivative $D\phi_t : \mathbb{R} \to \mathbb{R}^3$ maps the positive basis vector $e_1 \in \mathbb{R}$ to

$$D\phi_t(e_1) = \phi'(t) = (-\sin t, -\cos t, -1) \tag{450}$$

Since $\phi'(t) \neq 0$ for all t, this gives a basis of the tangent space $T_{\phi(t)}E$ for every point on E. Therefore, this parameterization declares

$$\big(\phi'(t)\big) \tag{451}$$

to be a positively oriented basis of $T_{\phi(t)}E$. Geometrically, this means that the positive direction on the helix is the direction in which $\phi(t)$ travels as t increases, which is downward.

Example 4.9 (Orientation of Surface in $\mathbb{R}^3$)

Let $E \subset \mathbb{R}^3$ be a regular surface with parameterization

$$\varphi : D \subset \mathbb{R}^2 \to E. \tag{452}$$

The parameter domain lies in $\mathbb{R}^2$, and we give $\mathbb{R}^2$ its standard orientation by declaring the ordered basis

$$(e_1, e_2) \tag{453}$$

to be positive. For each $(u, v) \in D$, the derivative

$$D\varphi_{(u,v)} : \mathbb{R}^2 \to \mathbb{R}^3 \tag{454}$$

maps the positive basis vectors e_1, e_2 to

$$D\varphi_{(u,v)}(e_1) = \varphi_u(u, v), \qquad D\varphi_{(u,v)}(e_2) = \varphi_v(u, v). \tag{455}$$

Since φ is regular, the vectors

$$\varphi_u(u, v), \varphi_v(u, v) \tag{456}$$

are linearly independent, so they form an ordered basis of the tangent space

$$T_{\varphi(u,v)}E. \tag{457}$$

Therefore, the parameterization declares

$$\big(\varphi_u(u, v), \varphi_v(u, v)\big) \tag{458}$$

to be a positively oriented basis of $T_{\varphi(u,v)}E$.
In $\mathbb{R}^3$, the orientation of the ordered tangent basis is equivalently described by the corresponding normal direction

$$\varphi_u(u, v) \times \varphi_v(u, v). \tag{459}$$

If we reverse the order of the tangent basis, then the cross product changes sign:

$$\varphi_v(u, v) \times \varphi_u(u, v) = -\big(\varphi_u(u, v) \times \varphi_v(u, v)\big). \tag{460}$$

Thus, a surface in $\mathbb{R}^3$ has two possible orientations, corresponding to the two possible continuous normal directions. The associated unit normal vector field is

$$n(\varphi(u, v)) = \pm \frac{\varphi_u(u, v) \times \varphi_v(u, v)}{|\varphi_u(u, v) \times \varphi_v(u, v)|}. \tag{461}$$

For example, consider the graph of a smooth function

$$z = f(x, y). \tag{462}$$

A natural parameterization is

$$\varphi(x, y) = (x, y, f(x, y)). \tag{463}$$

Here the parameter domain is again $\mathbb{R}^2$, with positive basis (e_1, e_2). The derivative maps this basis to the tangent vectors

$$\varphi_x(x, y) = (1, 0, f_x(x, y)), \qquad \varphi_y(x, y) = (0, 1, f_y(x, y)). \tag{464}$$

Hence the parameterization declares

$$\big(\varphi_x(x, y), \varphi_y(x, y)\big) \tag{465}$$

to be a positively oriented basis of the tangent space. The corresponding normal vector is

$$\varphi_x(x, y) \times \varphi_y(x, y) = (-f_x(x, y), -f_y(x, y), 1). \tag{466}$$

Since this vector has positive z-component, it points upward. Therefore, the unit normal

$$n(x, y) = \frac{(-f_x(x, y), -f_y(x, y), 1)}{\sqrt{1 + f_x(x, y)^2 + f_y(x, y)^2}} \tag{467}$$

gives the upward orientation of the graph. The opposite choice

$$-n(x, y) = \frac{(f_x(x, y), f_y(x, y), -1)}{\sqrt{1 + f_x(x, y)^2 + f_y(x, y)^2}} \tag{468}$$

gives the downward orientation.

Thus, for a surface in $\mathbb{R}^3$, choosing an orientation means choosing which ordered tangent bases are positive, or equivalently, choosing one of the two continuous normal directions. Intuitively, this is just choosing which side of the surface is the positive side.

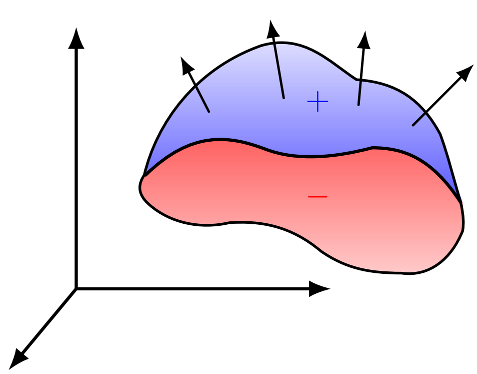


Figure 26: Choosing an orientation of a surface in $\mathbb{R}^3$ is equivalent to choosing one of the two continuous normal directions.

Example 4.10 (Determining Orientation of an Ellipsoid)

Consider the ellipsoid

$$E = \{(x, y, z) \in \mathbb{R}^3 : \tfrac{x^2}{a^2} + \tfrac{y^2}{b^2} + \tfrac{z^2}{c^2} = 1\}, \tag{469}$$

where $a, b, c > 0$. Let's look at three parameterizations.

1. *Original Parameterization.* Let

$$\varphi(u, v) = \big(a \sin u \cos v, b \sin u \sin v, c \cos u\big), \tag{470}$$

where $0 \leq u \leq \pi$ and $0 \leq v \leq 2\pi$. We give $\mathbb{R}^2$ the standard orientation by declaring (e_1, e_2) to be positive. For each (u, v), the derivative maps the positively oriented basis to what we define as the positively oriented tangent basis

$$D\varphi_{(u,v)}(e_1) = \varphi_u(u, v) = \big(a \cos u \cos v, b \cos u \sin v, -c \sin u\big) \tag{471}$$

$$D\varphi_{(u,v)}(e_2) = \varphi_v(u, v) = \varphi_v(u, v) = \big(-a \sin u \sin v, b \sin u \cos v, 0\big) \tag{472}$$

Their cross product is

$$\varphi_u(u, v) \times \varphi_v(u, v) = \big(bc \sin^2 u \cos v, ac \sin^2 u \sin v, ab \sin u \cos u\big). \tag{473}$$

Since

$$(\varphi_u \times \varphi_v) \cdot \varphi = abc \sin u > 0 \qquad \text{for } 0 < u < \pi, \tag{474}$$

the normal points outward. Therefore, this parameterization declares (φ_u, φ_v) to be a positively oriented basis of $T_{\varphi(u,v)}E$, and it induces the outward orientation on the ellipsoid.

2. *Same Orientation.* Now let

$$\psi(s, t) = \varphi(2s, t) = \big(a \sin(2s) \cos t, b \sin(2s) \sin t, c \cos(2s)\big), \tag{475}$$

where $0 \leq s \leq \frac{\pi}{2}$ and $0 \leq t \leq 2\pi$. Then $\psi_s(s, t) = 2\varphi_u(2s, t)$ and $\psi_t(s, t) = \varphi_v(2s, t)$, so

$$(\psi_s, \psi_t) = (\varphi_u, \varphi_v) \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \tag{476}$$

where the right-hand side is evaluated at $(2s, t)$. Since the determinant is $2 > 0$, (ψ_s, ψ_t) has the same orientation as (φ_u, φ_v). Equivalently,

$$\psi_s(s, t) \times \psi_t(s, t) = 2\big(\varphi_u(2s, t) \times \varphi_v(2s, t)\big), \tag{477}$$

so the normal still points outward. Hence ψ induces the same outward orientation as φ.

3. *Reverse Orientation.* Finally, let

$$\eta(u, v) = \varphi(u, -v) = \big(a \sin u \cos v, -b \sin u \sin v, c \cos u\big). \tag{478}$$

Then $\eta_u(u, v) = \varphi_u(u, -v)$ and $\eta_v(u, v) = -\varphi_v(u, -v)$, so

$$(\eta_u, \eta_v) = (\varphi_u, \varphi_v) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{479}$$

where the right-hand side is evaluated at $(u, -v)$. Since the determinant is $-1 < 0$, (η_u, η_v) has the opposite orientation from (φ_u, φ_v). Equivalently,

$$\eta_u(u, v) \times \eta_v(u, v) = -\big(\varphi_u(u, -v) \times \varphi_v(u, -v)\big), \tag{480}$$

so the normal points inward instead of outward. Hence η induces the inward orientation on the ellipsoid.

Example 4.11 (Orientation of a Torus in $\mathbb{R}^3$)

Consider the torus

$$E = \left\{ \big((R + r\cos u)\cos v, (R + r\cos u)\sin v, r\sin u \big) : 0 \leq u, v \leq 2\pi \right\}, \tag{481}$$

where $R > r > 0$. Let's look at three parameterizations.

1. *Original Parameterization.* Let

$$\varphi(u, v) = \big((R + r\cos u)\cos v, (R + r\cos u)\sin v, r\sin u \big). \tag{482}$$

We give $\mathbb{R}^2$ the standard orientation by declaring (e_1, e_2) to be positive. The tangent vectors are

$$\varphi_u(u, v) = \big(-r\sin u\cos v, -r\sin u\sin v, r\cos u \big) \tag{483}$$

and

$$\varphi_v(u, v) = \big(-(R + r\cos u)\sin v, (R + r\cos u)\cos v, 0 \big). \tag{484}$$

Their cross product is

$$\varphi_u(u, v) \times \varphi_v(u, v) = -r(R + r\cos u)\big(\cos u\cos v, \cos u\sin v, \sin u \big). \tag{485}$$

The vector $\big(\cos u\cos v, \cos u\sin v, \sin u \big)$ points outward from the central circle of the torus. Therefore, $\varphi_u \times \varphi_v$ points inward. Hence this parameterization declares (φ_u, φ_v) to be positively oriented and induces the inward orientation of the torus.

2. *Same Orientation.* Now let

$$\psi(s, t) = \varphi(s, 2t). \tag{486}$$

Then $\psi_s = \varphi_u(s, 2t)$ and $\psi_t = 2\varphi_v(s, 2t)$, so

$$(\psi_s, \psi_t) = (\varphi_u, \varphi_v) \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \tag{487}$$

where the right-hand side is evaluated at $(s, 2t)$. Since the determinant is $2 > 0$, (ψ_s, ψ_t) has the same orientation as (φ_u, φ_v). Equivalently,

$$\psi_s \times \psi_t = 2(\varphi_u \times \varphi_v), \tag{488}$$

so ψ induces the same inward orientation as φ.

3. *Reverse Orientation.* Finally, let

$$\eta(u, v) = \varphi(v, u). \tag{489}$$

Then $(\eta_u, \eta_v) = (\varphi_v, \varphi_u)$. Relative to (φ_u, φ_v), this corresponds to the matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{490}$$

whose determinant is -1. Therefore, η induces the opposite orientation. Equivalently, $\eta_u \times \eta_v = -(\varphi_u \times \varphi_v)$, so η gives the outward orientation.

Example 4.12 (Orientation of the Clifford Torus in $\mathbb{R}^4$)

Consider the Clifford torus

$$E = \left\{ \frac{1}{\sqrt{2}}(\cos u, \sin u, \cos v, \sin v) : 0 \leq u, v \leq 2\pi \right\} \subset \mathbb{R}^4. \tag{491}$$

Let's look at three parameterizations.

1. *Original Parameterization.* Let
$$\varphi(u,v) = \tfrac{1}{\sqrt{2}}(\cos u, \sin u, \cos v, \sin v). \tag{492}$$
We give $\mathbb{R}^2$ the standard orientation by declaring (e_1, e_2) to be positive. The derivative maps this basis to
$$D\varphi_{(u,v)}(e_1) = \varphi_u(u,v), \qquad D\varphi_{(u,v)}(e_2) = \varphi_v(u,v). \tag{493}$$
The tangent vectors are
$$\varphi_u(u,v) = \tfrac{1}{\sqrt{2}}(-\sin u, \cos u, 0, 0) \tag{494}$$
and
$$\varphi_v(u,v) = \tfrac{1}{\sqrt{2}}(0, 0, -\sin v, \cos v). \tag{495}$$
Since these vectors are linearly independent for all (u,v), they form a basis of the tangent space. Therefore, this parameterization declares
$$\big(\varphi_u(u,v), \varphi_v(u,v)\big) \tag{496}$$
to be a positively oriented basis of $T_{\varphi(u,v)}E$.
2. *Same Orientation.* Now let
$$\psi(s,t) = \varphi(s, 2t). \tag{497}$$
Then $\psi_s = \varphi_u(s,2t)$ and $\psi_t = 2\varphi_v(s,2t)$, so
$$(\psi_s, \psi_t) = (\varphi_u, \varphi_v)\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \tag{498}$$
where the right-hand side is evaluated at $(s,2t)$. Since the determinant is $2 > 0$, the ordered tangent basis (ψ_s, ψ_t) has the same orientation as (φ_u, φ_v).
3. *Reverse Orientation.* Finally, let
$$\eta(s,t) = \varphi(t,s). \tag{499}$$
Then $(\eta_s, \eta_t) = (\varphi_v, \varphi_u)$. Relative to (φ_u, φ_v), this corresponds to the matrix
$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{500}$$
whose determinant is -1. Therefore, η induces the opposite orientation.

Example 4.13 (Orientation of the 3-Sphere in $\mathbb{R}^4$)

Consider the unit 3-sphere
$$S^3 = (x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_1^2 + x_2^2 + x_3^2 + x_4^2 = 1. \tag{501}$$
One local parameterization of S^3 is given by
$$\varphi(\alpha,\beta,\gamma) = \big(\cos\alpha, \sin\alpha\cos\beta, \sin\alpha\sin\beta\cos\gamma, \sin\alpha\sin\beta\sin\gamma\big), \tag{502}$$
where $0 < \alpha < \pi$, $0 < \beta < \pi$, and $0 < \gamma < 2\pi$. The parameter domain lies in $\mathbb{R}^3$, and we give $\mathbb{R}^3$ its standard orientation by declaring (e_1, e_2, e_3) to be positive.
The derivative maps the positive basis vectors to
$$D\varphi_{(\alpha,\beta,\gamma)}(e_1) = \varphi_\alpha, \qquad D\varphi_{(\alpha,\beta,\gamma)}(e_2) = \varphi_\beta, \qquad D\varphi_{(\alpha,\beta,\gamma)}(e_3) = \varphi_\gamma. \tag{503}$$
Therefore, this parameterization declares
$$\big(\varphi_\alpha, \varphi_\beta, \varphi_\gamma\big) \tag{504}$$

to be a positively oriented basis of $T_{\varphi(\alpha,\beta,\gamma)}S^3$.
The tangent vectors are

$$\varphi_\alpha = \big(-\sin\alpha, \cos\alpha\cos\beta, \cos\alpha\sin\beta\cos\gamma, \cos\alpha\sin\beta\sin\gamma\big), \tag{505}$$
$$\varphi_\beta = \big(0, -\sin\alpha\sin\beta, \sin\alpha\cos\beta\cos\gamma, \sin\alpha\cos\beta\sin\gamma\big), \tag{506}$$
$$\varphi_\gamma = \big(0, 0, -\sin\alpha\sin\beta\sin\gamma, \sin\alpha\sin\beta\cos\gamma\big). \tag{507}$$

These vectors form a basis of the three-dimensional tangent space wherever $0 < \alpha < \pi$ and $0 < \beta < \pi$. Since S^3 is a hypersurface in $\mathbb{R}^4$, its orientation can also be described by choosing a continuous unit normal vector field. The outward unit normal at $\varphi(\alpha,\beta,\gamma)$ is simply

$$n(\varphi(\alpha,\beta,\gamma)) = \varphi(\alpha,\beta,\gamma), \tag{508}$$

because the sphere is centered at the origin.
To check whether the ordered tangent basis $(\varphi_\alpha, \varphi_\beta, \varphi_\gamma)$ agrees with the outward orientation, form the 4×4 matrix whose columns are the outward normal followed by the tangent basis:

$$A = \begin{pmatrix} | & | & | & | \\ n & \varphi_\alpha & \varphi_\beta & \varphi_\gamma \\ | & | & | & | \end{pmatrix}. \tag{509}$$

A direct computation gives

$$\det A = \sin^2\alpha\sin\beta. \tag{510}$$

Since $\sin^2\alpha\sin\beta > 0$ on the given domain, the ordered basis $(\varphi_\alpha, \varphi_\beta, \varphi_\gamma)$ agrees with the outward orientation of S^3.
Now consider the parameterization

$$\psi(r,s,t) = \varphi(2r,s,t), \tag{511}$$

where the domain is restricted so that $0 < 2r < \pi$. Then $\psi_r = 2\varphi_\alpha$, $\psi_s = \varphi_\beta$, and $\psi_t = \varphi_\gamma$. Hence

$$(\psi_r, \psi_s, \psi_t) = (\varphi_\alpha, \varphi_\beta, \varphi_\gamma)\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{512}$$

The determinant of this change-of-basis matrix is $2 > 0$, so ψ induces the same orientation.
Finally, define

$$\eta(r,s,t) = \varphi(r,t,s). \tag{513}$$

Then $(\eta_r, \eta_s, \eta_t) = (\varphi_\alpha, \varphi_\gamma, \varphi_\beta)$. Relative to $(\varphi_\alpha, \varphi_\beta, \varphi_\gamma)$, this corresponds to swapping the second and third basis vectors, so the change-of-basis matrix has determinant -1. Therefore, η induces the opposite orientation.
Thus, for a 3-parameterization in $\mathbb{R}^4$, orientation is determined by the ordered tangent basis $(\varphi_\alpha, \varphi_\beta, \varphi_\gamma)$. For hypersurfaces such as $S^3 \subset \mathbb{R}^4$, this can also be compared with an outward normal by checking the sign of a 4×4 determinant.

Example 4.14 (Möbius Strip is Not Orientable)

The Möbius strip can be parameterized by

$$\varphi(u,v) = \left(\left(1 + \frac{v}{2}\cos\frac{u}{2}\right)\cos u, \left(1 + \frac{v}{2}\cos\frac{u}{2}\right)\sin u, \frac{v}{2}\sin\frac{u}{2}\right), \tag{514}$$

where

$$0 \le u \le 2\pi, \qquad -1 \le v \le 1. \tag{515}$$

Here u moves around the central circle, while v moves across the width of the strip. The factors $\cos(u/2)$ and $\sin(u/2)$ encode the half-twist.
Notice that when $u = 0$,
$$\varphi(0, v) = \left(1 + \frac{v}{2}, 0, 0\right), \tag{516}$$
while when $u = 2\pi$,
$$\varphi(2\pi, v) = \left(1 - \frac{v}{2}, 0, 0\right) = \varphi(0, -v). \tag{517}$$
Thus, after going once around the strip, the transverse coordinate v is reversed.
This reversal is the geometric reason the Möbius strip is not orientable. If we start with a normal vector n at some point and move it continuously once around the strip, then after returning to the same point, the normal vector points in the opposite direction, namely $-n$. Therefore, there is no continuous way to choose a unit normal vector on the entire Möbius strip. Hence the Möbius strip is not orientable.

4.3 Line Integrals

We are interested in finding out the mass of a weighted curve.

Definition 4.15 (Scalar Line Integral)

Let $f : \mathbb{R}^n \to \mathbb{R}$ and $C \subset \mathbb{R}^n$ be a curve. Then, the **scalar line integral** of f along C is
$$\int_C f\, ds = \int_a^b f\big(\gamma(t)\big) \|\gamma'(t)\|\, dt \tag{518}$$
where $\gamma : [a, b] \to C$ is any C^1 path.

Proof. It suffices to prove that the line integral is invariant under paths. Let $\gamma_1 : [a, b] \to C$ be a path and $\gamma_2 = \gamma_1 \circ h : [\alpha, \beta] \to C$ be a reparamaterization.

If c is piecewise C^1, then the integral is the sum of the line integrals along all piecewise paths.
$$\int_a^b f\big(c(t)\big) ||c'(t)||\, dt = \sum_{i=0}^{n-1} \int_{\alpha_i}^{\alpha_{i+1}} f\big(c(t)\big) ||c'(t)||\, dt \tag{519}$$

Lemma 4.9 (Arc Length)

The arc length of a curve $C \subset \mathbb{R}^n$ is
$$\int_a^b \|\gamma'(t)\|\, dt \tag{520}$$
where $\gamma : [a, b] \to C$ is any C^1 parameterization of C.

Proof. Just plug in $f = 1$.
$$\int_C 1\, ds = \int_1^b \|\gamma'(t)\|\, dt \tag{521}$$

Lemma 4.10 (Line Integral of Graph)

Definition 4.16 (Vector Line Integral)

Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be a continuous vector field on $\mathbb{R}^n$ and $C \subset \mathbb{R}^n$ be a curve. Then, the **vector line integral** of f along C is

$$\int_C F \cdot d\gamma \coloneqq \int_a^b F\big(\gamma(t)\big) \cdot c'(t)\, dt \tag{522}$$

where $\gamma : [a, b] \to C$ is any C^1 path of a given orientation and $\cdot$ is the Euclidean dot product.

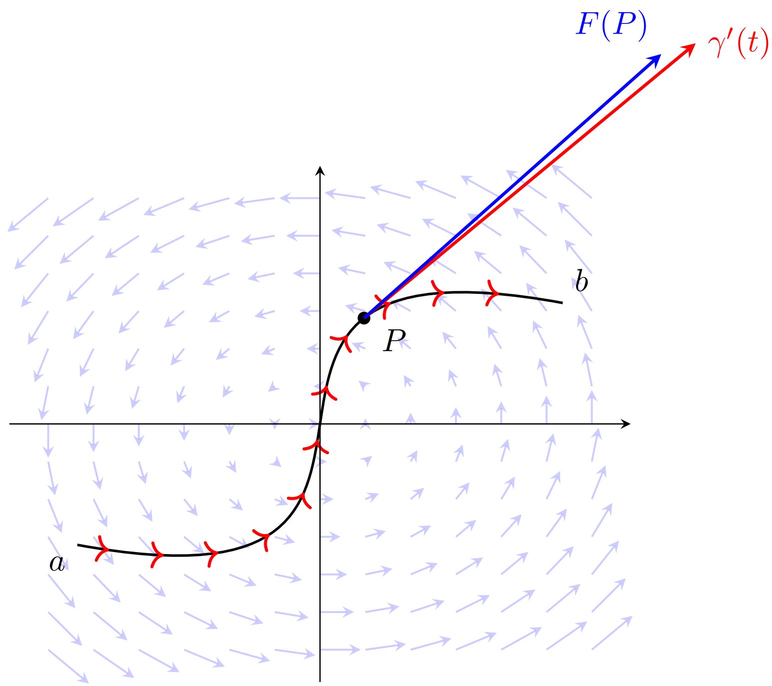


Figure 27: Visualizing a path integral. At a specific point P along the oriented curve, the red tangent vector $\gamma'(t)$ points along the path, while the solid blue vector $F(P)$ represents the local vector field at that exact location. The path integral $\int_C \vec{F} \cdot d\vec{r}$ accumulates the dot product of these two vectors at every point along the curve.

Proof. It suffices to show that if γ_1, γ_2 are two paths onto C of the same orientation, then the vector line integrals are equal.
If γ_1, γ_2 have different orientations, then

$$\int_C F \cdot d\gamma_1 = -\int_C F \cdot d\gamma_2 \tag{523}$$

Again, if C is piecewise C^1, then the integral is the sum of the vector line integrals along all piecewise paths.

$$\int_C F \cdot ds = \sum_{i=1}^k \int_{C_i} F \cdot ds \tag{524}$$

Note that a vector line integral is a generalization of scalar line integrals, so any results holding for vector line integrals also holds for their scalar counterpart.

We now introduce a fundamental theorem about line integrals over gradient fields. Recall the fundamental theorem of calculus. We can extend this to line integrals for functions mapping $\mathbb{R}^n$ to $\mathbb{R}$.

Theorem 4.11 (Invariance of Line Integrals in Conservative Vector Fields)

Given that $F : \mathbb{R}^n \to \mathbb{R}^n$ is a C^1 conservative vector field with $\nabla f = F$ for C^2 function $f : \mathbb{R}^n \to \mathbb{R}$ and path function $p : [a, b] \to \mathbb{R}^n$ is a piecewise C^1 path, then

$$\int_p F \cdot ds = \int_p \nabla f \cdot ds = f\big(p(b)\big) - f\big(p(a)\big) \tag{525}$$

That is, the line integral of any path in a conservative vector field is dependent on the value of f at the endpoints $p(a)$ and $p(b)$.

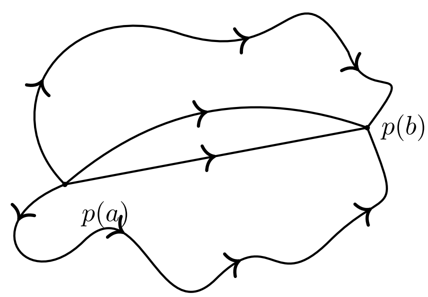


Figure 28

Example 4.15 (Work)

In mechanics, the force F is modeled as a vector field, and the *work* done by F on a particle traveling along a path $\gamma : [a, b] \to \mathbb{R}^n$ is defined as

$$W = \int_\gamma F \cdot ds := \int_a^b F\big(\gamma(t)\big) \cdot \gamma'(t)\, dt \tag{526}$$

Often, F is assumed to be conservative, so it is only necessary that we find the displacement of the particle from its endpoints, resulting in the simplification of the formula.

$$W = \int_p \nabla f \cdot ds = f\big(p(b)\big) - f\big(p(a)\big) \tag{527}$$

Corollary 4.12 (Equivalent Conditions for Vector Field to be Conservative)

The following conditions are equivalent:

1. $F : \mathbb{R}^n \to \mathbb{R}^n$ is a conservative vector field.
2. The line integral of $F : \mathbb{R}^n \to \mathbb{R}^n$ in curve C is path independent; that is, if C_1 and C_2 are two paramaterizations of C,
$$\int_{C_1} F \cdot ds = \int_{C_2} F \cdot ds$$
3. Given that C is a closed loop, the line integral of $F : \mathbb{R}^n \to \mathbb{R}^n$ across C is 0.
$$\oint_C F \cdot ds = 0$$

4. The curl of $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ vanishes
$$\text{curl}\, F = \nabla \times F = \begin{pmatrix} \frac{\partial F}{\partial x} \\ \frac{\partial F}{\partial y} \\ \frac{\partial F}{\partial z} \end{pmatrix} \times \begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} = 0$$
5. The following partial derivatives of $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ are equal
$$\frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x}$$

Proof.

The vector line integral is like starting on at a point and climbing the hills and valleys, creating work as you go up a hill (proportional to the steepness and thus the dot product of your motion vector with the gradient vector field in the path integral) and decreasing the work you put in by going down a hill. Since the path is closed, it is like you are going up and down the same amount overall, so the path integral is zero. Following this analogy, the vector field determined by this function (marked as arrows in the x, y plane) is conservative. If we can construct a closed loop around F where the line integral is nonzero, then it means that we have ended up at a 'higher or lower altitude at the same point.

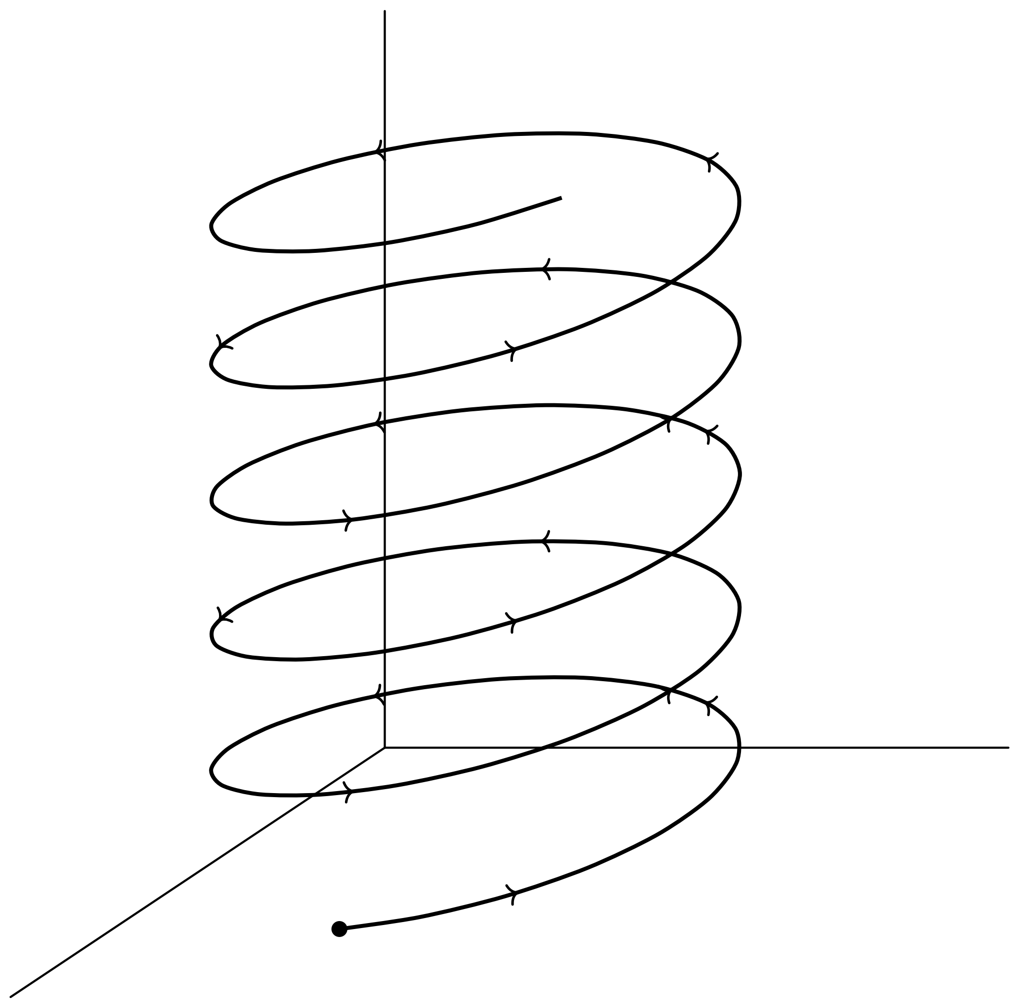


Figure 29: In the solenoidal vector field, we can construct closed loop (a circle going around the origin counterclockwise). There is no "surface" that can be defined such that it contains the solenoid. This means that rather than being a certain landscape, there exist different "levels" of values at one point, like a spiraling staircase. Clearly, as a particle travels through the vector field along the path, it does positive work while it has zero displacement, and clearly, there exists no function that can output both these values as determined by vector field F.

Definition 4.17 (Curvature at a Point)

Let $c : [a, b] \to C \subset \mathbb{R}^3$ be a unit-speed paramaterization of C, meaning that $||c'(t)|| = 1$ for all $t \in [a, b]$, and let $p = c(t_0)$ be a point in C. The *curvature* $\kappa(p)$ at p is a mapping defined

$$\kappa : C \to \mathbb{R}, \;\; \kappa(p) := ||c''(t_0)|| \tag{528}$$

Notice that since we require a unit speed paramaterization of C, we do not need to worry about how a given curve is paramaterized.

Since the curvature is defined pointwise for each point in curve C, we can integrate over all the curvatures in C to define the total curvature.

Definition 4.18 (Total Curvature)

The *total curvature* of a curve $c : [a, b] \to C \subset \mathbb{R}^3$ is the scalar line integral

$$\int_C \kappa \, ds \tag{529}$$

Theorem 4.13 (Fary-Milnor Theorem)

Given a unit speed paramaterization $c : [a, b] \to C \subset \mathbb{R}^3$, if C is closed (that is, $c(a) = c(b)$), then

$$\oint_C \kappa \, ds \geq 2\pi \tag{530}$$

and equals 2π only when C is a circle. Furthermore, if C is a closed space curve with

$$\oint_C \kappa \, ds \leq 4\pi \tag{531}$$

then C is "unknotted." That is, C can be continuously deformed without every intersecting itself into a planar circle. Therefore, for knotted curves C, we have

$$\oint_C \kappa \, ds > 4\pi \tag{532}$$

4.4 Surface Integrals

Surface integrals are the 2-dimensional analogue, or the double integral version, of line integrals. It is the integration of surfaces. A physical interpretation of a scalar surface integral is the weighted surface area of a certain surface.

Definition 4.19 (Scalar Surface Integrals)

Let $f : \mathbb{R}^3 \to \mathbb{R}$ be a C^1 scalar field defined on a paramaterized surface $S \subset \mathbb{R}^3$ with paramaterization $\varphi : D \subset \mathbb{R}^2 \to \mathbb{R}^3$. That is, $\varphi(D) = S$. We define the integral f over S to be

$$\iint_S f \; dS = \iint_S f(x, y, z) \; dS \tag{533}$$

$$= \iint_D f\big(\varphi(u, v)\big) \bigg|\bigg| \frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v} \bigg|\bigg| \, du \, dv \tag{534}$$

Note that this will require us to transform f, a function of x, y, z, into the function $f \circ \varphi$ of u, v. Additionally, if the paramaterization of the surface S is not defined, then it one must be constructed. It is also clear that if S is a union of surfaces S_i, then its surface integral is the sum of the surface integrals of the S_i's.

Letting the scalar field f be the constant field equal to 1, the scalar surface integral measures the surface area of S.

$$A(S) = \iint_S \; dS = \iint_D \bigg|\bigg| \frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v} \bigg|\bigg| \, du \, dv$$

It is easy to see that the orientation of the paramaterization φ does not affect scalar surface integrals, since the sign of the orientation gets nullified by the absolute value sign over $||\frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v}||$.

Its physical interpretation is to measure the rate at which a fluid (determined by a vector field F) is crossing a given surface S. It also has many applications in electromagnetism.

Definition 4.20 (Vector Surface Integrals)

Let $S \subset \mathbb{R}^3$ be an oriented surface, $\varphi : D \subset \mathbb{R}^2 \to \mathbb{R}^3$, and $F : S \subset \mathbb{R}^3 \to \mathbb{R}^3$ be a vector field. The **surface integral** of F over S is defined

$$\iint_S F \cdot dS = \iint_S (F \cdot n)\, dS = \iint_D F\big(\varphi(u,v)\big) \cdot \left(\frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v}\right) du\, dv \tag{535}$$

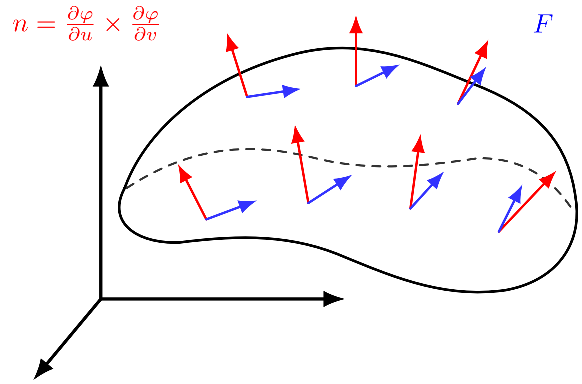


Figure 30: We can visualize it as summing up the dot product of the vector field and the normal vector to the surface.

Since we are now talking about vector fields, the orientation of the paramaterization is now significant. Visually, if the orientation of the surface S generally aligns with the vector field F, then the integral will be positive (since two vectors α, β generally pointing in the same direction implies that $\alpha \cdot \beta > 0$). The orientation of the paramaterization, which is dependent on $\frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v}$, determines the direction of the normal vector n (since it is defined to be $(\frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v}) / \big|\big|\frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v}\big|\big|$. Therefore, changing the orientation of φ will reverse the direction of n, which will then reverse the sign of the integral since n now points in the opposite direction of the vector field F than it previously did (by reversing the sign of the dot products). This is formalized in the theorem below.

Theorem 4.14 (Invariance Vector Surface Integrals Under Parameterization)

Let S be an oriented surface and let φ_1 and φ_2 be two regular parameterizations with F a continuous vector field defined on S. Then, assuming φ_1 is orientation preserving,

$$\varphi_2 \text{ is orientation preserving} \implies \iint_{\varphi_1} F \cdot dS = \iint_{\varphi_2} F \cdot dS \tag{536}$$

$$\varphi_2 \text{ is orientation reversing} \implies -\iint_{\varphi_1} F \cdot dS = \iint_{\varphi_2} F \cdot dS \tag{537}$$

Theorem 4.15 (Surface Integral Formula for Graphs)

Given that we have the graph of a function $g : \mathbb{R}^2 \to \mathbb{R}$ rather than a general surface, we can paramaterize it simply as

$$\varphi(u,v) := \big(u, v, g(u,v)\big) \tag{538}$$

This means that

$$\frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v} = \begin{pmatrix} -\frac{\partial g}{\partial u} \\ -\frac{\partial g}{\partial v} \\ 1 \end{pmatrix} \implies \bigg|\bigg| \frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v} \bigg|\bigg| = \sqrt{1 + \Big(\frac{\partial g}{\partial u}\Big)^2 + \Big(\frac{\partial g}{\partial v}\Big)^2} \tag{539}$$

So we can simplify the equation for the surface area S of the graph of g over the region D in the xy-plane, as

$$A(S) = \iint_S dS = \iint_D \bigg|\bigg| \frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v} \bigg|\bigg| \, dA \tag{540}$$

$$= \iint_D \sqrt{1 + \Big(\frac{\partial g}{\partial u}\Big)^2 + \Big(\frac{\partial g}{\partial v}\Big)^2} \, du \, dv \tag{541}$$

With the same g, we can find the weighed surface area of S over the scalar function $f : \mathbb{R}^3 \to \mathbb{R}$ with the formula

$$\iint_S f \, dS = \iint_D f\big(u, v, g(u,v)\big) \sqrt{1 + \Big(\frac{\partial g}{\partial u}\Big)^2 + \Big(\frac{\partial g}{\partial v}\Big)^2} \, du \, dv \tag{542}$$

Finally, with the same graph g, the surface integral over the vector field F is

$$\iint_S F \cdot dS = \iint_D F\big(\varphi(u,v)\big) \cdot \Big(\frac{\partial \varphi}{\partial u} \times \frac{\partial \varphi}{\partial v}\Big) \, du \, dv \tag{543}$$

$$= \iint_D \bigg(F_1(u,v)\Big(-\frac{\partial g}{\partial u}\Big) + F_2(u,v)\Big(-\frac{\partial g}{\partial v}\Big) + F_3(u,v) \bigg) \, du \, dv \tag{544}$$

4.5 Green Theorem

Recall the differential notation for writing line integrals. For 2 and 3 dimensions, it is written as

$$\int_C F \cdot ds = \int_C F \cdot (dx, dy) = \int_C F_1 \, dx + F_2 \, dy \tag{545}$$

$$\int_C F \cdot ds = \int_C F \cdot (dx, dy, dz) = \int_C F_1 \, dx + F_2 \, dy + F_3 \, dz \tag{546}$$

Green's Theorem gives the relationship between a line integral around a simple closed curve C and a double integral over the plane region D bounded by C.

Theorem 4.16 (Green's Theorem in $\mathbb{R}^2$)

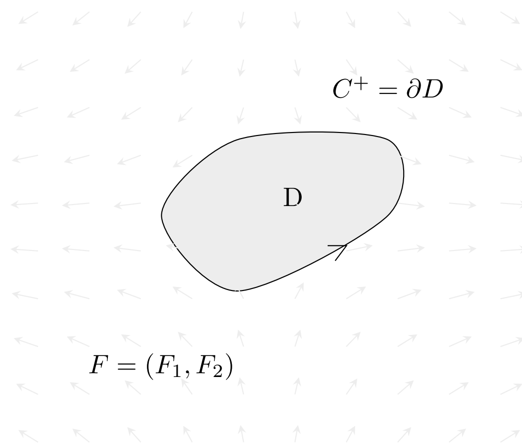
Let there be a 2-dimensional C^1 vector field F on $\mathbb{R}^2$ defined on a simple oriented closed piecewise-smooth curve C and its bounded region $D \subset \mathbb{R}^2$ (that is, $C = \partial D$). Let the orientation of the path of C be such that it is traveling *counterclockwise*, i.e. a point traveling through C would see the region D to its *left*, denoted as C^+ and the clockwise orientation as C^-. Then,

$$\oint_{C^+} F_1 \, dx + F_2 \, dy = \iint_D \Big(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\Big) \, dx \, dy \tag{547}$$

By reversing the orientation, it is clear that we have

$$\oint_{C^-} F_1 \, dx + F_2 \, dy = -\iint_D \Big(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\Big) \, dx \, dy \tag{548}$$

Note that this theorem is expressed in terms of the components of the vector field F.



Green's theorem has many applications in physics. For example, in order to solve two-dimensional flow integrals measuring the sum of fluid outflowing from a volume, Green's theorem allows us to calculate the total outflow summed about an enclosing area .

Corollary 4.17

Let D be a region for which Green's theorem applies with positively oriented boundary ∂D. Then, the area of D can be computed with the formula

$$A(D) = \frac{1}{2} \oint_{\partial D} x\, dy - y\, dx \tag{549}$$

Green's theorem can be used to determine the area of centroid of plane figures solely by integrating over the perimeter.

4.6 Stokes' Theorem

Green's theorem relates line integrals to double integrals. Stokes' theorem generalizes Green's theorem by relating line integrals to surface integrals of 2-dimensional surfaces embedded in $\mathbb{R}^3$.

Theorem 4.18 (Stokes' Theorem)

Let S be an oriented regular surface defined by paramaterization $\varphi : D \subset \mathbb{R}^2 \to \mathbb{R}^3$, and let the image of the boundary ∂D under φ be the boundary ∂S of S.

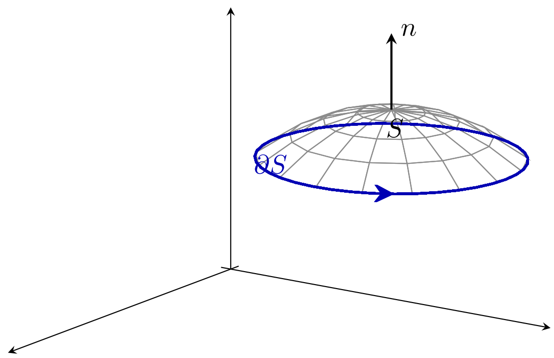


Figure 31: The orientation unit vector n of S induces the positive orientation of ∂S, denoted ∂S^+. Visually, if you are walking along the curve with your head is pointing in the same direction as the unit normal vectors while the surface is on the left then you are walking in the positive direction on ∂S.

Given that F is a C^1 vector field defined on S, then

$$\iint_S \operatorname{curl} F \cdot dS = \iint_S \big(\nabla \times F\big) \cdot dS = \oint_{\partial S^+} F \cdot ds \tag{550}$$

If S has no boundary, that is, if the image of $p' = \partial S$ is not a simple closed curve, then the integral is 0.

The above theorem implies that the vector surface integral of a surface without a boundary (i.e. a closed graph, such as a sphere) is always 0 along the curl of any C^1 field. Geometrically, this means that given a closed solid S with field $\nabla \times F$, the rate of flow of the vector field into S is equal to the flow out of S.

4.7 Gauss' Theorem

The divergence theorem relates the flux of a vector field through a closed surface to the divergence of the field in the volume enclosed.

Theorem 4.19 (Gauss' Divergence Theorem)

Let V be a subset of $\mathbb{R}^3$. Denote by ∂V the oriented closed surface that bounds V (with outward pointing normal orientation vectors), and let F be a C^1 vector field defined on a neighborhood of V. Then,

$$\iiint_V \operatorname{div} F \, dV = \iiint_V (\nabla \cdot F)\, dV = \oiint_{\partial V} F \cdot dS = \oiint_{\partial V} (F \cdot n)\, dS \tag{551}$$

where the two left-most integrals are volume integrals, and the two right-most integrals are surface integrals.

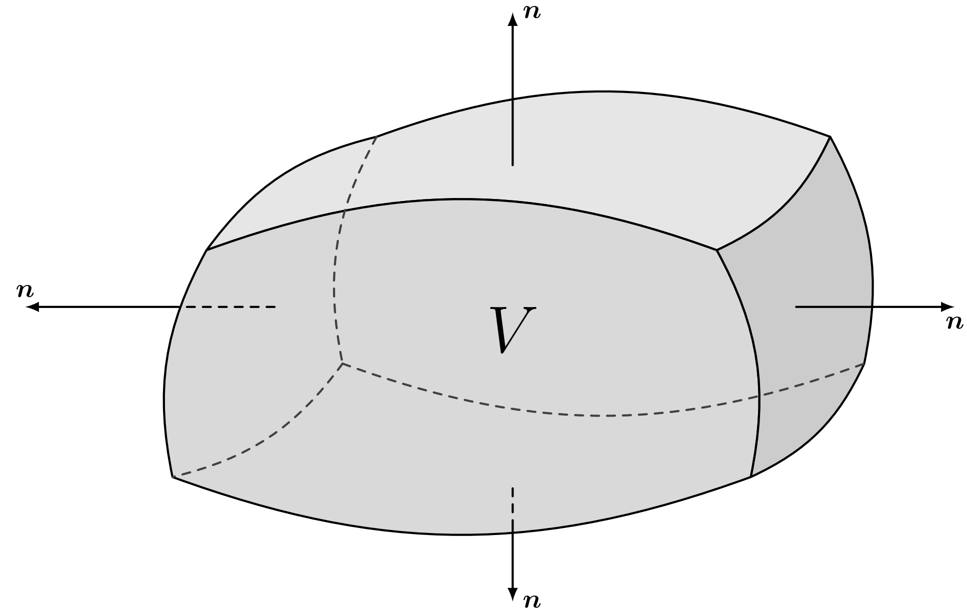


Figure 32

Intuitively, this makes sense; the volume integrals represent the total of the sources in volume V, and the right hand side represents the total flow across the boundary ∂V.

5 Sequences of Functions